RDK-CKKS: A Framework for Rigorous Error Analysis of CKKS Computation

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Abstract. The CKKS scheme allows approximate arithmetic on encrypted data, and has found widespread use in privacy-preserving computation. A rigorous control of its homomorphic error is critical for both precision and security guarantees. In particular, in some settings such as $\mathsf{IND\text{-}CPA}^D$ or threshold settings, the latter computation error should be hidden at the time of decryption. This is achieved by noise flooding techniques, which require a precise upper bound on the computational error magnitude. When dealing with discrete data and a single decryptor, this is addressed by ensuring statistical correctness, which also requires a precise error management. Yet, current approaches to CKKS error analysis provide either rigorous worst-case bounds that are far off empirical measures, or average-case bounds that rely on various heuristics and do not provide strong enough guarantees for security.

In this work, we develop a framework based on so-called randomization keys (RDKs), which are encryptions of zero injected at key-switching and rescaling steps to render the homomorphic error amenable to rigorous probabilistic analysis. Combined with tail bounds, this yields, to our knowledge, the first fully rigorous error analysis framework for CKKS computations. Our framework is modular: RDKs make the error of each elementary operation amenable to rigorous bounds, after which the error of a circuit follows by standard numerical error propagation.

Our rigorous bounds are comparable to heuristic ones and closely match empirical measures, but cost additional evaluation material (RDKs) generated by the secret key holder. In the context of rigorous bootstrapping, this induces a communication overhead of 27% over the bootstrapping keys for a single evaluation, and of a further 8% for each additional evaluation. We show that, when a circuit is evaluated repeatedly, most of these keys can be reused across evaluations.

As our main application, we derive a complete and rigorous analysis of binary CKKS bootstrapping, and a full, rigorous, parameter set for the bootstrap-and-clean technique. Compared to a standard Gaussian-type heuristic approach, our parameters only require a modulus increase for bootstrapping of $\approx 3\%$, while providing rigorous precision estimates.

1 Introduction

Background and motivation. Fully homomorphic encryption (FHE) enables a party to evaluate a computation on encrypted data [22], making it a central tool for privacy-preserving computation, from outsourced machine-learning inference to threshold cryptography. Among practical FHE schemes, CKKS [13] supports approximate SIMD arithmetic over large vectors of complex numbers, while its efficient bootstrapping permits computations of unbounded depth. These properties have made CKKS particularly effective for computations over real-valued data, but also for a number of exact computations (bits [19,1], small integers [2], large integers [26,9,21]) emulated with CKKS approximate arithmetic.

Approximation is both the main feature of CKKS and the source of a fundamental security difficulty. A CKKS ciphertext does not decrypt exactly to its intended message m, but rather to a noisy version $\widetilde{m} = m + e$, where $e \in \mathbb{C}^{N/2}$, the *computation error*, mixes numerical and cryptographic errors accumulated during encoding, encryption and homomorphic computation. In particular, it depends on the secret key and on the rounding operations performed by the homomorphic algorithms. For correctness in the approximate setting, it suffices that e remains small enough for the application. In deployment, however, e can itself leak information about the secret key. This issue is modeled by the $\mathsf{IND\text{-}CPA}^D$ security notion [29], first identified for CKKS approximate decryption [29] and subsequently extended to exact schemes when the decryption failure probability is non-negligible [12,10]. The same phenomenon is the main challenge when building threshold variants of FHE schemes [4,18,8,27,15], where distributed decryption reveals information about the computation error.

In the exact single decryptor setting, ensuring statistical correctness by a rigorous error bound addresses this security issue: the exact result is recovered at decryption and no information about the computation error leaks. For approximate computation or threshold decryption, the standard countermeasure is *noise flooding* [28]. Assuming $\|e\|_\infty \leq B$, a classical argument shows that adding a uniform, independent error e_{fl} of magnitude $B \cdot 2^\lambda$ hides e up to λ bits of statistical security[4]: decryption returns $m + e + e_{\mathsf{fl}}$, which is statistically close to $m + e_{\mathsf{fl}}$, and security follows as the latter no longer depends on the computation error and can be used to simulate decryption given only the output m, in both approximate single decryptor and (exact or approximate) threshold settings. In all cases, a rigorous error bound B is therefore crucial to achieve security. Notably, noise flooding converts a quantitative upper bound on the computation error directly into a security guarantee. If the bound itself fails with probability $\varepsilon_{\mathrm{err}}$, this must be included in the final security loss; and if flooding is calibrated on an estimate $\widehat{B} = 2^{-k}B$, the resulting guarantee only provides about $\lambda - k$ bits of statistical security. Thus, every bit by which the computation error is underestimated costs one bit of effective security, while every bit of overestimation introduces one unnecessary bit of flooding noise, reducing the precision of the released result and inflating parameters. Error estimation in FHE is therefore

[4] For simplicity, we ignore multiplicative factors here, e.g., the ring dimension N.

not merely a correctness problem, but a central part of the security analysis. The required estimate must hold for the complete computation, simultaneously over every released slot, and with a failure probability such as 2^{-128}, far beyond what can be validated by experiments: these tails must follow from a mathematical analysis, and yet remain close to the actual error for the system to be efficient.

The principal computation errors in CKKS arise from rounding. After a ciphertext multiplication, the scaling factor must be reduced by $\mathsf{Rescale}$, which maps $(b,a) \in \mathcal{R}_{qQ}^2$ to $\left(\lfloor \frac{b}{q} \rceil, \lfloor \frac{a}{q} \rceil\right) \in \mathcal{R}_Q^2$. Similarly, key switching (used for relinearization and slot rotations) lifts a ciphertext to a larger modulus PQ and then goes back to Q via $\mathsf{ModDown}$, a sub-case of $\mathsf{Rescale}$ for $q = P$; the error inherited from the switching key is shrunk by $\mathsf{ModDown}$ and, for a suitable auxiliary modulus P, is negligible compared to the rounding error. Rigorous estimation of the computation error thus boils down to bounding rounding errors, which, when dividing the modulus by q, are of the form

$$E_{\mathrm{rnd}} := \left[\frac{a}{q}\right]_1 \cdot s + \left[\frac{b}{q}\right]_1 \quad , \tag{1}$$

where $[\cdot]_1$ is the centered modulo 1 function applied coefficient-wise. After decoding, such a plaintext error results in the slot error

$$\mathbf{e}_{\mathrm{rnd}} = \frac{1}{\Delta}\left(\mathsf{DFT}\left(\left[\frac{a}{q}\right]_1\right) \odot \widehat{\mathbf{s}} + \mathsf{DFT}\left(\left[\frac{b}{q}\right]_1\right)\right) \quad , \tag{2}$$

with DFT the negacyclic Discrete Fourier Transform used for decoding and defining $\widehat{\mathbf{s}} = \mathsf{DFT}(s)$.

Prior works. A fully deterministic analysis treats every rounded coefficient adversarially: each is bounded by $1/2$, a slot sums N of them through the canonical embedding, and $\|\widehat{\mathbf{s}}\|_\infty \leq h$ for an h-sparse key, giving elementary error bounds of order Nh/Δ. These bounds are rigorous for arbitrary intermediate ciphertexts, but much larger than the errors observed in practice; using them throughout a circuit forces large scales and modulus chains and substantially reduces the available depth. Much sharper estimates are obtained by modeling the rounded coefficients as independent, centered uniform variables: their contributions to a slot largely cancel, the typical magnitude grows as $\sqrt{N}$ rather than N, and the Fourier coefficients of a random sparse key have typical magnitude $\sqrt{h}$ rather than h. This suggests elementary estimates of order $\sqrt{Nh}/\Delta$, up to factors accounting for the number of slots and the target tail probability. More elaborate approaches track variances or individual noise components through the circuit and use Gaussian approximations to estimate the resulting precision loss [16,5], while purely empirical approaches sample the error at the end of a computation.

Costache et al. [16] derive average-case estimates for CKKS by tracking coefficient variances and applying Gaussian tail formulas, invoking a central limit argument for individual coefficients of polynomial products [16, Theorem 3] and modeling the whole product as jointly Gaussian with diagonal covariance [16,

Heuristic 1 and the Small-S assumption]. Bossuat et al. [5] develop a more detailed model that tracks noise separately in the ciphertext components and propagates concrete message and noise values in the canonical embedding; the conversion from variances to failure probabilities, however, still relies on the expression $\mathrm{erf}(\alpha/\sqrt{2})^N$ of their Lemma 4, which amounts to assuming that the noise coefficient vector is jointly Gaussian with independent coordinates. Both sets of authors explicitly describe their analyses as heuristic, and the latter acknowledge that they do not support a proof of $\mathsf{IND\text{-}CPA}^D$ security.

As the canonical embedding turns polynomial multiplication into pointwise multiplication, whenever the noise contains a product of two independently sampled ring elements, as is unavoidable after any homomorphic multiplication, the slots are products of random variables whose law is never Gaussian regardless of N, and a jointly Gaussian coefficient vector would have a jointly Gaussian image under this fixed invertible linear map. Moreover, a qualitative central limit theorem does not specify how accurately a Gaussian approximation predicts tail probabilities at finite N: an additive error of order $1/\sqrt{N} = 2^{-8}$ at $N = 2^{16}$ leaves an uncertainty far larger than the 2^{-128} one seeks to guarantee, and agreement with empirical variances or observed maxima does not provide any guarantee regarding the large-deviation regime required for security. These models therefore provide practical guidance for circuit design and parameter selection, while leaving the cryptographic guarantee of tails unresolved.

A further difficulty concerns the inputs to rounding operations. The RLWE assumption provides pseudorandomness for fresh samples, but does not imply that the components of an evaluated ciphertext are independent uniform variables: multiplication, rescaling, key switching, and repeated use of evaluation keys introduce algebraic structure and correlations inherited from the preceding computation, so one cannot simply assume at each rounding step that the ciphertext components are fresh. This difficulty has been addressed, in a different setting, by Bernard et al. [3]. To control the modulus switching error of FHEW/TFHE/FINAL, they add an encryption of zero to the ciphertext before the rounding, and retry with a different one until the resulting drift vector passes a public quality test. The noise is then reset by the blind rotation that follows, and a single such test per bootstrapping suffices. In CKKS the rounding errors of rescaling and key switching are never reset and accumulate across the whole circuit, so that certifying one rounding step says nothing about the accumulated error. Finally, the treatment of the secret key matters: an averaged guarantee over key generation does not establish the same bound with the same failure probability for every fixed key, and rounding errors need not be identically distributed across slots, since a rounding contribution us has variance at slot i proportional to $|\widehat{s}_i|^2$. Slot-wise bounds, or a uniform bound using $\|\widehat{\mathbf{s}}\|_\infty$, account for this variability.

This leaves a substantial gap. Deterministic worst-case bounds are proven but too conservative for efficient deployment, whereas the much smaller bounds used in practice depend on distributional assumptions that are difficult to justify for non-fresh ciphertexts and on heuristics that do not provide strong enough

guarantees for security. When these estimates are used to ensure statistical correctness or to calibrate noise flooding, the missing tail guarantees become a gap in the security argument [20]. This leads to the central question of this work:

Can one obtain bounds for the CKKS computation error that are as tight as the heuristic bounds used in practice and that provide sufficient guarantees for security?

1.1 Our Contributions

We answer this question positively through a rerandomization framework based on *randomization keys* (RDKs). An RDK is a fresh encryption of zero that is injected immediately before the rounding steps of Rescale and ModDown. This refresh makes each relevant ciphertext component marginally uniform at the rounding modulus, independently of the preceding ciphertext, and thereby supplies the exact randomness needed for a rigorous probabilistic analysis. Combining this mechanism with sub-Gaussian tail bounds in the canonical embedding gives high-probability slot-wise error estimates of order $\widetilde{O}(\sqrt{Nh}/\Delta)$, matching the heuristic analyses while making no Gaussian assumption on the accumulated computation error. We also show that the keys consumed at the ModDown steps can be safely reused across repeated evaluations of a circuit, whereas those consumed at rescaling cannot. We extend the elementary bounds to matrix–vector multiplication, polynomial evaluation, binary bootstrapping, and the subsequent cleaning procedure.

As a concrete application, we derive a full set of parameters for binary CKKS. Compared to a worst-case error analysis, our parameters reduce the required modulus by 151 bits, increasing the number of available levels after bootstrapping from 12 to 20. The resulting parameters are only about 3% larger than those obtained from a Gaussian heuristic, while the RDK material amounts to roughly 27% of the switching-key storage.

1.2 Technical overview

Randomizing rounding inputs. We operate a simple tweak to randomize Rescale and KeySwitch in order to enable rigorous error analysis. Before the coefficient-wise rounding, we add an RDK $(B, A) = (-As + E, A)$, where A is uniform and E is a small, independently sampled RLWE error, for a secret key s invertible modulo the working modulus. When the RDK is independent of the pair (b, a) being rounded, adding it makes each of the two ciphertext components marginally uniform, regardless of the distribution of the input, while the decrypted value changes only by E. The two refreshed components need not be independent of one another: their individual uniformity is sufficient for the rounding analysis. We provide precise independence conditions required at each operation in Section 4.

Bounds for atomic operations. Within either uniform component, the coefficients of the centered rounding residual are independent, centered, and bounded.

Each Fourier coordinate is a weighted sum of these coefficients, so concentration bounds give a magnitude proportional to $\sqrt{N}$, with an explicit factor for the desired failure probability. In the decrypted rounding error, the contribution from the a-component is multiplied by $\widehat{s}_i$ (Equation 2), which explains the form of our slot-wise bounds for rescaling and key switching:

$$\Pr\left[|e_{\mathrm{op},i}| > \frac{\lambda_{\mathrm{op}}(\varepsilon)}{\Delta}\left(|\widehat{s}_i| + 1 + \alpha_{\mathrm{op}}\right)\right] \leq \varepsilon \ , \qquad \mathrm{op} \in \{\mathsf{Rescale}, \mathsf{KeySwitch}\} \ .$$

Here Δ is the output scaling factor, $\lambda_{\mathrm{op}}(\varepsilon) = \sqrt{(N/6)\log(c_{\mathrm{op}}/\varepsilon)}$ with c_{op} explicit, and α_{op} accounts for the RDK and switching-key errors, attenuated by the rescaling divisor or the auxiliary modulus. A union bound over slots gives a uniform estimate in terms of $\|\widehat{\mathbf{s}}\|_\infty$; enforcing a suitable bound on this norm at key generation yields elementary error bounds of order $\widetilde{O}(\sqrt{Nh}/\Delta)$ for an h-sparse secret key. The estimates hold for each fixed key satisfying this bound.

A numerical model for CKKS. These atomic bounds give CKKS multiplication and rotation a precise numerical semantics, analogous to fixed-point arithmetic with explicit additive error bounds under clear conditions (failure probability, uniformity and independence requirements). For a given circuit, we assign failure budgets ε_j to the initial error bounds and to the bounds for successive operations; with probability at least $1 - \sum_j \varepsilon_j$ these bounds hold simultaneously, and conditioned on this event, error propagation through the circuit becomes a deterministic numerical-analysis problem. Unlike a first-order analysis, which discards products of errors and higher-order remainders and therefore cannot by itself certify the flooding parameters, our model accounts for the full error. In Section 5, we use our framework to rigorously analyze classical CKKS operations, namely linear algebra (matrix–vector multiplication), polynomial evaluation, and binary bootstrapping. We track errors through the weighted rotations and sums of matrix–vector multiplication, rigorously analyze the Paterson-Stockmeyer algorithm in the Chebyshev basis retaining the nonlinear terms, and combine both stages in binary bootstrapping. Our approach increases the modulus cost by only about 3% compared to heuristic analyses, at the price of generating and storing randomization keys.

From numerical bounds to security. The outcome of our rigorous analysis is a precise bound B on the slot-wise decryption error (from which one can also derive a precise bound for the error in coeffs) together with an explicit failure probability ε. This is the quantitative information required to ensure statistical correctness for exact computations, or to calibrate noise flooding in an $\mathsf{IND\text{-}CPA}^D$ security argument (for approximate computation) or a threshold FHE simulation-security argument. Our framework yet imposes a relaxation of these security models: RDKs are sampled by the challenger and consumed internally when evaluation queries are answered, so that the hypotheses of our error analysis hold by construction. Our framework does not guarantee security against an adversary holding the RDKs, such as the evaluator itself.

Paper Organization. The rest of this paper is organized as follows. Section 2 provides basic reminders about CKKS. In Section 3, we define randomization keys and our randomized Rescale and KeySwitch algorithms, and provide basic tools for the analysis of atomic operations, which is described in Section 4. We then analyze linear algebra, polynomial evaluation, and binary bootstrapping in Section 5. Section 6 bridges our error analysis with IND-CPA^D (and threshold) security based on noise flooding. We conclude with detailed parameters for exact Binary CKKS and experiments supporting the tightness of our bounds in Section 7.

2 Preliminaries

We let N denote a power of two and $\mathcal{R}$ denote the cyclotomic ring $\mathbb{Z}[X]/(X^N+1)$. For an integer $q > 0$, we define $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$. For $u \in \mathbb{R}[X]$, we denote by $[u]_1$ the polynomial obtained by applying the centered modulo 1 function, with values in $(-\frac{1}{2}, \frac{1}{2}]$, to coefficients of u. For a finite set S, $U(S)$ denotes the uniform distribution over S; for an odd integer r, $\mathcal{U}_r$ denotes the uniform distribution over $\frac{1}{r}\mathbb{Z} \cap (-\frac{1}{2}, \frac{1}{2}]$; and $\mathcal{DG}_\sigma$ denotes the discrete Gaussian distribution over $\mathbb{Z}$, with parameter σ, i.e. $\mathcal{DG}_\sigma(x) \propto \exp(-x^2/(2\sigma^2))$. A polynomial is said to follow one of these distributions when its coefficients are i.i.d. with that distribution.

We let ζ_{2N} denote a primitive $2N$-th root of unity, and we define the order-$2N$ negacyclic DFT as

$$\begin{aligned}\mathsf{DFT}_N \colon \mathcal{R} &\to \mathbb{C}^{N/2} \\ a &\mapsto \left(a(\zeta_{2N}^{5^k})\right)_{0 \le k < N/2}.\end{aligned}$$

The inverse map is denoted iDFT_N. We drop the index N to ease notation as N does not vary. We use the following convention for notation: for polynomial $A \in \mathcal{R}$, we let $\widehat{\mathbf{a}} \in \mathbb{C}^{N/2}$ refer to $\mathsf{DFT}(A)$, and conversely.

For a real value x, we let $\lfloor x \rceil$ denote its nearest integer, and extend this notation to vectors and polynomials component-wise. For $Q, P \ge 0$, for $a \in \mathcal{R}_Q$, we let $\overline{a} \in \mathcal{R}_{PQ}$ denote the lifting of a to $\mathcal{R}_{PQ}$, i.e. the unique polynomial in $\mathcal{R}_{PQ}$ with coefficients in $(-Q/2, Q/2]$ which satisfies $\overline{a} = a \bmod Q$.

2.1 Reminders about CKKS

We recall the elements of CKKS [13] that our framework modifies.

Encoding, decoding, encryption and decryption. For a scaling factor Δ, $\mathsf{Ecd}(\mathbf{m}) = \lfloor \Delta\, \mathsf{iDFT}(\mathbf{m}) \rceil \in \mathcal{R}$ and $\mathsf{Dcd}(\mathsf{pt}) = \mathsf{DFT}(\mathsf{pt})/\Delta$. A ciphertext at modulus Q under s is a pair $(b, a) \in \mathcal{R}_Q^2$ with $b = -a \cdot s + \mathsf{pt} + E$, where $a \sim U(\mathcal{R}_Q)$ and $E \sim \mathcal{DG}_\sigma$; decryption computes $\mathsf{Dec}_s(b, a) = b + a \cdot s$ followed by Dcd. For a plaintext-level error $E \in \mathcal{R}$, we call $\widehat{\mathbf{e}}^{(\Delta)} := \mathsf{DFT}(E)/\Delta$ for the associated *message-level* error, and $\widehat{\mathbf{e}}_i^{(\Delta)}$ its i-th slot.

Invertible secret key. We shall assume throughout the paper that all secret keys are invertible in $\mathcal{R}_Q$. For $Q = \prod_{i=0}^{L} q_i$, for an s sampled uniformly from $\mathcal{R}_Q$, the probability of this event is $\geq 1 - N\sum_{i=0}^{L} q_i^{-1}$, a quantity which is $\geq 1/2$ as soon as $q_i \geq 2NL$ for all i. Under the heuristic assumption that this probabilistic estimate still holds for uniformly sampled ternary keys with fixed Hamming weight, the inversibility assumption can thus be made with a loss of at most one bit of security.

The two rounding steps. $\mathsf{Rescale}$ maps $(b, a) \in \mathcal{R}_{qQ}^2$ to $(\lfloor b/q \rceil, \lfloor a/q \rceil) \in \mathcal{R}_Q^2$; $\mathsf{ModDown}$ is the case $q = P$, mapping $\mathcal{R}_{PQ}^2$ to $\mathcal{R}_Q^2$. Key switching uses a gadget decomposition: let $Q = D_0 \cdots D_{\mathsf{dnum}-1}$ be a decomposition of Q into dnum pairwise coprime factors, and let $(\overline{a}_0, \ldots, \overline{a}_{\mathsf{dnum}-1})$ be the lifted CRT decomposition of a, i.e. $\overline{a}_i$ is the lift to $\mathcal{R}_{PQ}$ of $a \bmod D_i$, with $\|\overline{a}_i\|_\infty \leq D_i/2$; we denote this decomposition by $\mathsf{ModUp}(a)$. Let $\mathsf{swk}^{(i)} = (B_{\mathsf{swk}}^{(i)}, A_{\mathsf{swk}}^{(i)}) \in R_{PQ}^2$ satisfy $B_{\mathsf{swk}}^{(i)} + A_{\mathsf{swk}}^{(i)} \cdot s = P g_i s' + E_{\mathsf{swk}}^{(i)}$, where $g_i = 1 \bmod D_i$ and $0 \bmod D_j$ for $j \neq i$. Key switching from s' to s returns

$$\mathsf{KeySwitch}\big((b, a), \mathsf{swk}\big) = (b, 0) + \mathsf{ModDown}\Big(\textstyle\sum_i \overline{a}_i \cdot B_{\mathsf{swk}}^{(i)},\ \sum_i \overline{a}_i \cdot A_{\mathsf{swk}}^{(i)}\Big) \in \mathcal{R}_Q^2,$$

so that $\mathsf{Dec}_s(ct') = \mathsf{Dec}_{s'}(ct) + E_{\mathsf{ks}}$, where E_{ks} collects the error of the switching key and the rounding error of $x \mapsto \lfloor x/P \rceil$.

Multiplication, rotation, levels. Ciphertext multiplication tensors $(b, a), (b', a')$ into $(d_0, d_1, d_2) = (bb',\ ab' + a'b,\ aa')$, key switches $(0, d_2)$ from s^2 to s using a relinearization key, and rescales the result by $q \approx \Delta$ to restore the scaling factor. Plaintext–ciphertext multiplication is the case $a' = 0$ and requires no relinearization. Rotation Rot_j applies a key switch to $(b(X^{5^j}), a(X^{5^j}))$ and induces a key-switching error. In RNS-CKKS we fix primes $q_0, \ldots, q_L$, set $Q_\ell = \prod_{i \leq \ell} q_i$, call ℓ the level of a ciphertext in $R_{Q_\ell}^2$, and define scaling factors by $\Delta_L = \Delta$, $\Delta_{\ell-1} = \Delta_\ell^2/q_\ell$.

2.2 Probabilistic toolbox

We review probabilistic tools which lie at the core of our approach. The first one is the following uniformity lemma, on which the rerandomization argument of Section 3 rests.

Lemma 1 (Uniformity Lemma). *Let $\mathbb{G}$ be a finite additive group, and let X, Y be independent random variables over $\mathbb{G}$. If X is uniformly distributed over $\mathbb{G}$, then $X + Y$ is uniform over $\mathbb{G}$ and independent of Y.*

Proof. For every $y \in G$ and every $x \in G$, $\Pr[X + Y = x | Y = y] = \Pr[X = x - y] = 1/|G|$, which doesn't depend on y. Thus, $X + Y$ is uniform over G and independent of Y.

To turn average-case CKKS analysis into rigorous arguments require tail bounds for the negacyclic DFT of various sources of randomness occurring in

FHE (uniform in $\mathcal{R}_Q$, discrete Gaussian). We group these results in the following theorem, whose proof is provided in Appendix B.2.

Theorem 1. *Let $(A_0, \ldots, A_{n-1}) \in \mathcal{R}^n$ be a vector of i.i.d. random polynomials. We assume that each A_i has i.i.d. coefficients distributed according to $\mathcal{D}$ being either*

- *the uniform distribution $\mathcal{U}_r$ over $\frac{1}{r}\mathbb{Z} \cap \left(-\frac{1}{2}, \frac{1}{2}\right]$, for r an odd integer;*
- *the discrete Gaussian distribution $\mathcal{DG}_\sigma$ with parameter σ.*

Let $\mathbf{c}^{(0)}, \ldots, \mathbf{c}^{(n-1)} \in \mathbb{C}^{N/2}$ be vectors of complex numbers, and $\langle \cdot, \cdot \rangle$ be the canonical Hermitian product on $\mathbb{C}^{N/2}$. Then, for any $0 < \epsilon < 1$,

$$\Pr\left[\left|\sum_{j=0}^{n-1} \left\langle \mathbf{c}^{(j)}, \widehat{\mathbf{a}}_j \right\rangle\right| \geq \alpha_{\mathcal{D}} \left(\sum_{j=0}^{n-1} \|\mathbf{c}^{(j)}\|_2^2\right)^{1/2} \sqrt{N \ln \frac{4}{\epsilon}}\right] \leq \epsilon \ ,$$

where $\widehat{\mathbf{a}}_j = \mathsf{DFT}(A_j)$, $\alpha_{U_r} = \sqrt{1/6}$ and $\alpha_{\mathcal{DG}_\sigma} = \sigma\sqrt{2}$.

In particular, with $n = 1$ and $\mathbf{c}^{(0)}$ the i-th canonical basis vector, we have $\Pr\left[|a_{0,i}| \geq \alpha_{\mathcal{D}} \sqrt{N \ln \frac{4}{\epsilon}}\right] \leq \epsilon$, where $\widehat{\mathbf{a}}_0 = (a_{0,0}, \ldots, a_{0,N/2-1})$.

3 RDK-CKKS

In this section, we explain how we randomize the Rescale and KeySwitch algorithms in order to enable a rigorous analysis of the error which is roughly as tight as the heuristic one. We operate the same simple tweak to both algorithms: we add an encryption of zero to the input ciphertext just before the rounding step. We term such an encryption of zero a *Randomization Key* (RDK for short), which we define below. The effect of adding a randomization key is to make the ciphertext components uniform, as we explain in Lemma 2 below. As a consequence, the error distribution becomes amenable to a rigorous analysis.

3.1 Randomization Keys

Compared with the same computation in the clear, a homomorphic computation has exactly two sources of cryptographic errors:[5] (i) the errors of the RLWE encryptions that are part of the (fresh) input ciphertexts and of the evaluation keys, and (ii) the errors resulting from the rounding steps in Rescale and KeySwitch. Every cryptographic error term in our analysis is built from these two components. The first form of errors does not cause particular issues: RLWE errors are small and follow a fixed, known distribution, so they can be analyzed directly. The second form is the difficult one.

[5] Note that the computation also involves numerical errors, such as the rounding error of encoding, or polynomial approximation: these are deterministic and are accounted for separately.

As already illustrated in the introduction with Equations 1 and 2, rounding errors are larger than the RLWE encryption errors and depends on the distribution of the ciphertext entering the rounding step, in particular of its a-part, which gets multiplied by the secret key. This distribution is uniform for a fresh ciphertext, but a ciphertext that has gone through prior homomorphic operations is no longer uniform. To address the issue, we randomize the ciphertext before these rounding steps by relying on randomization keys (RDKs), which we now define.

Definition 1 (Randomization Key). *A* randomization key *(RDK) associated to a secret key s is a fresh secret-key encryption of zero, or equivalently an RLWE sample of the form:*

$$\mathsf{rdk} = (B_{\mathsf{rdk}},\ A_{\mathsf{rdk}}) = (-A_{\mathsf{rdk}} \cdot s + E_{\mathsf{rdk}},\ A_{\mathsf{rdk}}) \in \mathcal{R}_Q^2 \ ,$$

where $A_{\mathsf{rdk}} \sim U(\mathcal{R}_Q)$ and $E_{\mathsf{rdk}} \sim \mathcal{DG}_\sigma$.

3.2 RDK-Rescale and RDK-KeySwitch Algorithms

The purpose of RDKs is to randomize a ciphertext before rounding occurs in Rescale and KeySwitch. Hence, we integrate randomization keys in exactly two places: immediately before the rounding step of Rescale, and immediately before the rounding step of KeySwitch, i.e. before the ModDown step.

This leads to randomized variants of Rescale and KeySwitch algorithms for CKKS, which we call RDK-Rescale and RDK-KeySwitch, and detail in Algorithms 1 and 2; apart from the addition of a randomization key, they coincide with the classical Rescale and KeySwitch procedures.

Algorithm 1 RDK-Rescale: rescaling with RDK refresh

Require: Ciphertext $\mathsf{ct} = (b, a) \in \mathcal{R}_{qQ}^2$ at modulus qQ; randomization key $\mathsf{rdk} = (B_{\mathsf{rdk}}, A_{\mathsf{rdk}}) \in \mathcal{R}_{qQ}^2$.
Ensure: Ciphertext $\mathsf{ct}' \in \mathcal{R}_Q^2$ at modulus Q.
1: $(b', a') \leftarrow (b, a) + (B_{\mathsf{rdk}}, A_{\mathsf{rdk}}) \pmod{qQ}$ ▷ RDK refresh
2: $b'' \leftarrow \lfloor b'/q \rceil$ ▷ standard rescale
3: $a'' \leftarrow \lfloor a'/q \rceil$
4: **return** $\mathsf{ct}' = (b'', a'') \in R_Q^2$

3.3 A Refresh Lemma

Both Algorithms 1 and 2 end with a rounding step: a ciphertext (b, a) at modulus qQ (resp. PQ) is divided by q (resp. P) and rounded coefficient-wise. On the plaintext side, this rounding introduces the error

$$E_{\mathrm{rnd}} = \left[\frac{a}{q}\right]_1 \cdot s + \left[\frac{b}{q}\right]_1 \ .$$

Algorithm 2 RDK-KeySwitch: key switching with RDK refresh

Require: Ciphertext $\mathsf{ct} = (b, a) \in \mathcal{R}_Q^2$ under key s'; switching key $\{\mathsf{swk}^{(i)} = (B_{\mathsf{swk}}^{(i)}, A_{\mathsf{swk}}^{(i)})\}_{0 \le i < \mathsf{dnum}}$ from s' to s; randomization key $\mathsf{rdk} = (B_{\mathsf{rdk}}, A_{\mathsf{rdk}}) \in \mathcal{R}_{PQ}^2$.
Ensure: Ciphertext $\mathsf{ct}' \in \mathcal{R}_Q^2$ under key s.
1: $(\overline{a_0}, \ldots, \overline{a_{\mathsf{dnum}-1}}) \leftarrow \mathsf{ModUp}(a)$ ▷ lift to $\mathcal{R}_{PQ}$(Section 2.1)
2: $(B', A') \leftarrow (B_{\mathsf{rdk}}, A_{\mathsf{rdk}}) + \sum_{i=0}^{\mathsf{dnum}-1} \overline{a_i} \cdot \mathsf{swk}^{(i)} \pmod{PQ}$ ▷ RDK refresh
3: $b'' \leftarrow b + \lfloor B'/P \rceil \pmod{Q}$ ▷ ModDown on b-side
4: $a'' \leftarrow \lfloor A'/P \rceil \pmod{Q}$ ▷ ModDown on a-side
5: **return** $\mathsf{ct}' = (b'', a'') \in \mathcal{R}_Q^2$

Its size is governed by the distribution of a and b modulo q: when both are uniform over $\mathcal{R}_{qQ}$, the coefficients of $[a/q]_1$ and $[b/q]_1$ are i.i.d. uniform on $\frac{1}{q}\mathbb{Z} \cap (-\frac{1}{2}, \frac{1}{2}]$, and Theorem 1 gives tail bounds of order $\sqrt{N}$ on every slot of $\mathsf{DFT}(E_{\mathrm{rnd}})$. However, without uniformity, only the worst-case bound of order N is guaranteed.

While fresh ciphertexts are uniform under the RLWE assumption, nothing guarantees that a ciphertext produced by a sequence of homomorphic operations is; randomization keys restore this property exactly where it is needed, and we obtain the following Lemma which is at the core of our analysis.

Lemma 2 (Refresh Lemma). *Let $s \in \mathcal{R}_Q$ be invertible and $\mathsf{rdk} = (B_{\mathsf{rdk}}, A_{\mathsf{rdk}}) = (-A_{\mathsf{rdk}} \cdot s + E_{\mathsf{rdk}}, A_{\mathsf{rdk}})$ be a randomization key with $A_{\mathsf{rdk}} \sim U(\mathcal{R}_Q)$ and $E_{\mathsf{rdk}} \sim \mathcal{DG}_\sigma$. Let $(b, a) \in \mathcal{R}_Q^2$ be random variables.*

Then, $(b', a') = (b, a) + \mathsf{rdk}$ satisfies the following:

1. *Error growth: $b' + a's = (b + as) + E_{\mathsf{rdk}}$; on the message-level error grows by $\widehat{\mathbf{e}}_{\mathsf{rdk}}^{(\Delta)} = \frac{1}{\Delta}\mathsf{DFT}(E_{\mathsf{rdk}})$.*
2. *Statistical properties:*
   (a) *If rdk is independent of a, then $a' = a + A_{\mathsf{rdk}}$ is uniform over $\mathcal{R}_Q$ and independent of a.*
   (b) *If rdk is independent of (b, a), then $a' = a + A_{\mathsf{rdk}}$ is uniform over $\mathcal{R}_Q$ and independent of (b, a) and $b' = b + B_{\mathsf{rdk}}$ is uniform over $\mathcal{R}_Q$.*

Proof. Fix an invertible secret key $s \in \mathcal{R}_Q$, and let $A := A_{\mathsf{rdk}}$, $E := E_{\mathsf{rdk}}$, and $B := B_{\mathsf{rdk}} = -As + E$, where A and E are sampled independently.

For the first claim, we have, in $\mathcal{R}_Q$,

$$b' + a's = (b + B) + (a + A)s = b + as + E \ .$$

Thus, the refresh adds E to the plaintext error, corresponding to an additional message-side error $\widehat{\mathbf{e}}^{(\Delta)} = \mathsf{DFT}(E)/\Delta$.

For (a), independence of rdk and a implies independence of A and a. Since A is uniform over $\mathcal{R}_Q$, Lemma 1 shows that $a' = a + A$ is uniform over $\mathcal{R}_Q$ and independent of a.

For (b), suppose that rdk is independent of (b, a). For every (u, v) in the support of (b, a) and every $x \in \mathcal{R}_Q$,

$$\Pr[a' = x \mid (b, a) = (u, v)] = \Pr[A = x - v] = \frac{1}{|\mathcal{R}_Q|} \ .$$

Hence, a' is uniform over $\mathcal{R}_Q$ and independent of (b, a).

It remains to establish the uniformity of b'. Since s is invertible, the map $A \mapsto -As + e$ is a bijection of $\mathcal{R}_Q$ for every fixed e. As A is uniform and independent of E, the conditional distribution of $B = -As + E$ given $E = e$ is uniform. Averaging over E shows that B is uniform over $\mathcal{R}_Q$. Moreover, independence of rdk and (b, a) implies independence of B and (b, a). Therefore,

$$\Pr[b' = x \mid (b, a) = (u, v)] = \Pr[B = x - u] = \frac{1}{|\mathcal{R}_Q|} \ .$$

In particular, b' is uniform over $\mathcal{R}_Q$. This completes the proof of Lemma 2. □

Remark 1. When rdk is freshly sampled and has not been used in the computation of (b, a), it is independent of (b, a) and (2b) applies. The situation is different once a key is reused, as in Section 4.3: if rdk has already been consumed in the computation of (b, a), adding a fresh new rdk', not used so far, yields $(b, a) + \mathsf{rdk}'$ whose a-part is independent of rdk. The pair $(b, a) + \mathsf{rdk}'$, however, is not independent of rdk: its decryption carries the error accumulated so far, which depends on rdk. This case, we can only apply (2a).

4 Analysis of Elementary Homomorphic Operations

Lemma 2 proves that ciphertext components become uniformly distributed by adding a single randomization key. We now leverage this property to rigorously analyze the computation error of elementary homomorphic operations, including rescaling and key switching.

Probability Space. Throughout this section, the probabilities are taken over the randomness of the input ciphertext, of the switching keys, and of the randomization keys. The secret key s and the circuit are fixed: all bounds below are stated conditionally on s and depend either on $|\widehat{s}_i|$ or on $\|\widehat{\mathbf{s}}\|_\infty$. Section 4.2 explains how to bound $\|\widehat{\mathbf{s}}\|_\infty$ at key generation time.

About Independence of RDKs. Lemma 2 makes a ciphertext component uniform only when the randomization key is independent of the right object, and the two items of its statement ask for different things: independence from a makes the a-part uniform, whereas making *both* parts uniform requires independence from the pair (b, a). Which of the two is needed depends on the rounding step. In RDK-KeySwitch, the pair (B', A') to which ModDown is applied only depends on the a-part of the input ciphertext and on the switching key, so independence from (a, swk) suffices. RDK-Rescale, on the other hand, rounds (b', a'), which

depends on both parts, so uniformity is needed for both: independence from a alone does not suffice, since $b = -as + m + E$ carries the accumulated error E, which may depend on rdk even when a does not.

This minimal requirement is the only property of a randomization key that our analysis uses, so we name it; Section 4.3 exploits precisely the fact that it is weaker than freshness to share keys across iterations.

Definition 2. *Let* $\mathsf{rdk} = (-A_{\mathsf{rdk}}s + E_{\mathsf{rdk}}, A_{\mathsf{rdk}})$ *denote a randomization key. Then we say that:*

1. $\mathsf{rdk} \in \mathcal{R}_{qQ}^2$ *is admissible for a rescaling of* $(b, a) \in \mathcal{R}_{qQ}^2$ *if* rdk *is independent of* (b, a).
2. $\mathsf{rdk} \in \mathcal{R}_{PQ}^2$ *is admissible for a key switching of* $(b, a) \in \mathcal{R}_{Q}^2$ *if* rdk *is independent of* (a, swk).

4.1 Error Analysis of Elementary RDK-based Operations

In this section, we provide explicit bounds for elementary homomorphic operations. We provide a concrete expression for the (message-level) error $\widehat{\mathbf{e}}^{(\Delta)}$ at the end of RDK-Rescale, RDK-KeySwitch, as well as after a ciphertext-ciphertext multiplication. We then give a precise bound for the magnitude of each slot $\widehat{\mathbf{e}}_i^{(\Delta)}$ of these errors. These slot-specific bounds depend on the magnitude of $\widehat{s}_i$. In addition, we extend these slot-specific bounds into bounds for $\|\widehat{\mathbf{e}}^{(\Delta)}\|_\infty$ via a union bound argument. The latter bounds depend on $\|\widehat{\mathbf{s}}\|_\infty$.

All proofs of this section are provided in Appendix B.3.

Theorem 2 (Rescaling with RDK). *Let* $\mathsf{ct} = (b, a) \in \mathcal{R}_{qQ}^2$ *and assume that* s *is invertible in* $\mathcal{R}_{qQ}$. *Let* $\mathsf{rdk} = (-A_{\mathsf{rdk}}s + E_{\mathsf{rdk}}, A_{\mathsf{rdk}}) \in \mathcal{R}_{qQ}^2$ *be admissible for its rescaling. Let* $\mathsf{ct}'' = (b'', a'') \in \mathcal{R}_{Q}^2$, *with scaling factor* Δ, *be the output of Algorithm 1, and write* $(b', a') = (b, a) + \mathsf{rdk}$. *The message-level rescaling error is defined as:*

$$\widehat{\mathbf{e}}_{\mathsf{rs}}^{(\Delta)} = \frac{1}{\Delta}\mathsf{DFT}\left(b'' + a'' \cdot s - \frac{1}{q} \cdot (b + a \cdot s)\right).$$

Then b'' *and* a'' *are uniform over* $\mathcal{R}_Q$, a'' *is independent of* (b, a) *and the following holds:*

1. *The rescaling error satisfies:*

$$\widehat{\mathbf{e}}_{\mathsf{rs}}^{(\Delta)} = \frac{1}{\Delta}\left(\frac{1}{q}\mathsf{DFT}(E_{\mathsf{rdk}}) - \mathsf{DFT}\left(\left[\frac{b'}{q}\right]_1\right) - \mathsf{DFT}\left(\left[\frac{a'}{q}\right]_1\right) \odot \widehat{\mathbf{s}}\right) \ .$$

2. *For every* $0 < \epsilon < 1$ *and every* $0 \le i < N/2$,

$$\Pr\left[|\widehat{\mathbf{e}}_{\mathsf{rs},\,i}^{(\Delta)}| > \frac{\lambda_{\mathsf{rs}}(\epsilon)}{\Delta}\left(|\widehat{s}_i| + 1 + \alpha_{\mathsf{rs}}(q)\right)\right] < \epsilon \ ,$$

where

$$\lambda_{\mathsf{rs}}(\epsilon) = \sqrt{\frac{N}{6}\ln\frac{12}{\epsilon}} \quad \textit{and} \quad \alpha_{\mathsf{rs}}(q) = \frac{2\sqrt{3}\sigma}{q} \ .$$

Furthermore, defining $B(\epsilon) := \frac{\lambda_{\mathsf{rs}}(\epsilon)}{\Delta}\left(\|\widehat{\mathbf{s}}\|_\infty + 1 + \alpha_{\mathsf{rs}}(q)\right)$ *and setting* $\epsilon' = 2\epsilon/N$*, we then obtain:*

$$\Pr\left[\|\widehat{\mathbf{e}}_{\mathsf{rs}}^{(\Delta)}\|_\infty > B(\epsilon')\right] < \epsilon \ .$$

Proof (Sketch; see Appendix B). Uniformity. As rdk is admissible, it is independent of (b, a), so Lemma 2(2b) applies: a' is uniform over $\mathcal{R}_{qQ}$ and independent of (b, a), and b' is uniform over $\mathcal{R}_{qQ}$. Hence b'' and a'' are uniform over $\mathcal{R}_Q$, and a'' is independent of (b, a).

Error identity. With $m = b + a \cdot s$ and $m' = \lfloor b'/q \rceil + \lfloor a'/q \rceil \cdot s$, the relation $b' + a's = (b + as) + E_{\mathsf{rdk}}$ gives

$$m' - \frac{m}{q} \ = \ \frac{1}{q}\big(b' + a's - b - as\big) - \left[\tfrac{b'}{q}\right]_1 - \left[\tfrac{a'}{q}\right]_1 \cdot s \ = \ \frac{1}{q}E_{\mathsf{rdk}} - \left[\tfrac{b'}{q}\right]_1 - \left[\tfrac{a'}{q}\right]_1 \cdot s,$$

and applying DFT and dividing by Δ yields the claimed formula.

Tail bound. Since b' and a' are uniform over $\mathcal{R}_{qQ}$, $q \mid qQ$ and q is odd, the coefficients of $[b'/q]_1$ and $[a'/q]_1$ are i.i.d. uniform over $\frac{1}{q}\mathbb{Z} \cap (-\frac{1}{2}, \frac{1}{2}]$. Theorem 1, applied with failure probability $\epsilon/3$ to each of $\frac{1}{q}\mathsf{DFT}(E_{\mathsf{rdk}})_i$, $\mathsf{DFT}([b'/q]_1)_i$ and $\mathsf{DFT}([a'/q]_1)_i \cdot \widehat{s_i}$, bounds them by $\lambda_{\mathsf{rs}}(\epsilon)\alpha_{\mathsf{rs}}(q)$, $\lambda_{\mathsf{rs}}(\epsilon)$ and $\lambda_{\mathsf{rs}}(\epsilon)|\widehat{s_i}|$ respectively; no independence between the three terms is required, as they are combined by Boole's inequality. The uniform bound follows by a union bound over the $N/2$ slots with $\epsilon' = 2\epsilon/N$.

A randomization key is not always required: when the input of the rescaling is already uniform, for instance when it has been just produced by a key switching whose randomization key is independent of the whole input and of swk(Theorem 3), the same bound holds with no key consumed and no $\alpha_{\mathsf{rs}}(q)$ term.

Corollary 1 (Rescaling a uniform ciphertext). *Let* $(b, a) \in \mathcal{R}_{qQ}^2$ *be such that* b *and* a *are each uniform over* $\mathcal{R}_{qQ}$*, and let* $\mathsf{ct}'' = (b'', a'') = (\lfloor b/q \rceil, \lfloor a/q \rceil) \in \mathcal{R}_Q^2$ *be its plain rescaling, with scaling factor* Δ*. Then* b'' *and* a'' *are uniform over* $\mathcal{R}_Q$*, and items* 1. *and* 2. *of Theorem 2 hold with* $E_{\mathsf{rdk}} = 0$ *and* $\alpha_{\mathsf{rs}}(q)$ *replaced by 0; in particular no randomization key is consumed.*

The same mechanism extends to $\mathsf{KeySwitch}$. In addition to the rounding error of $\mathsf{ModDown}$, key switching additionally introduces error from the switching key, as illustrated in the following Theorem.

Theorem 3 (Key switching with RDK). *Let* $\mathsf{ct} = (b, a) \in \mathcal{R}_Q^2$ *be a ciphertext encrypted under* s'*, let* swk *switch from* s' *to* s*, assume that* s, s' *are invertible in* $\mathcal{R}_{PQ}$*, and let* $\mathsf{rdk} \in \mathcal{R}_{PQ}^2$ *be admissible for this key switching i.e. independent of* (a, swk)*. Let* (b'', a'') *be the output of Algorithm 2. Then the message-level key switching error is defined as:*

$$\widehat{\mathbf{e}}_{\mathsf{ks}}^{(\Delta)} = \frac{1}{\Delta}\mathsf{DFT}\left(b'' + a'' \cdot s - (b + a \cdot s')\right) \ .$$

Then, a'' is uniform over $\mathcal{R}_Q$ and independent of (a, swk); if moreover rdk *is independent of (b, a, swk), then b'' is uniform over $\mathcal{R}_Q$ as well, so that Corollary 1 could be applied to (b'', a''). Furthermore, the following holds:*

1. *Let $B' = B_{\mathsf{rdk}} + \sum_{0 \le i < \mathsf{dnum}} \overline{a_i} \cdot B^{(i)}_{\mathsf{swk}}$ and $A' = A_{\mathsf{rdk}} + \sum_{0 \le i < \mathsf{dnum}} \overline{a_i} \cdot A^{(i)}_{\mathsf{swk}}$, The message-level error introduced by key switching with RDK is:*

$$\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{ks}} = \frac{1}{\Delta}\left(\frac{1}{P}\ \left(\mathsf{DFT}(E_{\mathsf{rdk}}) + \textstyle\sum_{0 \le i < \mathsf{dnum}} \mathsf{DFT}(\overline{a_i}) \odot \mathsf{DFT}(E^{(i)}_{\mathsf{swk}})\right) - \mathsf{DFT}\left(\left[\tfrac{B'}{P}\right]_1\right)\ -\ \mathsf{DFT}\left(\left[\tfrac{A'}{P}\right]_1\right) \odot \widehat{\mathbf{s}}\right).$$

2. *For every $0 < \epsilon < 1$ and every $0 \le i < N/2$,*

$$\Pr\left[|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{ks},i}| > \frac{\lambda_{\mathsf{ks}}(\epsilon)}{\Delta}\left(|\widehat{s}_i| + 1 + \alpha_{\mathsf{ks}}(P)\right)\right] < \epsilon\ ,$$

where $\lambda_{\mathsf{ks}}(\epsilon) := \sqrt{\frac{N}{6}\ln\frac{4(\mathsf{dnum}+3)}{\epsilon}}$ and

$$\alpha_{\mathsf{ks}}(P) := \frac{2\sqrt{3}\sigma}{P}\left(1 + \max_{0 \le i < \mathsf{dnum}} D_i \cdot \frac{N \cdot \mathsf{dnum}}{2}\right).$$

Furthermore, $B(\epsilon) := \frac{\lambda_{\mathsf{ks}}(\epsilon)}{\Delta}\left(\|\widehat{\mathbf{s}}\|_\infty + 1 + \alpha_{\mathsf{ks}}(P)\right)$, and setting $\epsilon' = 2\epsilon/N$, we obtain:

$$\Pr\left[\|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{ks}}\|_\infty > B(\epsilon')\right] < \epsilon\ .$$

Suitably choosing the auxiliary modulus P allows one to make $\alpha_{\mathsf{ks}}(P)$ negligible. In order to match the orders of magnitude of α_{ks} and α_{rs}, one can use $P \approx \Delta \cdot \max_{0 \le i < \mathsf{dnum}} Q_i \cdot \frac{N \cdot \mathsf{dnum}}{2}$.

The following theorem exhibits averaging effects when several key-switching outputs are combined; we shall use it in the context of the analysis of linear algebra algorithms.

Theorem 4 (Independent sum of weighted key switching error). *For $t = 0, \ldots, n-1$, let $\widehat{\mathbf{e}}^{(\Delta)(t)}_{\mathsf{ks}}$ be the error of a key switching using $(\mathsf{swk}^{(t)}, \mathsf{rdk}^{(t)})$, and assume that:*

1. *the switching keys $\mathsf{swk}^{(0)}, \ldots, \mathsf{swk}^{(n-1)}$ are distinct and independently generated.*
2. *the randomization keys $\mathsf{rdk}^{(0)}, \ldots, \mathsf{rdk}^{(n-1)}$ are mutually independent, and independent of the a-parts of the n inputs and of $\mathsf{swk}^{(0)}, \ldots \mathsf{swk}^{(n-1)}$; in particular, each $\mathsf{rdk}^{(t)}$ is admissible for its key switching*
3. *the a-parts of the n inputs are independent of the switching-key errors.*

Let $\mathbf{c}^{(0)}, \ldots, \mathbf{c}^{(n-1)} \in \mathbb{C}^{N/2}$. Then, for every $0 < \epsilon < 1$,

$$\Pr\left[\left|\sum_{t=0}^{n-1} \langle \mathbf{c}^{(t)}, \widehat{\mathbf{e}}^{(\Delta)(t)}_{\mathsf{ks}} \rangle\right| > \frac{\lambda_{\mathsf{ks}}(\epsilon)}{\Delta}\left(\sqrt{\sum_{t=0}^{n-1} \|\mathbf{c}^{(t)} \odot \widehat{\mathbf{s}}^{(t)}\|_2^2} + \sqrt{\sum_{t=0}^{n-1} \|\mathbf{c}^{(t)}\|_2^2 \cdot (1 + \alpha_{\mathsf{ks}}(P))}\right)\right] < \epsilon,$$

Here $\mathbf{s}^{(t)}$ denotes the DFT of the target secret key used in the t-th key switching. In the usual case where all key switchings output under the same secret key, one has $\widehat{\mathbf{s}}^{(t)} = \widehat{\mathbf{s}}$ for all t.

Combining Theorems 2 and 3, we obtain the following error bounds for ciphertext-ciphertext multiplication.

Theorem 5 (Homomorphic multiplication error). *Let $\mathsf{ct} = (b, a)$, $\mathsf{ct}' = (b', a')$ be two ciphertexts in $\mathcal{R}_{qQ}^2$ under the same secret key s, and let $(d_0, d_1, d_2) = (bb',\, ab' + a'b,\, aa')$. Let swk be a relinearization key, let $\mathsf{rdk}_{\mathsf{rl}} \in \mathcal{R}_{PqQ}^2$, and let $\mathsf{ct}'' = (b'', a'') \in \mathcal{R}_Q^2$ be obtained by running Algorithm 2 on $(0, d_2)$ with $(\mathsf{swk}, \mathsf{rdk}_{\mathsf{rl}})$, adding (d_0, d_1), and rescaling by q. We denote by Δ the scaling factor of ct'' and define message-level multiplication error as*

$$\widehat{\mathbf{e}}_{\mathsf{mult}}^{(\Delta)} = \frac{1}{\Delta}\mathsf{DFT}\left(b'' + a'' \cdot s - \frac{1}{q}\left(b + a \cdot s\right)\left(b' + a' \cdot s\right)\right) \ .$$

Assume s is invertible in $\mathcal{R}_{PqQ}$ and that $\mathsf{rdk}_{\mathsf{rl}}$ is independent of (a, a', swk). Then:

1. *If moreover $\mathsf{rdk}_{\mathsf{rl}}$ is independent of (b, b'), both components of the relinearized ciphertext are uniform over $\mathcal{R}_{qQ}$, the rescaling requires no randomization key (Corollary 1) and the bound below holds with $\alpha = 0$.*
2. *Otherwise, the rescaling is performed by Algorithm 1 with a fresh $\mathsf{rdk}_{\mathsf{rs}} \in \mathcal{R}_{qQ}^2$ independent of $(b, a, b', a', \mathsf{swk}, \mathsf{rdk}_{\mathsf{rl}})$, and the bound holds with $\alpha = \alpha_{\mathsf{rs}}(q)$.*

In both cases b'' and a'' are uniform over $\mathcal{R}_Q$, and for every $0 < \epsilon < 1$ and every $0 \le i < N/2$,

$$\Pr\left[|\widehat{\mathbf{e}}_{\mathsf{mult},i}^{(\Delta)}| > \frac{1}{\Delta}\left(\lambda_{\mathsf{rs}}(\epsilon/2)\left(|\widehat{s}_i| + 1 + \alpha\right) + \frac{1}{q}\,\lambda_{\mathsf{ks}}(\epsilon/2)\left(|\widehat{s}_i| + 1 + \alpha_{\mathsf{ks}}(P)\right)\right)\right] < \epsilon \ .$$

Case 1 is the situation of a circuit evaluated once with fresh randomization keys, and is the form used for the parameters of Section 7; Case 2 arises as soon as $\mathsf{rdk}_{\mathsf{rl}}$ is shared across several evaluations, since b then carries the error accumulated by the previous ones(Section 4.3).

4.2 Making the bound independent from the secret key

As s has Hamming weight h, each value $\widehat{s}_i$ is a sum of h complex numbers of modulus one, so is at most h; a deterministic worst-case bound is thus $\|\widehat{\mathbf{s}}\|_\infty \le h$. The following analysis allows to turn this bound into the more familiar $\|\widehat{\mathbf{s}}\|_\infty = O(\sqrt{h})$.

Theorem 6 (DFT tail of an h-sparse secret key). *Let $s \in \mathcal{R}$ be chosen uniformly among ternary polynomials with Hamming weight h. Then,*

$$\Pr\left[\|\widehat{\mathbf{s}}\|_\infty > \sqrt{2h \ln \frac{2N}{\epsilon}}\right] \le \epsilon \ .$$

For instance, for $h = 1024, \epsilon = 2^{-128}$, $N = 2^{16}$, we obtain the bound $\|\widehat{\mathbf{s}}\|_\infty \leq 454$, which saves a factor 2 over the worst case analysis.

This bound can be made sharper by using rejection sampling – as suggested in [24]). We fix a parameter $\beta_h < \sqrt{2h\ln(2N)}$, sample an h-sparse secret key s, compute $\|\widehat{\mathbf{s}}\|_\infty$ by an FFT, and accept the key only if $\|\widehat{\mathbf{s}}\|_\infty \leq \beta_h$. If the acceptance probability is p_{acc}, this conditioning costs $-\log_2 p_{\mathrm{acc}}$ bits of security. The rejection sampling of $\epsilon = \frac{1}{2}$ thus loses at most one bit of security, and leads to reduce β_h from h to $\sqrt{2h\ln 4N}$. For the most commonly used ring degree $N = 2^{16}$, $\sqrt{2h\ln 4N} < 5\sqrt{h}$. We can thus obtain a key for which $\beta_h = 5\sqrt{h}$ by rejection sampling with, on average, 2 trials. The same bound can also be obtained in constant time with overwhelming probability by sampling 128 independent keys and keeping the key with the minimal $\|\widehat{\mathbf{s}}\|_\infty$.

Remark 2. We replace $\|\widehat{\mathbf{s}}\|_\infty$ by β_h. For $h = 1024$, over $10,000$ sampled keys the median $\|\widehat{\mathbf{s}}\|_\infty/\sqrt{h}$ is approximately 3.3, so that the replacing $\|\widehat{\mathbf{s}}\|_\infty$ by β_h in bounds costs $\log_2(5/3.3) \approx 0.6$ bits.

Notation. For an integer $R \geq 1$ and $\mathsf{op} \in \{\mathsf{rs}, \mathsf{ks}\}$, we write $\lambda_{\mathsf{op}}^{(R)}(\epsilon) := \lambda_{\mathsf{op}}(\epsilon/R)$ for the constant obtained after a union bound over R events; in particular, bounds uniform over the $N/2$ slots use $\lambda_{\mathsf{op}}^{(N/2)}(\epsilon) = \lambda_{\mathsf{op}}(2\epsilon/N)$, and $\beta_s := \|\widehat{\mathbf{s}}\|_\infty$ is replaced by β_h.

4.3 Reusing randomization keys across iterations

Before moving to the larger circuits, we address an important practical question: must a fresh RDK be stored for every invocation of a circuit? Let C be a circuit with k_{md} ModDown steps and k_{rs} Rescale steps, and suppose C is evaluated n times in sequence. The results above assume the use of a fresh randomization key for every rounding step, which costs $n(k_{\mathsf{md}} + k_{\mathsf{rs}})$ keys in total. We show that the keys of the ModDown steps can be shared by all n evaluations, while the keys of the Rescale steps must remain fresh. The number of required RDKs thus decreases to $k_{\mathsf{md}} + nk_{\mathsf{rs}} + (n-1)$, the last term accounting for one small fresh randomization key which is added to the input of every i-th iteration ($i \geq 2$). We quantify the resulting footprint in Section 7.2.

Why reusing is possible, and where it breaks After the first evaluation, the decryption $b + as = m + E$ carries an error E that depends on the randomization keys consumed so far. Adding a fresh key to the output refreshes its a-part (Statement 2. (a) of Lemma 2), but not its b-part: $b = as + m + E$ remains correlated with the shared keys through E. This suffices for key switching, which only requires rdk to be independent of (a, swk), but not for rescaling, which requires independence from the pair (b, a) (Definition 2 and Theorem 2).

Theorem 7 (RDK reuse). *Let C be the fixed circuit in which every multiplication is performed as in Theorem 5.*[6] *Number the* ModDown *steps of C as*

[6] The rescaling may be deferred past additions and plaintext multiplications, as in the giant step of Section 5.2; the proof is unchanged.

$1, \ldots, k_{\mathsf{md}}$ *in the evaluation order, and let* $K = \{\mathsf{rdk}_1, \ldots, \mathsf{rdk}_{k_{\mathsf{md}}}\}$ *be randomization keys. For* $n \geq 2$*, evaluation* n *times sequentially the circuit* C*, where in every iteration* ModDown *step* t *consumes* rdk_t*, every* Rescale *consumes a fresh randomization key, and a fresh randomization key* ρ_i *is added to the input of the* i*-th evaluation for* $i \geq 2$*. Then every rounding step of every evaluation consumes an admissible key.*

5 Application to Error Analysis: Linear Algebra, Polynomial Evaluation, and Binary Bootstrapping

The bounds derived in Section 4 can be combined to derive bounds for more advanced circuits. In order to demonstrate the versatility of our work, we perform a full analysis of the bootstrapping circuit of binary CKKS. Along the way, we provide analyses for Plaintext-Ciphertext Matrix-Vector (PCMv) multiplication and polynomial evaluation, that are used largely beyond bootstrapping when performing computation with CKKS.

5.1 Plaintext-Ciphertext Matrix-Vector Multiplication

We analyze the error of plaintext–ciphertext matrix-vector multiplication using the diagonal method with the baby-step giant-step (BSGS) approach [23]. Let M be an $N/2 \times N/2$ matrix with ℓ-th diagonal $M^{(\ell)} = (M_{i,i+\ell})_{0 \leq i < N/2}$ (indices modulo $N/2$), and let $L \subseteq \mathbb{Z}_{N/2}$ be the set of indices of its nonzero diagonals. For $\mathbf{v} \in \mathbb{C}^{N/2}$ we have $M\mathbf{v} = \sum_{\ell \in L} M^{(\ell)} \odot \mathsf{rot}_\ell(\mathbf{v})$, where $\mathsf{rot}_\ell(v)_j = v_{j+\ell}$. Given $\mathsf{bs}, \mathsf{gs} \subseteq \mathbb{Z}_{N/2}$ with $L \subseteq \mathsf{bs} + \mathsf{gs} := \{i + j : i \in \mathsf{bs}, j \in \mathsf{gs}\}$, the BSGS reformulation reads

$$M\mathbf{v} = \sum_{j \in \mathsf{gs}} \mathsf{rot}_j \left(\sum_{i \in \mathsf{bs}} \mathsf{rot}_i(\mathbf{v}) \odot \mathsf{rot}_{-j}(M^{(i+j)}) \right) . \tag{3}$$

Let ct encrypt $\mathbf{v}$ with scale $\Delta_{\mathbf{v}}$, and let the diagonals of M be encoded with a (possibly different) scale Δ_M; after evaluating Eq. (3) the result has scale $\Delta_M \cdot \Delta_{\mathbf{v}}$, which is restored by rescaling by $q \approx \Delta_M$, $q|Q$. The homomorphic evaluation introduces two sources of error: the deterministic encoding error of the plaintext diagonals, which makes the algorithm compute $(M + E)\mathbf{v}$ rather than $M\mathbf{v}$ and is discussed in Appendix E, and the key-switching error $\widehat{\mathbf{e}}_{\mathsf{mv}}^{(\Delta_M \Delta_v)}$ of the homomorphic rotations, so that the decrypted result is $(M+E)\mathbf{v} + \widehat{\mathbf{e}}_{\mathsf{mv}}^{(\Delta_M \Delta_v)}$. We now bound the latter.

Theorem 8 (Error of the BSGS algorithm for PCMv). *Let* ct *be a ciphertext with the underlying message* $\mathbf{v}$ *and scaling factor* $\Delta_{\mathbf{v}}$*. For each* $\ell \in L$*, let* diag_ℓ *be the plaintext encoding* $M^{(\ell)}$ *with scale* Δ_M*, and write:*

$$\widetilde{M}^{(\ell)} = \frac{1}{\Delta_M} \mathsf{DFT}(\mathsf{diag}_\ell).$$

Then the homomorphic error introduced by the BSGS algorithm can be represented as:

$$\widehat{\mathbf{e}}_{\mathsf{mv}}^{(\Delta_M \Delta_v)} = \sum_{j \in \mathsf{gs}} \widehat{\mathbf{e}}_{\mathsf{gs},j}^{(\Delta)} + \sum_{i \in \mathsf{bs}, j \in \mathsf{gs}} \mathsf{rot}_j(\widehat{\mathbf{e}}_{\mathsf{bs},i}^{(\Delta)}) \odot \widetilde{M}^{(i+j)},$$

where $\widehat{\mathbf{e}}_{\mathsf{bs},i}^{(\Delta)}$ *and* $\widehat{\mathbf{e}}_{\mathsf{gs},j}^{(\Delta)}$ *are the key switching errors with scale* Δ_v *and* $\Delta_M \cdot \Delta_v$ *respectively. The tail satisfies:*

$$\Pr\left[|\widehat{\mathbf{e}}_{\mathsf{mv},k}^{(\Delta_M \Delta_v)}| > \frac{\lambda_{\mathsf{ks}}(\epsilon/2)}{\Delta_{\mathbf{v}}} \left(B_{\mathsf{bs},k} + B_{\mathsf{gs},k}\right)\right] < \epsilon,$$

where

$$B_{\mathsf{bs},k} = \sqrt{\sum_{i \in \mathsf{bs}, j \in \mathsf{gs}} \left|\widetilde{M}_k^{(i+j)} \cdot \widehat{s}_{j+k}\right|^2} + \sqrt{\sum_{i \in \mathsf{bs}, j \in \mathsf{gs}} \left|\widetilde{M}_k^{(i+j)}\right|_2^2} \cdot (1 + \alpha_{\mathsf{ks}}(P)),$$

$$B_{\mathsf{gs},k} = \frac{\sqrt{|\mathsf{gs}|}}{\Delta_M} \left(|\widehat{s}_k| + 1 + \alpha_{\mathsf{ks}}(P)\right).$$

Proof. See Appendix B.4.

In practice, the $B_{\mathsf{bs},k}$ term dominates by far, because of the division by Δ_M in $B_{\mathsf{gs},k}$. A uniform version of this bound is given in Appendix B.4.

Remark 3 (Giant step hoisting). Hoisting the giant step [6] accumulates the giant-step ciphertexts before applying a single ModDown, so that a single RDK suffices for all giant steps. This substantially reduces the RDK storage cost; see Section 7.2. It also improves the bound: Corollary 3 in Appendix B.4 shows that the rounding part of $B_{\mathsf{gs},k}$ loses its $\sqrt{|\mathsf{gs}|}$ factor, while $B_{\mathsf{bs},k}$ is unchanged.

5.2 Polynomial evaluation

As a second example of complex homomorphic circuit and component of the bootstrapping procedure, we analyze the error introduced during the evaluation of a polynomial S on a ciphertext ct using the Paterson-Stockmeyer algorithm [31] in a simple form: the degree is $2^k - 1$, the baby-step parameter is 2^t, and $k+1$ multiplicative levels are used. Following [11], the polynomial is written in the Chebyshev basis for better numerical properties:

$$S = \sum_{i=0}^{2^t-1} \sum_{j=0}^{2^{k-t}-1} p_{i+2^t j} T_i(X) T_{j \cdot 2^t}(X), \tag{4}$$

where $T_0 = 1, T_1 = X, T_{n+1} = 2T_n(X) - T_{n-1}(X)$.

On input $x \in [-1, 1]$, the Chebyshev terms $T_i(x)$, $2 \le i \le 2^t$, are computed by the induction $T_i(x) = 2T_{\lfloor i/2 \rfloor}(x)T_{\lceil i/2 \rceil}(x) - T_{i \bmod 2}(x)$, and the $T_{j \cdot 2^t}(x)$ by the same induction started from T_0, T_{2^t}. If the input level is ℓ' and the output

level $\ell = \ell' - k - 1$, all monomials are brought to level $\ell + 2$; the input $x = T_1(x)$ itself is never rescaled but brought to lower levels by (exact) modulus dropping, the scaling-factor mismatch being absorbed into the constants it is multiplied by. One then computes the *baby steps* $\mathrm{BS}_j = \sum_{i=0}^{2^t-1} p_{i+j2^t} T_i(x)$, $0 \le j < 2^{k-t}$, and recombines them in the *giant step* $\sum_{j=0}^{2^{k-t}-1} \mathrm{BS}_j \cdot T_{j2^t}(x)$. Rescaling is skipped during the baby steps and postponed to a double rescale of the final sum. We assume the coefficients p_{i+j2^t} to be rationals with denominators dividing $\Delta_{\ell+2}$; replacing p_{i+j2^t} by $\lfloor \Delta_{\ell+2} p_{i+j2^t} \rceil / \Delta_{\ell+2}$ achieves this with an induced error on $S(x)$ of at most $2^{k-1}/\Delta_{\ell+2}$.

We let $q = \min_{\ell+1 \le j \le \ell'} q_j$, $\Delta = \min_{\ell \le j \le \ell'-1} \Delta_\ell$. For R an integer, we define

$$\Theta_R(\epsilon) = \frac{1}{\Delta}\left(\lambda_{\mathsf{rs}}^{(RN/2)}(\epsilon)\,(\beta_s + 1) + \frac{1}{q}\lambda_{\mathsf{ks}}^{(RN/2)}(\epsilon)\,(\beta_s + 1 + \alpha_{\mathsf{ks}}(P))\right). \tag{5}$$

Theorem 9 (Polynomial evaluation with RDK). *Let* ct *be a ciphertext with scaling factor* $\Delta_{\ell'}$ *decrypting to a plaintext* pt*, which encrypts real-valued message such that* $\|\mathsf{DFT}_N(\mathsf{pt})\|_\infty \le 1$.

Let S *be a degree* $2^k - 1$ *polynomial as in (4),* $R = 2^t + 2^{k-t}$*,* $\epsilon \in (0, 1)$*. We assume that* $\Theta_R(\epsilon) < 4^{-k}$*. Then, using the homomorphic Paterson-Stockmeyer's algorithm with baby-step parameter* t *and RDK, one obtains an output ciphertext* ct' *decrypting to* pt' *such that the message-level error*

$$\widehat{\mathbf{e}}_{\mathsf{poleval}}^{(\Delta)} = \mathsf{DFT}_N(\mathsf{pt}') - (S(\mathsf{DFT}_N(\mathsf{pt})_i))_{0 \le i < N/2}$$

verifies

$$\Pr\left[\|\widehat{\mathbf{e}}_{\mathsf{poleval}}^{(\Delta)}(S)\|_\infty > \Theta_R(\epsilon)\ \cdot A_S + \Theta_R^2(\epsilon) \cdot B_S\right] < \epsilon,$$

where

$$A_S := 1 + \sum_{j=0}^{2^{k-t}-1}\Big(\sum_{i=0}^{2^t-1} |p_{i+j2^t}|\frac{i^2}{2} + |BS_j|_\infty (j2^t)^2\Big), \quad B_S := \sum_{j,i} |p_{i+j2^t}|\frac{(ij2^t)^2}{2}.$$

Proof. See Appendix B.4.

This bound can be made uniform by replacing β_s with β_h in the definition of $\Theta_R(\epsilon)$. For polynomial approximations of very regular (e.g. analytic) functions, the coefficients p_{i+j2^t} decrease quickly with j and the first baby step dominates, so that the bound has magnitude $\approx \Theta_R(\epsilon) \sum_{i=0}^{2^t-1} |p_i| \frac{i^2}{2}$.

5.3 Double angle formula for EvalMod

To reduce the evaluation cost, rather than evaluating $f_0(x) = \cos(2\pi x)$ with a large degree polynomial, one uses a smaller degree approximation $\tilde{S}_r$ of $f_r(x) = \cos\left(\frac{2\pi}{2^r}x\right)$ and, since $f_{r-1} = 2f_r^2 - 1$, iterates $S_{r-1} = 2S_r^2 - 1$; the final value $S_1^2 = (1 + \cos(2\pi x))/2$ is returned.

Theorem 10 (Double angle formula with RDK). *Under the same notations and assumptions as in Theorem 9 with* $R = 2^t + 2^{k-t} + r$, *and* $\Theta_R(\epsilon) + 3\|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{poleval}}(S)\|_\infty < 4^{4-r}$, *when the polynomial evaluation is followed by* r *iterations of the double angle formula the total error* $\mathbf{e}_{\mathsf{EM}}$ *can be estimated as*

$$\Pr\left[\|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{EM}}\|_\infty > 3 \cdot 2^{2r-1} \cdot (\Theta_R(\epsilon) \cdot A_S + \Theta_R(\epsilon)^2 \cdot B_S) + \Theta_R(\epsilon) \cdot (2^{2r-1} + 1)\right] < \epsilon.$$

Proof. See Appendix B.4.

The first term again dominates, so that the total EM error is expected to be of the order of $3 \cdot 2^{2r-2}\Theta_R(\epsilon) \sum_{i=0}^{2^t-1} |p_i| \frac{i^2}{2}$.

5.4 Binary Bootstrapping

Binary bootstrapping [1] consists of the four steps StC, $\mathsf{ModRaise}$, CtS and $\mathsf{EvalMod}$, specified in Algorithm 3 and analyzed in Theorem 11. We denote by $U_{N/2}$ the matrix realizing the StC linear transform on the slot-message vector, by Δ_{In} and Δ_{Out} the scaling factors of the input and the output ciphertext, by Δ_{StC} and Δ_{CtS} the encoding scales of the diagonals used in the two linear transforms, by q_0 the base modulus, by Δ_{EM} the scale used during $\mathsf{EvalMod}$, and by d_{StC}, d_{CtS}, d_{EM} the number of levels consumed by the corresponding steps, so that the total modulus consumption is $d_{\mathsf{StC}} \log_2 \Delta_{\mathsf{StC}} + d_{\mathsf{CtS}} \log_2 \Delta_{\mathsf{CtS}} + d_{\mathsf{EM}} \log_2 \Delta_{\mathsf{EM}}$ bits.

Throughout this and the following subsections, message-side errors are written $\mathbf{e}$, dropping the scale superscript of the notation $\widehat{\mathbf{e}}^{(\Delta)}$ used so far: each error is understood to be normalized by the scale of the ciphertext at the step where it arises, and this scale is made explicit whenever it enters a bound.

Two scalar corrections are needed: the scaling factor at the bottom level must be $q_0/2$, and the input of $\mathsf{EvalMod}$ must be normalized to $[-1, 1]$ at scale Δ_{EM}. Both come for free, as they are fused into the plaintext diagonals: starting from the Cooley–Tukey decompositions $U_{N/2} = M_{d_{\mathsf{StC}}-1} \cdots M_0$ and $U_{N/2}^{-1} = N_{d_{\mathsf{CtS}}-1} \cdots N_0$ [11,25], we define, for $0 \le i < d_{\mathsf{StC}}$ and $0 \le j < d_{\mathsf{CtS}}$,

$$\widetilde{M}_i := \Big(\frac{q_0}{2\Delta_{\mathsf{In}}}\Big)^{1/d_{\mathsf{StC}}} M_i, \qquad \widetilde{N}_j := \Big(\frac{\Delta_{\mathsf{EM}}}{q_0(K+1)}\Big)^{1/d_{\mathsf{CtS}}} N_j,$$

so that $\widetilde{\mathsf{StC}} = \frac{q_0}{2\Delta_{\mathsf{In}}} U_{N/2}$ and $\widetilde{\mathsf{CtS}} = \frac{\Delta_{\mathsf{EM}}}{q_0(K+1)} U_{N/2}^{-1}$. Here K bounds the vector $\mathbf{I}$ introduced by $\mathsf{ModRaise}$,[7] and the factor $1/(K+1)$ normalizes the message, as $\|\frac{1}{K+1}(\mathbf{I} + \frac{\mathbf{z}}{2})\|_\infty < 1$ as soon as the maximum error on the coordinates of $\mathbf{z}$ is $\le 1/2$. All scales are to be understood as idealized values: the prime removed at each rescaling is not exactly the nominal scale of that level, which is compensated by folding constants close to 1 into the diagonals and into the coefficients of the $\mathsf{EvalMod}$ polynomial, as in Section 5.2.

[7] We use $K = \lfloor(\widetilde{h}+3)/2\rfloor$, which always holds: each entry of $q_0(\mathbf{I} + \frac{\mathbf{z}+\mathbf{e}_{\mathsf{In}}}{2})$ is a coefficient of $b + a \cdot \widetilde{s}$, hence at most $\frac{q_0}{2}(\widetilde{h}+1)$ in absolute value, while every entry of $\mathbf{z} + \mathbf{e}_{\mathsf{In}}$ is less than 2. For instance, $\widetilde{h} = 32$ yields $K = 17$.

Algorithm 3 Binary bootstrapping.

Require: A ciphertext ct at scale Δ_{In} encrypting $\mathbf{z} + \mathbf{e}_{\mathsf{In}}$, where $\mathbf{z} \in \{0,1\}^{N/2}$ and $\mathbf{e}_{\mathsf{In}}$ is the input message-side error; the layers $\widetilde{M}_i$, $\widetilde{N}_j$; a sparse secret key $\widetilde{s}$ of Hamming weight $\widetilde{h}$ and $K := \lfloor (\widetilde{h}+3)/2 \rfloor$; a polynomial p of degree $\leq 2^{d_{\mathsf{EM}}-1}$ with $\|p - f\|_\infty \leq \epsilon_{\mathsf{apx}}$, where $f(x) := \frac{1}{2}(1 - \cos(2(K+1)\pi x))$.

Ensure: A ciphertext at scale $\Delta_{\mathsf{Out}} \approx \Delta_{\mathsf{In}}$ encrypting $\mathbf{z}_{\mathsf{EM}} \approx \mathbf{z}$.

for $i = 0$ **to** $d_{\mathsf{StC}} - 1$ **do** ▷ StC; out: scale q_0, message $U_{N/2}\mathbf{z}/2$
  $\mathsf{ct} \leftarrow \mathsf{PCMv}(\widetilde{M}_i, \mathsf{ct})$ ▷ PCMv error $\mathbf{e}_{\mathsf{StC},i}$
end for
$\mathsf{ct} \leftarrow \mathsf{DtS}(\mathsf{ct})$ ▷ key switch $s \to \widetilde{s}$, error $\mathbf{e}'_{\mathsf{DtS}}$
$\mathsf{ct} \leftarrow \mathsf{ModRaise}(\mathsf{ct})$ ▷ SSE [7]; $\mathsf{pt} \mapsto \mathsf{pt} + \mathbf{I}$, $\|\mathbf{I}\|_\infty \leq K$
$\mathsf{ct} \leftarrow \mathsf{StD}(\mathsf{ct})$ ▷ key switch $\widetilde{s} \to s$, error $\mathbf{e}'_{\mathsf{StD}}$
for $j = 0$ **to** $d_{\mathsf{CtS}} - 1$ **do** ▷ CtS; out: scale Δ_{EM}, message $(\mathbf{I} + \mathbf{z}/2)/(K+1)$
  $\mathsf{ct} \leftarrow \mathsf{PCMv}(\widetilde{N}_j, \mathsf{ct})$ ▷ PCMv error $\mathbf{e}_{\mathsf{CtS},j}$
end for
$\mathsf{ct} \leftarrow \mathsf{PS}(p, \mathsf{ct})$ ▷ EvalMod; error $\mathbf{e}_{\mathsf{EM}}$
return $\frac{\Delta_{\mathsf{In}}}{\Delta_{\mathsf{EM}}} \cdot \mathsf{ct}$ ▷ scale correction, one level

Theorem 11 (Error of binary bootstrapping). *Considering the above construction, for $0 \leq i < d_{\mathsf{StC}}$ and $0 \leq j < d_{\mathsf{CtS}}$, let $\mathbf{e}_{\mathsf{StC},i}$ and $\mathbf{e}_{\mathsf{CtS},j}$ be the message-side PCMv errors of the corresponding steps, bounded by Theorem 8, and let $\mathbf{e}'_{\mathsf{DtS}}$ and $\mathbf{e}'_{\mathsf{StD}}$ be the message side key switching error from the* DtS *step and* StD *step, bounded by Theorem 3. We let $\mathbf{e}_{\mathsf{EM}}$ denote the homomorphic error introduced by the* EvalMod *step (independent of the input message). Defining the accumulated* StC *and* CtS *errors by*

$$\mathbf{e}_{\mathsf{StC}} := \sum_{i=0}^{d_{\mathsf{StC}}-1} \left(\widetilde{M}_i \widetilde{M}_{i-1} \cdots \widetilde{M}_0\right)^{-1} \mathbf{e}_{\mathsf{StC},i}, \quad \mathbf{e}_{\mathsf{DtS}} := 2U_{N/2}^{-1}\mathbf{e}'_{\mathsf{DtS}},$$

$$\mathbf{e}_{\mathsf{CtS}} := \frac{2q_0(K+1)}{\Delta_{\mathsf{EM}}} \sum_{j=0}^{d_{\mathsf{CtS}}-1} \widetilde{N}_{d_{\mathsf{CtS}}-1} \cdots \widetilde{N}_{j+1}\, \mathbf{e}_{\mathsf{CtS},j}, \quad \mathbf{e}_{\mathsf{StD}} = 2U_{N/2}^{-1}\mathbf{e}'_{\mathsf{StD}}$$

and set $\mathbf{e} := \mathbf{e}_{\mathsf{In}} + \mathbf{e}_{\mathsf{StC}} + \mathbf{e}_{\mathsf{DtS}} + \mathbf{e}_{\mathsf{StD}} + \mathbf{e}_{\mathsf{CtS}}$. Then, the element-wise bootstrapping error $\mathbf{e}_{\mathsf{BTS}} := \mathbf{z}_{\mathsf{EM}} - \mathbf{z}$ satisfies, for every $0 \leq k < N/2$,

$$|\mathbf{e}_{\mathsf{BTS},k}| \;\leq\; |\mathbf{e}_{\mathsf{EM},k}| + \|f - p\|_\infty + \frac{\pi^2}{4}|\mathbf{e}_k|^2$$

Proof. See Appendix B.4.

A uniform version, Corollary 4 in Appendix B.4, follows by a union bound over the $d_{\mathsf{StC}} + d_{\mathsf{CtS}} + 3$ error terms, using the uniform PCMv bounds and the operator norms of the partial products of the $\widetilde{M}_i$ and $\widetilde{N}_j$.

5.5 Cleaning polynomial $f_{\mathsf{cln}}(x) = 3x^2 - 2x^3$

As a final example, we analyze the **cleaning** circuit [19], which reduces the error of a ciphertext whose underlying message is already close to a binary vector. The cleaning circuit is a key building block of $\mathsf{IND}-\mathsf{CPA}^D$ security in CKKS computations with binary output.

The circuit is built with the polynomial $f_{\mathsf{cln}}(x) = 3x^2 - 2x^3$, which satisfies

$$|f_{\mathsf{cln}}(b+e) - b| \;\le\; 3|e|^2 + 2|e|^3,$$

for $b \in \{0,1\}$. We exploit this to clean a ciphertext encrypting $\mathbf{z} = \mathbf{b} + \mathbf{e}$ with $\mathbf{b} \in \{0,1\}^N$.

We denote the scaling factor of the input ciphertext by Δ, let $q \approx \Delta$ be a prime, and denote $\widetilde{\Delta} := \lfloor \Delta \rceil$ for the integer rounding of Δ. The cleaning circuit consumes one multiplicative level, and raises the scale from Δ to $\widetilde{\Delta}^3/q \approx \Delta^2$.

Cleaning algorithm. We recall the modulus-saving implementation of the cleaning circuit introduced in [14]. The input ciphertext encrypting $\mathbf{z} = \mathbf{b} + \mathbf{e}$ at scale Δ is viewed as a ciphertext encrypting $\widetilde{\mathbf{z}} := \frac{\Delta}{\widetilde{\Delta}}\mathbf{z}$ with scaling factor $\tilde{\Delta}$. Terms $e_{\mathsf{ks},i}$ correspond to key-switching errors, while terms $e_{\mathsf{rs},j}$ correspond to rescaling errors.

1. $\widetilde{\mathbf{z}}_2 \leftarrow \widetilde{\mathbf{z}}^{\odot 2} + \mathbf{e}_{\mathsf{ks},1}$. // Tensor + Relin, sc. fact. $\widetilde{\Delta}^2$:
2. $\widetilde{\mathbf{z}}_3 \leftarrow \widetilde{\mathbf{z}}_2 \odot \widetilde{\mathbf{z}} + \mathbf{e}_{\mathsf{ks},2}$. // Tensor + Relin, sc. fact. $\widetilde{\Delta}^3$
3. $\widetilde{\mathbf{z}}_4 \leftarrow 3\Delta\widetilde{\mathbf{z}}_2 - 2\widetilde{\mathbf{z}}_3 = f_{\mathsf{cln}}(\widetilde{\mathbf{z}}) + (3 - 2\widetilde{\mathbf{z}}) \odot \mathbf{e}_{\mathsf{ks},1} - 2\mathbf{e}_{\mathsf{ks},2}$. // Sc. fact. $\widetilde{\Delta^3}$.
4. $\widetilde{\mathbf{z}}_5 \leftarrow \mathsf{Rescale}_q(\widetilde{\mathbf{z}}_4) = f_{\mathsf{cln}}(\widetilde{\mathbf{z}}) + (3 - 2\widetilde{\mathbf{z}}) \odot \mathbf{e}_{\mathsf{ks},1} - 2\mathbf{e}_{\mathsf{ks},2} + \mathbf{e}_{\mathsf{rs},3}$. // Sc. fact. $\widetilde{\Delta}^3/q$.

Theorem 12 (Cleaning error). *Let* ct *encrypt* $\mathbf{z} = \mathbf{b} + \mathbf{e}$ *with scaling factor* Δ *with* $\mathbf{b} \in \{0,1\}^{N/2}$. *Define* $\widetilde{\Delta} = \lfloor \Delta \rceil$ *and*

$$\widetilde{\mathbf{e}} = \frac{\Delta}{\widetilde{\Delta}}\mathbf{z} - \mathbf{b} = \frac{\Delta}{\widetilde{\Delta}}\mathbf{e} + \frac{\Delta - \widetilde{\Delta}}{\widetilde{\Delta}}\mathbf{b}.$$

so that

$$|\widetilde{\mathbf{e}}_k| \le \frac{\Delta}{\widetilde{\Delta}}|\mathbf{e}_k| + \frac{1}{2\widetilde{\Delta}}.$$

Let $\mathbf{e}_{\mathsf{ks},1}, \mathbf{e}_{\mathsf{ks},2}$ *denote the key switching errors of steps 1–2 and* $\mathbf{e}_{\mathsf{rs},3}$ *the rescale error of step 4. Then the output of the cleaning algorithm encrypts* $\mathbf{b} + \mathbf{e}_{\mathsf{cln}}$ *at scale* $\widetilde{\Delta}^3/q$ *where*

$$|(\mathbf{e}_{\mathsf{cln}})_k| \le 3|(\widetilde{\mathbf{e}})_k|^2 + 2|(\widetilde{\mathbf{e}})_k|^3 + (3 + 2|(\widetilde{\mathbf{e}})_k|) \cdot |(\mathbf{e}_{\mathsf{ks},1})_k| + 2|(\mathbf{e}_{\mathsf{ks},2})_k| + |(\mathbf{e}_{\mathsf{rs},3})_k|$$

A uniform version, Corollary 5 in Appendix B.4, follows by bounding the three key-switching and rescaling terms with Theorems 3 and 2 at failure probability $\epsilon/3$ each.

6 Application to IND-CPA^D and Threshold Security

In this section, we explain how our error analysis translates into rigorous noise-flooding arguments for IND-CPA^D security of approximate CKKS [29] and threshold decryption [4]. In both settings, decryption must hide the computation error, which is addressed by noise flooding. Calibrating the flooding noise requires a bound on this error with a cryptographically small failure probability.

Accounting for the RDKs. Our bounds rely on the randomness introduced by the RDKs and on the independence and admissibility conditions of Sections 4 and 5. In the IND-CPA^D game, the RDKs are accounted for on the challenger side: the evaluation key given to the adversary consists of the switching keys, while the RDKs are sampled by the challenger and consumed internally when it answers evaluation queries. They are thus on the same footing as the coins used by encryption, and the adversary's view (public key, switching keys, evaluated ciphertexts, and flooded decryption responses) is that of the standard game, with the admissibility conditions of Definition 2 guaranteed by construction.

Since every rounding step consumes an RDK, the challenger must know in advance how many keys to sample, so a bound on the amount of required RDKs is fixed ahead, which bounds the amount of computation.

As the evaluator has access to the RDKs, our argument does not establish security against the evaluator. Similarly, in the threshold setting, corrupted decryptors must neither possess the RDKs nor obtain them through collusion with the evaluator. Hence, our analysis does not provide guarantees to achieve IND-CPA^D or threshold security in their most general form, but only in these relaxed variants.

Noise-Flooding in Slots for IND-CPA^D Security. Suppose we have a ciphertext $(b, a) \in \mathcal{R}_Q^2$ to decrypt, which satisfies:

$$\mathsf{Dcd}(as + b) = \frac{1}{\Delta} \cdot (\mathsf{DFT}(a) \odot \mathsf{DFT}(s) + \mathsf{DFT}(b)) = \mathbf{m} + \widehat{\mathbf{e}}^{(\Delta)} \ ,$$

where $\mathbf{m}$ is the exact output and $\widehat{\mathbf{e}}^{(\Delta)}$ is the message-level computation error. Further assume that, via our analysis, we have:

$$\Pr\left[\|\widehat{\mathbf{e}}^{(\Delta)}\|_\infty > B\right] \leq \epsilon \ ,$$

for some $\epsilon > 0$. Then, by a standard smudging argument, adding an error $\mathbf{f} \leftarrow U(\{\mathbf{f} \in \mathbb{C}^{N/2} : \|\mathbf{f}\|_\infty \leq B \cdot N \cdot \delta^{-1}\})$ guarantees that $\mathbf{m} + \mathbf{f}$ is within statistical distance at most δ from $\mathbf{m} + \widehat{\mathbf{e}}^{(\Delta)} + \mathbf{f}$, if the condition $\|\widehat{\mathbf{e}}^{(\Delta)}\|_\infty \leq B$ holds. This bounds the statistical error of simulating one decryption response by $\delta + \epsilon$. These losses must be budgeted over all permitted decryption queries and included in the overall security reduction.

Extension to Threshold Decryption. For threshold decryption, our bounds can be used to calibrate flooding in the standard coefficient-domain protocol, where each party adds a flooding error to its partial decryption before releasing

it. We have $E = \Delta\,\mathrm{iDFT}(\widehat{\mathbf{e}}^{(\Delta)})$, which satisfies $\|E\|_\infty \le \Delta\|\widehat{\mathbf{e}}^{(\Delta)}\|_\infty$, where the norms are respectively the norm over $\mathcal{R}_Q$ and over $\mathbb{C}^{N/2}$. Hence, our analysis provides a coefficient-error bound ΔB with failure probability at most ϵ. For instance, in the n-out-of-n case, the secret key is shared as $s = \sum_{i\in[n]} \mathsf{sh}_i$ over $\mathcal{R}_Q$, each party releases a partial decryption share of the form $p_i = a \cdot \mathsf{sh}_i + f_i$ where f_i is a fresh flooding error. Simulation security follows a similar argument as for $\mathsf{IND\text{-}CPA}^D$ security, while correctness must also account for the total flooding noise in the reconstruction, which computes $b + \sum_{i\in[n]} p_i \bmod Q$ (and decodes the result).

7 Constructing Secure Exact CKKS with Binary Bootstrapping and Cleaning

7.1 Constructing concrete Binary Bootstrapping parameter

We now instantiate the binary bootstrapping circuit of Section 5.4 and derive the minimal scales Δ_{StC}, Δ_{CtS}, and Δ_{EM} that achieve 9 bits of output precision at failure probability $\epsilon_{\mathsf{BTS}} \le 2^{-128}$.

Fixed inputs. We use RLWE parameters $N = 2^{16}$, $\sigma = 3.2$, secret-key Hamming weight $h = 1024$ and sparse-secret encapsulation weight $\widetilde{h} = 32$; input scaling factor $\Delta_{\mathsf{Eval}} \approx 2^{30}$ and base modulus $q \approx 2^{35}$; mod-raise bound $K = 17$, so that $\|\mathbf{I}\|_\infty \le K$; and StC, CtS each decomposed into three sparse linear maps ($d_{\mathsf{StC}} = d_{\mathsf{CtS}} = 3$). For $\mathsf{EvalMod}$, since $K = 17$, we approximate $\cos(36\pi x)$ by a degree-15 Chebyshev interpolant of $\cos(\frac{9}{4}\pi x)$ (baby-step parameter 2^3) followed by $r = 4$ double-angle iterations, giving $\|f - p\|_\infty \approx 2^{-10.6}$.

We provisionally assume P large enough that $\alpha_{\mathsf{ks}}(P) \le 1$, and verify in Table 2 that the remaining modulus budget admits such a P without violating RLWE security. For the diagonal-encoded plaintexts of StC and CtS, we use a probabilistic encoding-error estimate of order $O(\sqrt{N}/\Delta_M)$ to search for candidate Δ_{StC}, Δ_{CtS} keeping the encoding error below the rescaling error budget; once the parameters are fixed, we recompute the deterministic encoding error from the realized matrices, and it is this value that enters our rigorous analysis.

Minimal Δ_{StC}, Δ_{CtS}, and Δ_{EM}. Applying the per-step union bounds of Sections 4 and 5, we derive the minimal scale factors satisfying the resulting error bounds for the three subroutines; the resulting scales and slot-wise errors are summarized in Table 1.

Cleaning property of binary bootstrapping. The analysis above assumes a clean input ($\mathbf{e}_{\mathsf{In}} = 0$), whereas in practice the input to binary bootstrapping carries the error of the preceding computation, which binary bootstrapping itself *cleans*. For $\log_2\|\mathbf{e}_{\mathsf{In}}\|_\infty \in \{-15, -12, -9, -7.5, -6\}$ the certified output precision is 9.05, 9.04, 9.13, 8.81 and 8.27 bits respectively (Table 4 in Appendix D): it stays close to 9 bits up to $\|\mathbf{e}_{\mathsf{In}}\|_\infty \approx 2^{-7.5}$ and degrades gracefully beyond. This is the cleaning headroom that the binary CKKS pipeline exploits.

Table 1. Derived binary bootstrapping scaling factors and slot-wise error decomposition for the parameter set of Section 7 ($N = 2^{16}$, $h = 1024$, $\tilde{h} = 32$, $K = 17$, $\Delta_{\mathsf{In}} = 2^{30}$, $q_0 = 2^{35}$, $\epsilon_{\mathsf{BTS}} \leq 2^{-128}$).

Scales ($\log_2$)		Slot-wise error ($\log_2$)			
Δ_{StC}	23	$\|\mathbf{e}_{\mathsf{StC}}\|_\infty$	-7.40	$\|\mathbf{e}_{\mathsf{EM}}\|_\infty$	-9.95
Δ_{CtS}	25	$\|\mathbf{e}_{\mathsf{CtS}}\|_\infty$	-8.02	$\|\mathbf{e}\|_\infty$	-6.67
Δ_{EM}	40	$\|\mathbf{e}_{\mathsf{DtS}}\|_\infty$	-19.32	$\frac{\pi^2}{4}\|\mathbf{e}\|_\infty^2$	-12.04
Q_{BTS}	579	$\|\mathbf{e}_{\mathsf{StD}}\|_\infty$	-16.92	$\|f-p\|_\infty$	-10.60
		total			-9.05

Concrete parameter set with modulus chain. Table 2 gives the resulting modulus chain, which provides 12 bits of precision for key switching and rescaling at the Eval levels, and a 9-bit-precision binary bootstrapping accepting up to 7.5 bits of input error.

Table 2. Modulus chain composition. Each entry denotes $b \times k$ (k primes of b bits).

Base	StC	Mult	EvalMod	CtS	P	Total
35	23×3	30×20	40×10	25×3	61×8	1667

7.2 Memory Overhead of RDK

In RDK-CKKS, the randomization keys are part of the key material generated by the key holder (Section 6): they must be generated by the key holder and stored by the evaluator for the whole evaluation of the circuit. We quantify this overhead on the binary bootstrapping circuit of Table 1. Each decomposed step of StC and CtS uses the BSGS layout with $\mathsf{bs} = 8$ and $\mathsf{gs} = 4$, hence 10 rotation keys per step; with giant-step hoisting(Remark 3) the giant-step rotations share a single RDK, so each step needs only 8 RDKs. Table 3 in Appendix C compares the resulting RDK size to that of the switching keys (SWKs) already present in the evaluation key: the RDKs add roughly 27.3% to the SWK size for a single invocation. When the circuit is run n times, Theorem 7 lets the ModDown keys be shared, while each invocation consumes fresh keys at its Rescale steps, which a single invocation performs for free (Corollary 1); the overhead then reads $(27.3 + (n-1) \cdot 7.9)\,\%$ of the SWK size. This favorable ratio comes from two effects: each RDK is a single RLWE sample, much smaller than a decomposed switching key of dnum samples, and hoisting collapses many randomization keys into one per step.

7.3 Comparison with worst-case analysis

We compare the RDK bounds with those of a rigorous worst-case analysis at the binary bootstrapping parameters of Table 2 and $\epsilon = 2^{-128}$. The worst-case per-step rounding bound is $Nh/(2\Delta)$, whereas RDK yields $\lambda_{\mathsf{rs}}(\epsilon)(\beta_h + 1 + \alpha_{\mathsf{rs}}(q))/\Delta$;

at $N = 2^{16}$, $h = 1024$ the leading factors differ by $\log_2(\sqrt{Nh}/2^5) \approx 8$ bits per rescale step. In addition, the independent refresh provided by RDK turns the $\|c\|_1$ aggregation of the worst-case analysis into an $\|c\|_2$ aggregation (Theorem 4), saving a further $\frac{1}{2}\log_2(\mathsf{bs} \cdot \mathsf{gs}) = 2.5$ bits per BSGS layer in StC and CtS.

Altogether, RDK saves 10.5 bits per StC/CtS layer and 8 bits per EvalMod level and base prime, reducing the BTS portion of the modulus chain from $Q_{\mathsf{wc}}^{\mathsf{BTS}} = 730$ bits to $Q_{\mathsf{rdk}}^{\mathsf{BTS}} = 579$ bits, a 151-bit (roughly 25%) reduction. At fixed total modulus (Table 2), the freed budget supports $L_{\mathsf{rdk}} = 20$ evaluation levels against $L_{\mathsf{wc}} \approx 12$ under worst-case analysis, i.e. 8 additional levels.

7.4 Noise flooding with Cleaning Polynomial

We conclude by instantiating the cleaning circuit of Section 5.5 on the binary bootstrapping output to reach a precision above 100 bits, sufficient for statistical noise flooding of a binary CKKS pipeline. With the parameters of Table 2, four successive cleanings reach $\log_2\|\mathbf{e}_{\mathsf{cln}}\|_\infty \leq -120.21$ while consuming 240 bits of modulus (6 levels).

The binary bootstrapping output has scale $\Delta_{\mathsf{Eval}} = 2^{30}$ and 9 bits of precision. We apply four cleanings with input scales $2^{30}, 2^{60}, 2^{60}, 2^{90}$, choosing at each step the largest rescaling prime such that the rescale-error term $B_3(\epsilon')$ of Corollary 5 stays below the $3\widetilde{B}^2 + 2\widetilde{B}^3$ term left by the previous cleaning; this gives $q_1 = 2^{30}$, $q_2 = 2^{120}$, $q_3 = 2^{90}$, and the last step ends without rescaling at scale 2^{270}. Tables 5 and 6 in Appendix D give the resulting scales, output precision and per-term breakdown at every step.

7.5 Experimental validation

We compare the slot-wise bounds of Theorems 2, 3, 8, and 9 (with 10) against measurements, using the parameter set of Table 2 ($N = 2^{16}$, $h = 1024$, $\Delta \approx 2^{30}$); for PCMv we vary the scaling factors to observe the encoding and key-switching errors independently. All experiments are run with the HEaaN2 library [17], measuring the error of the first slot over iterations with fresh RDKs and fresh encryptions, for a fixed secret key (so as to keep $\widehat{s}_i$ fixed), switching keys, and input message. At experimentally accessible tail probabilities $\epsilon \in \{2^{-5}, 2^{-10}\}$, our bounds closely track the empirical tails. For $\epsilon \in \{2^{-64}, 2^{-128}\}$, where direct measurement is infeasible, we compare out bounds against Gaussian extrapolations based on the measured mean and variance. Full results are given in Tables 7, 8 and 9 in Appendix D.

Our tail bounds for the rescaling and key-switching errors lie within 0.5 bit of the empirical estimates at cryptographic-type tail probabilities. The linear-algebra error is dominated by the encoding error of the diagonals, so the corresponding bound is essentially independent of the target tail probability at the optimal scale setting. For EvalMod our bounds are loose by roughly 2.5 bits. Aggregated over the modulus chain of Table 2, this slack amounts to 0.5 bit per evaluation level and 2.5 bits per EvalMod level, 35 bits in total, i.e. only 3% of the modulus excluding the auxiliary modulus: the modulus overhead of our rigorous analysis is small in the evaluated parameter set.

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A AI tools use

We used Overleaf's AI assist, Claude Sonnet 4.6 and Fable 5.1 for spelling, grammar, and improving English. The proof of Theorem 4 was co-designed by authors and Claude Opus 4.7; it is similar to other proofs of the paper, which have been fed to the AI before the co-design step. The resulting proof has been fully human-checked.

B Proofs

B.1 Probabilistic toolbox

Lemma 3. *Let $\zeta_{2N} = e^{\pi i/N}$ be a primitive $2N$-th root of unity, and let $X_0, X_1, \ldots, X_{N-1}$ be i.i.d. real random variables satisfying*

$$\mathbb{E}\left[e^{tX_i}\right] \le e^{kt^2} \quad \textit{for all } t \in \mathbb{R}.$$

Define the complex random variable

$$X = \sum_{0 \le j < N/2} c_j \left(X_j + i\, X_{j+N/2}\right).$$

Then

$$\Pr\left[\, \|X\| \ge u \,\right] \le 4 \exp\left(-\frac{u^2}{8k\,|c|_2^2}\right),$$

where $|c|_2 = \sum_{0 \le j < N/2} |c_j|^2$

Proof. Let $c_j = a_j + ib_j$ for $a_j, b_j \in \mathbb{R}$. Expanding into real and imaginary parts,

$$\begin{aligned} X &= \sum_{j=0}^{N/2-1} (a_j + ib_j)\left(X_j + i\, X_{j+N/2}\right) \\ &= \underbrace{\sum_{j=0}^{N/2-1} \left(a_j X_j - b_j X_{j+N/2}\right)}_{=:\mathrm{Re}(X)} + i \underbrace{\sum_{j=0}^{N/2-1} \left(b_j X_j + a_j X_{j+N/2}\right)}_{=:\mathrm{Im}(X)}. \end{aligned}$$

Since $X_0, \ldots, X_{N-1}$ are independent and each satisfies $\mathbb{E}[e^{tX_i}] \le e^{kt^2}$, we have $\mathbb{E}[e^{t(aX_i)}] \le e^{ka^2t^2}$ for any $a \in \mathbb{R}$.

As the real part $\mathrm{Re}(X)$ is a sum of N independent terms, we obtain:

$$\begin{aligned} \mathbb{E}\left[e^{t\,\mathrm{Re}(X)}\right] &= \prod_{j=0}^{N/2-1} \mathbb{E}\left[e^{ta_jX_j}\right] \cdot \mathbb{E}\left[e^{-tb_jX_{j+N/2}}\right] \\ &\le \prod_{j=0}^{N/2-1} e^{ka_j^2t^2} \cdot e^{kb_j^2t^2} = e^{k|c|_2^2\, t^2}. \end{aligned}$$

Using the Chernoff bound and symmetry,

$$\Pr\left[|\mathrm{Re}(X)| \ge u\right] \le 2\exp\left(-\frac{u^2}{4k\,|c|_2^2}\right).$$

An identical argument gives the same bound for $|\mathrm{Im}(X)|$.

To pass from coordinate-wise bounds to the Euclidean norm, note that if $\|X\|^2 = \mathrm{Re}(X)^2 + \mathrm{Im}(X)^2 \ge t^2$, then at least one of $|\mathrm{Re}(X)|$ or $|\mathrm{Im}(X)|$ is at least $t/\sqrt{2}$. A union bound therefore gives

$$\Pr\left[\|X\| \ge t\right] \le \Pr\left[|\mathrm{Re}(X)| \ge \frac{t}{\sqrt{2}}\right] + \Pr\left[|\mathrm{Im}(X)| \ge \frac{t}{\sqrt{2}}\right] \le 4\exp\left(-\frac{t^2}{8k\,|c|_2^2}\right).$$

Lemma 4 (Sub-Gaussian property of $\mathcal{U}_r$ for odd r). *Let $r \ge 1$ be an odd integer, and let $X \sim \mathcal{U}_r$ be uniformly distributed on $\frac{1}{r}\mathbb{Z} \cap \left(-\frac{1}{2}, \frac{1}{2}\right]$. Then for all $s \in \mathbb{R}$,*

$$\mathbb{E}\left[e^{sX}\right] \le e^{s^2/24}.$$

Proof. The MGF of X is

$$\mathbb{E}\left[e^{sX}\right] = \frac{1}{r}\, e^{-s(r-1)/(2r)} \sum_{k=0}^{r-1} e^{sk/r} = \frac{1}{r}\, e^{-s(r-1)/(2r)} \cdot \frac{e^s - 1}{e^{s/r} - 1} = \frac{1}{r} \cdot \frac{\sinh(s/2)}{\sinh(s/(2r))}.$$

Rewriting as

$$\mathbb{E}\left[e^{sX}\right] = \frac{\sinh(s/2)/(s/2)}{\sinh\left(s/(2r)\right)/\left(s/(2r)\right)},$$

and using $\sinh(u)/u \geq 1$ for all $u \in \mathcal{R}$ to drop the denominator, followed by $\sinh(u)/u \leq e^{u^2/6}$ with $u = s/2$, we obtain

$$\mathbb{E}\left[e^{sX}\right] \leq \frac{\sinh(s/2)}{s/2} \leq e^{s^2/24}.$$

Lemma 5 (Sub-Gaussian property of discrete Gaussian). *Let $X \sim \mathcal{DG}_\sigma$ with $\mathcal{DG}_\sigma(x) \propto \exp(-x^2/(2\sigma^2))$. Then for all $s \in \mathbb{R}$,*

$$\mathbb{E}\left[e^{sX}\right] \leq e^{\sigma^2 s^2/2}.$$

Proof. The Gaussian parameter in the [30] is $r = \sigma\sqrt{2\pi}$, i.e. $D_{\mathbb{Z},\sigma}(x) \propto \exp(-\pi x^2/r^2)$. By [30, Lemma 2.8], the centered discrete Gaussian $D_{\mathbb{Z},r}$ is 0-subgaussian with parameter r for every $r > 0$:

$$\mathbb{E}\big[e^{2\pi u X}\big] \leq e^{\pi r^2 u^2}, \qquad \forall\, u \in \mathbb{R}.$$

Substituting $s = 2\pi u$ (i.e. $u = s/(2\pi)$) and $r = \sigma\sqrt{2\pi}$:

$$\mathbb{E}\big[e^{sX}\big] \leq \exp\Big(\pi \cdot 2\pi\sigma^2 \cdot \frac{s^2}{4\pi^2}\Big) = \exp\Big(\frac{\sigma^2 s^2}{2}\Big).$$

B.2 Proof of Theorem 1

Theorem 1. *Let $(A_0, \ldots, A_{n-1}) \in \mathcal{R}^n$ be a vector of i.i.d. random polynomials. We assume that each A_i has i.i.d. coefficients distributed according to $\mathcal{D}$ being either*

- *the uniform distribution $\mathcal{U}_r$ over $\frac{1}{r}\mathbb{Z} \cap \left(-\frac{1}{2}, \frac{1}{2}\right]$, for r an odd integer;*
- *the discrete Gaussian distribution $\mathcal{DG}_\sigma$ with parameter σ.*

Let $\mathbf{c}^{(0)}, \ldots, \mathbf{c}^{(n-1)} \in \mathbb{C}^{N/2}$ be vectors of complex numbers, and $\langle \cdot, \cdot \rangle$ be the canonical Hermitian product on $\mathbb{C}^{N/2}$. Then, for any $0 < \epsilon < 1$,

$$\Pr\left[\left|\sum_{j=0}^{n-1} \left\langle \mathbf{c}^{(j)}, \widehat{\mathbf{a}}_j \right\rangle\right| \geq \alpha_{\mathcal{D}} \left(\sum_{j=0}^{n-1} \|\mathbf{c}^{(j)}\|_2^2\right)^{1/2} \sqrt{N \ln \frac{4}{\epsilon}}\right] \leq \epsilon \ ,$$

where $\widehat{\mathbf{a}}_j = \mathsf{DFT}(A_j)$, $\alpha_{U_r} = \sqrt{1/6}$ and $\alpha_{\mathcal{DG}_\sigma} = \sigma\sqrt{2}$.

In particular, with $n = 1$ and $\mathbf{c}^{(0)}$ the i-th canonical basis vector, we have $\Pr\left[|a_{0,i}| \geq \alpha_{\mathcal{D}} \sqrt{N \ln \frac{4}{\epsilon}}\right] \leq \epsilon$, where $\widehat{\mathbf{a}}_0 = (a_{0,0}, \ldots, a_{0,N/2-1})$.

Proof. Write $A_j = \sum_{k=0}^{N-1} A_{j,k} X^k$. By definition of the negacyclic DFT,

$$\mathsf{DFT}(A_j)_t = \sum_{k=0}^{N-1} A_{j,k}\, \zeta_{2N}^{k \cdot 5^t}.$$

Substituting into the weighted sum,

$$Y := \sum_{j=0}^{n-1} \langle \mathbf{c}^{(j)}, \mathsf{DFT}(A_j) \rangle = \sum_{j=0}^{n-1} \sum_{k=0}^{N-1} d_k^{(j)} A_{j,k},$$

where

$$d_k^{(j)} := \sum_{t=0}^{N/2-1} \overline{c_t^{(j)}}\, \zeta_{2N}^{k \cdot 5^t}.$$

Since $\zeta_{2N}^{5^t \cdot N/2} = i$, we have $d_{k+N/2}^{(j)} = i \cdot d_k^{(j)}$, hence

$$Y = \sum_{j=0}^{n-1} \sum_{k=0}^{N/2-1} d_k^{(j)} \big(A_{j,k} + i\, A_{j,k+N/2} \big).$$

The nN coefficients $\{A_{i,k}\}$ are i.i.d. sub-Gaussian with parameter $k_{\mathcal{D}} = 1/24$ for $\mathcal{D} = \mathcal{U}_r$ (by Lemma 4), and $k_{\mathcal{D}} = \sigma^2/2$ for $\mathcal{D} = \mathcal{DG}_\sigma$ (by Lemma 5). Applying Lemma 3 to the $nN/2$ pairs $\{(A_{j,k},\, A_{j,k+N/2})\}_{j,k<N/2}$ with weights $\{d_k^{(j)}\}$,

$$\Pr[|Y| \geq u] \leq 4 \exp\left(- \frac{u^2}{8\, k_{\mathcal{D}} \sum_{j,\, k<N/2} |d_k^{(j)}|^2} \right).$$

It remains to evaluate $\sum_{j,k<N/2} |d_k^{(j)}|^2$. By the orthogonality,

$$\sum_{k=0}^{N-1} |d_k^{(j)}|^2 = N\, \|c^{(j)}\|_2^2.$$

Combined with $|d_{k+N/2}^{(j)}| = |d_k^{(j)}|$, this gives $\sum_{k<N/2} |d_k^{(j)}|^2 = \frac{N}{2} \|c^{(j)}\|_2^2$, hence

$$\sum_{j,\, k<N/2} |d_k^{(j)}|^2 = \frac{N}{2} \sum_{j=0}^{n-1} \|c^{(j)}\|_2^2.$$

Substituting back,

$$\Pr[|Y| \geq u] \leq 4 \exp\left(- \frac{u^2}{4\, k_{\mathcal{D}}\, N \sum_j \|c^{(j)}\|_2^2} \right).$$

Setting the right-hand side equal to ϵ and solving for u,

$$u = \sqrt{4\, k_{\mathcal{D}}}\, \sqrt{N \ln(4/\epsilon)}\, \sqrt{\textstyle\sum_j \|c^{(j)}\|_2^2},$$

with $\sqrt{4 \cdot 1/24} = \sqrt{1/6}$ for $\mathcal{D} = \mathcal{U}_r$, and $\sqrt{4 \cdot \sigma^2/2} = \sigma\sqrt{2}$ for $\mathcal{D} = \mathcal{DG}_\sigma$.

B.3 Proofs of Section 4

Theorem 2 (Rescaling with RDK). *Let* $\mathsf{ct} = (b, a) \in \mathcal{R}_{qQ}^2$ *and assume that* s *is invertible in* $\mathcal{R}_{qQ}$*. Let* $\mathsf{rdk} = (-A_{\mathsf{rdk}}s + E_{\mathsf{rdk}}, A_{\mathsf{rdk}}) \in \mathcal{R}_{qQ}^2$ *be admissible for its rescaling. Let* $\mathsf{ct}'' = (b'', a'') \in \mathcal{R}_Q^2$*, with scaling factor* Δ*, be the output of Algorithm 1, and write* $(b', a') = (b, a) + \mathsf{rdk}$*. The message-level rescaling error is defined as:*

$$\widehat{\mathbf{e}}_{\mathsf{rs}}^{(\Delta)} = \frac{1}{\Delta}\mathsf{DFT}\left(b'' + a'' \cdot s - \frac{1}{q} \cdot (b + a \cdot s)\right).$$

Then b'' *and* a'' *are uniform over* $\mathcal{R}_Q$*,* a'' *is independent of* (b, a) *and the following holds:*

1. *The rescaling error satisfies:*

$$\widehat{\mathbf{e}}_{\mathsf{rs}}^{(\Delta)} = \frac{1}{\Delta}\left(\frac{1}{q}\mathsf{DFT}(E_{\mathsf{rdk}}) - \mathsf{DFT}\left(\left[\frac{b'}{q}\right]_1\right) - \mathsf{DFT}\left(\left[\frac{a'}{q}\right]_1\right) \odot \widehat{\mathbf{s}}\right) \ .$$

2. *For every* $0 < \epsilon < 1$ *and every* $0 \le i < N/2$*,*

$$\Pr\left[|\widehat{\mathbf{e}}_{\mathsf{rs},\,i}^{(\Delta)}| > \frac{\lambda_{\mathsf{rs}}(\epsilon)}{\Delta}\left(|\widehat{s}_i| + 1 + \alpha_{\mathsf{rs}}(q)\right)\right] < \epsilon \ ,$$

where

$$\lambda_{\mathsf{rs}}(\epsilon) = \sqrt{\frac{N}{6}\ln\frac{12}{\epsilon}} \quad \textit{and} \quad \alpha_{\mathsf{rs}}(q) = \frac{2\sqrt{3}\sigma}{q} \ .$$

Furthermore, defining $B(\epsilon) := \frac{\lambda_{\mathsf{rs}}(\epsilon)}{\Delta}\left(\|\widehat{\mathbf{s}}\|_\infty + 1 + \alpha_{\mathsf{rs}}(q)\right)$ *and setting* $\epsilon' = 2\epsilon/N$*, we then obtain:*

$$\Pr\left[\|\widehat{\mathbf{e}}_{\mathsf{rs}}^{(\Delta)}\|_\infty > B(\epsilon')\right] < \epsilon \ .$$

Proof. Step 1: Uniformity and independence. Since rdk is admissible for the rescaling, it is independent of (b, a), so Lemma 2(2b) applies: a' is uniform over $\mathcal{R}_{qQ}$ and independent of (b, a), and b' is uniform over $\mathcal{R}_{qQ}$. Thus, b'' and a'' are uniform over $\mathcal{R}_Q$, and a'' is also independent of (b, a).

Step 2: The error identity. Let m denote the decryption of original ciphertext, and let m' denote the decryption of resulting ciphertext after rescaling. The message-level rescaling error is

$$\frac{1}{\Delta}\,\mathsf{DFT}\left(m' - \frac{m}{q}\right).$$

Since $m' = \lfloor\frac{b'}{q}\rceil + \lfloor\frac{a'}{q}\rceil \cdot s$ and $m = b + a \cdot s$, we compute

$$\begin{aligned} m' - \frac{m}{q} &= \left(\left\lfloor\frac{b'}{q}\right\rceil + \left\lfloor\frac{a'}{q}\right\rceil \cdot s\right) - \left(\frac{b}{q} + \frac{a}{q} \cdot s\right) \\ &= \frac{1}{q}\left(b' + a' \cdot s - b - a \cdot s\right) - \left[\frac{b'}{q}\right]_1 - \left[\frac{a'}{q}\right]_1 \cdot s \\ &= \frac{1}{q}E_{\mathsf{rdk}} - \left[\frac{b'}{q}\right]_1 - \left[\frac{a'}{q}\right]_1 \cdot s. \end{aligned}$$

Applying DFT to both sides yields the claimed formula.

Step 3: The tail bound. We now bound each of the terms separately. Since b' and a' are uniformly distributed over $\mathcal{R}_{qQ}$ (Step 1), $q \mid qQ$ and q is odd, the coefficients of $[b'/q]_1$ and of $[a'/q]_1$ are i.i.d uniform over $\frac{1}{q}\mathbb{Z} \cap (-1/2, 1/2]$, i.e. $[b'/q]_1, [a'/q]_1 \sim \mathcal{U}_q$. Note that no independence between the two is needed: the following three terms below are combined using a union bound.

Applying Theorem 1 with failure probability $\epsilon/3$ each, we obtain

$$\Pr\left[\left|\frac{1}{q}\mathsf{DFT}(E_{\mathsf{rdk}})_i\right| \geq \frac{1}{q}\sqrt{2N\sigma^2 \ln\frac{12}{\epsilon}}\,(= \lambda_{\mathsf{rs}}(\epsilon)\alpha_{\mathsf{rs}}(q))\right] \leq \frac{\epsilon}{3},$$

$$\Pr\left[\left|\mathsf{DFT}\left(\left[\frac{b'}{q}\right]_1\right)_i\right| \geq \sqrt{\frac{N}{6}\ln\frac{12}{\epsilon}}\,(= \lambda_{\mathsf{rs}}(\epsilon))\right] \leq \frac{\epsilon}{3},$$

$$\Pr\left[\left|\mathsf{DFT}\left(\left[\frac{a'}{q}\right]_1\right)_i\right| \cdot |\widehat{s}_i| \geq \sqrt{\frac{N}{6}\ln\frac{12}{\epsilon}} \cdot |\widehat{s}_i|\,(= \lambda_{\mathsf{rs}}(\epsilon)|\widehat{s}_i|)\right] \leq \frac{\epsilon}{3}.$$

Combining these terms via Boole's inequality gives

$$\Pr\left[|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{rs},i}| > \frac{\lambda_{\mathsf{rs}}(\epsilon)}{\Delta}\,(|\widehat{s}_i| + 1 + \alpha_{\mathsf{rs}}(q))\right] < \epsilon.$$

Corollary 1 (Rescaling a uniform ciphertext). *Let $(b, a) \in \mathcal{R}^2_{qQ}$ be such that b and a are each uniform over $\mathcal{R}_{qQ}$, and let $\mathsf{ct}'' = (b'', a'') = (\lfloor b/q \rceil, \lfloor a/q \rceil) \in \mathcal{R}^2_Q$ be its plain rescaling, with scaling factor Δ. Then b'' and a'' are uniform over $\mathcal{R}_Q$, and items 1. and 2. of Theorem 2 hold with $E_{\mathsf{rdk}} = 0$ and $\alpha_{\mathsf{rs}}(q)$ replaced by 0; in particular no randomization key is consumed.*

Proof. Since b, a are uniform over $\mathcal{R}_{qQ}$, b'' and a'' are uniform over $\mathcal{R}_Q$. Step 1 and Step 2 of the proof of Theorem 2 apply verbatim with $(b', a') = (b, a)$ and $E_{\mathsf{rdk}} = 0$: the error identity becomes:

$$\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{rs}} = -\frac{1}{\Delta}\left(\mathsf{DFT}\left(\left[\frac{b}{q}\right]_1\right) + \mathsf{DFT}\left(\left[\frac{a}{q}\right]_1\right) \odot \widehat{\mathbf{s}}\right),$$

and the coefficients of $[b/q]_1$ and $[a/q]_1$ are i.i.d. $\mathcal{U}_q$. Applying Theorem 1 with failure probability $\epsilon/3$ each of the two remaining terms and a union bound gives:

$$\Pr\left[|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{rs},i}| > \frac{\lambda_{\mathsf{rs}}(\epsilon)}{\Delta}(|\widehat{s}_i| + 1)\right] \leq \frac{2\epsilon}{3} < \epsilon.$$

Theorem 3 (Key switching with RDK). *Let $\mathsf{ct} = (b, a) \in \mathcal{R}^2_Q$ be a ciphertext encrypted under s', let swk switch from s' to s, assume that s, s' are invertible in $\mathcal{R}_{PQ}$, and let $\mathsf{rdk} \in \mathcal{R}^2_{PQ}$ be admissible for this key switching i.e. independent of (a, swk). Let (b'', a'') be the output of Algorithm 2. Then the message-level key switching error is defined as:*

$$\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{ks}} = \frac{1}{\Delta}\mathsf{DFT}\left(b'' + a'' \cdot s - (b + a \cdot s')\right) \quad .$$

Then, a'' is uniform over $\mathcal{R}_Q$ and independent of (a, swk); if moreover rdk is independent of (b, a, swk), then b'' is uniform over $\mathcal{R}_Q$ as well, so that Corollary 1 could be applied to (b'', a''). Furthermore, the following holds:

1. *Let $B' = B_{\mathsf{rdk}} + \sum_{0 \le i < \mathsf{dnum}} \overline{a_i} \cdot B^{(i)}_{\mathsf{swk}}$ and $A' = A_{\mathsf{rdk}} + \sum_{0 \le i < \mathsf{dnum}} \overline{a_i} \cdot A^{(i)}_{\mathsf{swk}}$, The message-level error introduced by key switching with RDK is:*

$$\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{ks}} = \frac{1}{\Delta}\left(\frac{1}{P}\ \left(\mathsf{DFT}(E_{\mathsf{rdk}}) + \sum_{0 \le i < \mathsf{dnum}} \mathsf{DFT}(\overline{a_i}) \odot \mathsf{DFT}(E^{(i)}_{\mathsf{swk}})\right) - \mathsf{DFT}\left(\left[\tfrac{B'}{P}\right]_1\right)\ -\ \mathsf{DFT}\left(\left[\tfrac{A'}{P}\right]_1\right) \odot \widehat{\mathbf{s}}\right).$$

2. *For every $0 < \epsilon < 1$ and every $0 \le i < N/2$,*

$$\Pr\left[|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{ks},i}| > \frac{\lambda_{\mathsf{ks}}(\epsilon)}{\Delta}\left(|\widehat{s}_i| + 1 + \alpha_{\mathsf{ks}}(P)\right)\right] < \epsilon\ ,$$

where $\lambda_{\mathsf{ks}}(\epsilon) := \sqrt{\frac{N}{6} \ln \frac{4(\mathsf{dnum}+3)}{\epsilon}}$ and

$$\alpha_{\mathsf{ks}}(P) := \frac{2\sqrt{3}\sigma}{P}\left(1 + \max_{0 \le i < \mathsf{dnum}} D_i \cdot \frac{N \cdot \mathsf{dnum}}{2}\right).$$

Furthermore, $B(\epsilon) := \frac{\lambda_{\mathsf{ks}}(\epsilon)}{\Delta}\left(\|\widehat{\mathbf{s}}\|_\infty + 1 + \alpha_{\mathsf{ks}}(P)\right)$, and setting $\epsilon' = 2\epsilon/N$, we obtain:

$$\Pr\left[\|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{ks}}\|_\infty > B(\epsilon')\right] < \epsilon\ .$$

Proof. Let m denote the plaintext encrypted by the input ciphertext, and m' the plaintext encrypted by the output ciphertext after key switching.

The key switching produces

$$\mathsf{ct}'' = \left(b + \left\lfloor \tfrac{B'}{P} \right\rceil,\ \left\lfloor \tfrac{A'}{P} \right\rceil\right) = \left(b + \tfrac{B'}{P} - \left[\tfrac{B'}{P}\right]_1,\ \tfrac{A'}{P} - \left[\tfrac{A'}{P}\right]_1\right),$$

Step 1-1: Uniformity under the admissibility hypothesis alone. By admissibility, rdk is independent of (a, swk); in particular A_{rdk} is uniform over $\mathcal{R}_{PQ}$ and independent of (a, swk), and, s being invertible in $\mathcal{R}_{PQ}$, thus $B_{\mathsf{rdk}} = -A_{\mathsf{rdk}}s + E_{\mathsf{rdk}}$ is uniform over $\mathcal{R}_{PQ}$ and independent of (a, swk) as well. Since $\sum_i \overline{a}_i A^{(i)}_{\mathsf{swk}}$ and $\sum_i \overline{a}_i B^{(i)}_{\mathsf{swk}}$ are functions of (a, swk), Lemma 1 gives that A' and B' are both uniform of $\mathcal{R}_{PQ}$ and independent of (a, swk). Consequently $a'' = [A'/P]$ is uniform over $\mathcal{R}_Q$ and independent of (a, swk).

Step 1-2: Uniformity of b''. In the case rdk is independent of (b, a, swk). Then, by the same argument, B' is uniform over $\mathcal{R}_{PQ}$ and independent of (b, a, swk), so $[B'/P]$ is uniform over $\mathcal{R}_Q$ and independent of b. Hence $b'' = b + [B'/P]$ is uniform over $\mathcal{R}_Q$ by Lemma 1.

Step 2: The error identity. Using the gadget relation $B^{(i)}_{\mathsf{swk}} + A^{(i)}_{\mathsf{swk}} s = P g_i s' + E^{(i)}_{\mathsf{swk}}$ with $\sum_i \overline{a}_i g_i = a$, and $B_{\mathsf{rdk}} + A_{\mathsf{rdk}} s = E_{\mathsf{rdk}}$,

$$b'' + a''s - (b + as') \;= \tfrac{1}{P}\Big(E_{\mathsf{rdk}} + \sum_{i=0}^{\mathsf{dnum}-1} \overline{a}_i \, E^{(i)}_{\mathsf{swk}}\Big) - \Big[\tfrac{B'}{P}\Big]_1 - \Big[\tfrac{A'}{P}\Big]_1 s.$$

Applying DFT and dividing by Δ yields the error formula in part (1).

Step 3: Tail bound. We now bound $|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{ks},j}|$ slot-wise. Set $\epsilon' := \epsilon/(\mathsf{dnum}+3)$ and split the message-level error into $\mathsf{dnum}+3$ contributions handled separately.

Probabilistic terms. Applying Theorem 1 (with parameter ϵ') slot-wise to each of E_{rdk}, $E^{(0)}_{\mathsf{swk}}, \ldots, E^{(\mathsf{dnum}-1)}_{\mathsf{swk}}$, $[B'/P]_1$ and $[A'/P]_1$:

$$\begin{aligned}
\Pr\Big[\big|\mathsf{DFT}(E_{\mathsf{rdk}})_j\big| > \sigma\sqrt{2N\ln(4/\epsilon')}\Big] &\le \epsilon', \\
\Pr\Big[\big|\mathsf{DFT}(E^{(i)}_{\mathsf{swk}})_j\big| > \sigma\sqrt{2N\ln(4/\epsilon')}\Big] &\le \epsilon', \quad \forall\, i \in [\mathsf{dnum}], \\
\Pr\Big[\big|\mathsf{DFT}\big([\tfrac{B'}{P}]_1\big)_j\big| > \sqrt{\tfrac{N}{6}\ln(4/\epsilon')}\Big] &\le \epsilon', \\
\Pr\Big[\big|\mathsf{DFT}\big([\tfrac{A'}{P}]_1\big)_j\big| \cdot |\widehat{s}_j| > \sqrt{\tfrac{N}{6}\ln(4/\epsilon')} \cdot |\widehat{s}_j|\Big] &\le \epsilon'.
\end{aligned}$$

Worst-case bound on $\overline{a}_i$. Since $\overline{a}_i$ has integer coefficients in $(-D_i/2,\, D_i/2]$:

$$\big|\mathsf{DFT}(\overline{a}_i)_j\big| \;\le\; \|\overline{a}_i\|_1 \;\le\; \tfrac{N D_i}{2}.$$

This bound is deterministic, so no probabilistic budget is consumed.

Combining. Conditioned on the $\mathsf{dnum}+3$ probabilistic events (total failure probability $\le \epsilon$ by union bound):

$$\Big|\sum_{i=0}^{\mathsf{dnum}-1} \mathsf{DFT}(\overline{a}_i)_j \cdot \mathsf{DFT}(E^{(i)}_{\mathsf{swk}})_j\Big| \;\le\; \sigma\sqrt{2N\ln\frac{4(\mathsf{dnum}+3)}{\epsilon}} \cdot \frac{N \cdot \mathsf{dnum} \cdot \max_i D_i}{2}.$$

Hence

$$|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{ks},j}| \le \tfrac{1}{\Delta}\Big[\frac{\sigma\sqrt{2N\ln\frac{4(\mathsf{dnum}+3)}{\epsilon}}}{P}\Big(1 + \frac{N \cdot \mathsf{dnum} \cdot \max_i D_i}{2}\Big) + \sqrt{\tfrac{N}{6}\ln\tfrac{4(\mathsf{dnum}+3)}{\epsilon}}\,(1 + |\widehat{s}_j|)\Big],$$

with probability at least $1-\epsilon$. Using the definitions of $\lambda_{\mathsf{ks}}(\epsilon)$ and $\alpha_{\mathsf{ks}}(P)$, we conclude

$$\Pr\Big[|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{ks},j}| > \frac{\lambda_{\mathsf{ks}}(\epsilon)}{\Delta}\big(|\widehat{s}_j| + 1 + \alpha_{\mathsf{ks}}(P)\big)\Big] < \epsilon.$$

Theorem 4 (Independent sum of weighted key switching error). *For $t = 0, \ldots, n-1$, let $\widehat{\mathbf{e}}^{(\Delta)(t)}_{\mathsf{ks}}$ be the error of a key switching using $(\mathsf{swk}^{(t)}, \mathsf{rdk}^{(t)})$, and assume that:*

1. *the switching keys* $\mathsf{swk}^{(0)}, \ldots, \mathsf{swk}^{(n-1)}$ *are distinct and independently generated.*
2. *the randomization keys* $\mathsf{rdk}^{(0)}, \ldots, \mathsf{rdk}^{(n-1)}$ *are mutually independent, and independent of the a-parts of the n inputs and of* $\mathsf{swk}^{(0)}, \ldots \mathsf{swk}^{(n-1)}$*; in particular, each* $\mathsf{rdk}^{(t)}$ *is admissible for its key switching*
3. *the a-parts of the n inputs are independent of the switching-key errors.*

Let $\mathbf{c}^{(0)}, \ldots, \mathbf{c}^{(n-1)} \in \mathbb{C}^{N/2}$*. Then, for every* $0 < \epsilon < 1$*,*

$$\Pr\left[\left|\sum_{t=0}^{n-1} \langle \mathbf{c}^{(t)}, \widehat{\mathbf{e}}_{\mathsf{ks}}^{(\Delta)(t)} \rangle\right| > \frac{\lambda_{\mathsf{ks}}(\epsilon)}{\Delta}\left(\sqrt{\sum_{t=0}^{n-1} \|\mathbf{c}^{(t)} \odot \widehat{\mathbf{s}}^{(t)}\|_2^2} + \sqrt{\sum_{t=0}^{n-1} \|\mathbf{c}^{(t)}\|_2^2} \cdot (1 + \alpha_{\mathsf{ks}}(P))\right)\right] < \epsilon,$$

Here $\mathbf{s}^{(t)}$ *denotes the DFT of the target secret key used in the t-th key switching. In the usual case where all key switchings output under the same secret key, one has* $\widehat{\mathbf{s}}^{(t)} = \widehat{\mathbf{s}}$ *for all t.*

Proof. By the independence assumption (independent RDKs and switching keys across invocations), the random objects $\{E_{\mathsf{rdk}}^{(t)}\}_t$, $\{E_{\mathsf{swk}}^{(t,i)}\}_{t,i}$, $\{[B'^{(t)}/P]_1\}_t$, $\{[A'^{(t)}/P]_1\}_t$ are i.i.d. within each family, across the t, with the coefficient distributions following $\mathcal{DG}_\sigma, \mathcal{DG}_\sigma, \mathcal{U}_P, \mathcal{U}_P$, respectively. This is all that is used below: they are combined by a union bound.

For each $t \in [n]$, part (1) of Theorem 3 gives

$$\widehat{\mathbf{e}^{(\mathbf{t})}}_{\mathsf{ks}}^{(\Delta)} = \frac{1}{\Delta}\left[\frac{1}{P}\Big(\mathsf{DFT}(E_{\mathsf{rdk}}^{(t)}) + \sum_{i=0}^{\mathsf{dnum}-1} \mathsf{DFT}(\overline{a}_i^{(t)}) \odot \mathsf{DFT}(E_{\mathsf{swk}}^{(t,i)})\Big) - \mathsf{DFT}\big([\tfrac{B'^{(t)}}{P}]_1\big) - \mathsf{DFT}\big([\tfrac{A'^{(t)}}{P}]_1\big) \odot \widehat{\mathbf{s}}^{(t)}\right].$$

Setting $Y := \sum_{t=0}^{n-1} \langle \mathbf{c}^{(t)}, \widehat{\mathbf{e}^{(\mathbf{t})}}_{\mathsf{ks}}^{(\Delta)} \rangle$, we decompose

$$\Delta \cdot Y \;=\; \tfrac{1}{P}\big(T_{\mathsf{rdk}} + T_{\mathsf{swk}}\big) - T_{B'} - T_{A'},$$

where

$$T_{\mathsf{rdk}} := \sum_t \langle \mathbf{c}^{(t)}, \mathsf{DFT}(E_{\mathsf{rdk}}^{(t)}) \rangle, \quad T_{\mathsf{swk}} := \sum_{t,i} \langle \mathbf{c}^{(t)} \odot \mathsf{DFT}(\overline{a}_i^{(t)}), \mathsf{DFT}(E_{\mathsf{swk}}^{(t,i)}) \rangle,$$

$$T_{B'} := \sum_t \langle \mathbf{c}^{(t)}, \mathsf{DFT}([B'^{(t)}/P]_1) \rangle, \quad T_{A'} := \sum_t \langle \mathbf{c}^{(t)} \odot \widehat{\mathbf{s}}^{(t)}, \mathsf{DFT}([A'^{(t)}/P]_1) \rangle.$$

Set $\epsilon' := \epsilon/(\mathsf{dnum} + 3)$. We bound each term so that the total failure probability is at most ϵ.

Bounding T_{rdk}*,* $T_{B'}$*,* $T_{A'}$*.* Each is a sum of independent inner products. Theorem 1 (with parameter ϵ') gives

$$\Pr\Big[\,|T_{\mathsf{rdk}}| > \sigma\sqrt{2N\ln(4/\epsilon')} \cdot \sqrt{\textstyle\sum_t \|\mathbf{c}^{(t)}\|_2^2}\,\Big] < \epsilon',$$

$$\Pr\Big[\,|T_{B'}| > \sqrt{\tfrac{N}{6}\ln(4/\epsilon')} \cdot \sqrt{\textstyle\sum_t \|\mathbf{c}^{(t)}\|_2^2}\,\Big] < \epsilon',$$

$$\Pr\Big[\,|T_{A'}| > \sqrt{\tfrac{N}{6}\ln(4/\epsilon')} \cdot \sqrt{\textstyle\sum_t \|\mathbf{c}^{(t)} \odot \widehat{\mathbf{s}}^{(t)}\|_2^2}\,\Big] < \epsilon'.$$

Bounding T_{swk} via worst-case $\overline{a}_i^{(t)}$. For each $i \in [\mathsf{dnum}]$, set

$$T_{\mathsf{swk}}^{(i)} \;:=\; \sum_t \langle \mathbf{c}^{(t)} \odot \mathsf{DFT}(\overline{a}_i^{(t)}),\, \mathsf{DFT}(E_{\mathsf{swk}}^{(t,i)}) \rangle,$$

so that $T_{\mathsf{swk}} = \sum_i T_{\mathsf{swk}}^{(i)}$. The variables $\{E_{\mathsf{swk}}^{(t,i)}\}_t$ are independent across t. Using the worst-case bound $|\mathsf{DFT}(\overline{a}_i^{(t)})_j| \le ND_i/2$ to treat the weights $\mathbf{w}^{(t,i)} := \mathbf{c}^{(t)} \odot \mathsf{DFT}(\overline{a}_i^{(t)})$ as deterministic:

$$\sum_t \|\mathbf{w}^{(t,i)}\|_2^2 \;\le\; \left(\tfrac{ND_i}{2}\right)^2 \sum_t \|\mathbf{c}^{(t)}\|_2^2.$$

Theorem 1 then yields

$$\Pr\Big[\, |T_{\mathsf{swk}}^{(i)}| > \sigma\sqrt{2N\ln(4/\epsilon')} \cdot \tfrac{ND_i}{2} \cdot \sqrt{\textstyle\sum_t \|\mathbf{c}^{(t)}\|_2^2} \Big] < \epsilon'.$$

A union bound over $i \in [\mathsf{dnum}]$ (total failure $\le \mathsf{dnum}\,\epsilon'$) gives

$$\Pr\Big[\, |T_{\mathsf{swk}}| > \sigma\sqrt{2N\ln(4/\epsilon')} \cdot \frac{N \cdot \mathsf{dnum} \cdot \max_i D_i}{2} \cdot \sqrt{\textstyle\sum_t \|\mathbf{c}^{(t)}\|_2^2} \Big] < \mathsf{dnum} \cdot \epsilon'.$$

Combining. A final union bound over all $\mathsf{dnum} + 3$ events (total failure $\le (\mathsf{dnum}+3)\,\epsilon' = \epsilon$) yields, with probability $\ge 1-\epsilon$:

$$\begin{aligned} |Y| \le \tfrac{1}{\Delta}\Big[&\tfrac{\sigma\sqrt{2N\ln(4/\epsilon')}}{P}\Big(1 + \frac{N \cdot \mathsf{dnum} \cdot \max_i D_i}{2}\Big)\sqrt{\textstyle\sum_t \|\mathbf{c}^{(t)}\|_2^2} \\ &+ \sqrt{\tfrac{N}{6}\ln(4/\epsilon')}\left(\sqrt{\textstyle\sum_t \|\mathbf{c}^{(t)}\|_2^2} + \sqrt{\textstyle\sum_t \|\mathbf{c}^{(t)} \odot \widehat{\mathbf{s}}^{(t)}\|_2^2}\right)\Big]. \end{aligned}$$

Using the definitions $\lambda_{\mathsf{ks}}(\epsilon) = \sqrt{\frac{N}{6}\ln\frac{4(\mathsf{dnum}+3)}{\epsilon}}$ and $\alpha_{\mathsf{ks}}(P) = \frac{2\sqrt{3}\,\sigma}{P}\big(1 + \frac{N\,\mathsf{dnum}\,\max_i D_i}{2}\big)$:

$$|Y| \;\le\; \tfrac{\lambda_{\mathsf{ks}}(\epsilon)}{\Delta}\Big(\sqrt{\textstyle\sum_t \|\mathbf{c}^{(t)} \odot \widehat{\mathbf{s}}^{(t)}\|_2^2} + \sqrt{\textstyle\sum_t \|\mathbf{c}^{(t)}\|_2^2}\,(1 + \alpha_{\mathsf{ks}}(P))\Big),$$

which is the claimed bound.

Theorem 5 (Homomorphic multiplication error). *Let $\mathsf{ct} = (b, a)$, $\mathsf{ct}' = (b', a')$ be two ciphertexts in $\mathcal{R}_{qQ}^2$ under the same secret key s, and let $(d_0, d_1, d_2) = (bb',\, ab' + a'b,\, aa')$. Let swk be a relinearization key, let $\mathsf{rdk}_{\mathsf{rl}} \in \mathcal{R}_{PqQ}^2$, and let $\mathsf{ct}'' = (b'', a'') \in \mathcal{R}_Q^2$ be obtained by running Algorithm 2 on $(0, d_2)$ with $(\mathsf{swk}, \mathsf{rdk}_{\mathsf{rl}})$, adding (d_0, d_1), and rescaling by q. We denote by Δ the scaling factor of ct'' and define message-level multiplication error as*

$$\widehat{\mathbf{e}}_{\mathsf{mult}}^{(\Delta)} = \frac{1}{\Delta}\mathsf{DFT}\left(b'' + a'' \cdot s - \frac{1}{q}\,(b + a \cdot s)\,(b' + a' \cdot s)\right) \;.$$

Assume s is invertible in $\mathcal{R}_{PqQ}$ and that $\mathsf{rdk}_{\mathsf{rl}}$ is independent of (a, a', swk). Then:

1. *If moreover* $\mathsf{rdk}_{\mathsf{rl}}$ *is independent of* (b, b')*, both components of the relinearized ciphertext are uniform over* $\mathcal{R}_{qQ}$*, the rescaling requires no randomization key (Corollary 1) and the bound below holds with* $\alpha = 0$*.*
2. *Otherwise, the rescaling is performed by Algorithm 1 with a fresh* $\mathsf{rdk}_{\mathsf{rs}} \in \mathcal{R}_{qQ}^2$ *independent of* $(b, a, b', a', \mathsf{swk}, \mathsf{rdk}_{\mathsf{rl}})$*, and the bound holds with* $\alpha = \alpha_{\mathsf{rs}}(q)$*.*

In both cases b'' *and* a'' *are uniform over* $\mathcal{R}_Q$*, and for every* $0 < \epsilon < 1$ *and every* $0 \le i < N/2$*,*

$$\Pr\left[|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{mult},i}| > \frac{1}{\Delta}\left(\lambda_{\mathsf{rs}}(\epsilon/2)\left(|\widehat{s}_i| + 1 + \alpha\right) + \frac{1}{q}\,\lambda_{\mathsf{ks}}(\epsilon/2)\left(|\widehat{s}_i| + 1 + \alpha_{\mathsf{ks}}(P)\right)\right)\right] < \epsilon \ .$$

Proof. Let $(b_{\mathsf{rl}}, a_{\mathsf{rl}})$ be the result of the relinearization step performed on $(0, aa')$. We write

$$\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{mult}} = \underbrace{\frac{1}{\Delta}\mathsf{DFT}\left(b'' + a'' \cdot s - q^{-1}((a_{\mathsf{rl}} + ab' + ba') \cdot s + bb' + b_{\mathsf{rl}})\right)}_{=:T_{\mathsf{rs}}}$$
$$\underbrace{-\frac{1}{q\Delta}\mathsf{DFT}\left(aa' \cdot s^2 - a_{\mathsf{rl}}s - b_{\mathsf{rl}}\right)}_{=:T_{\mathsf{ks}}}.$$

Here T_{rs} is the message-level rescaling error fo the relinearized ciphertext, and T_{ks} is $\frac{1}{q}$ times the message-level key switching error of the relinearization.

Key switching Since $\mathsf{rdk}_{\mathsf{rl}}$ is independent of (a, a', swk), it's admissible for key switching $(0, d_2) = (0, aa')$. Theorem 3, applied with target probability $\epsilon/2$, bounds T_{ks}:

$$\Pr\left[|T_{\mathsf{ks},i}| > \frac{1}{q\Delta}\lambda_{\mathsf{ks}}(\epsilon/2)(|\widehat{s}_i| + 1 + \alpha_{\mathsf{ks}}(P))\right] < \epsilon/2.$$

Rescaling, case 1. Assume $\mathsf{rdk}_{\mathsf{rl}}$ is independent of $(b, a, b', a', \mathsf{swk})$. Similar approach used in the proof of Theorem 3 can be applied. $A' = A_{\mathsf{rdk}_{\mathsf{rl}}} + \sum_i \overline{(d_2)}_i A^{(i)}_{\mathsf{swk}}$ and $B' = B_{\mathsf{rdk}_{\mathsf{rl}}} + \sum_i \overline{(d_2)}_i B^{(i)}_{\mathsf{swk}}$ are uniform over $\mathcal{R}_{PqQ}$ and is independent of $(b, a, b', a', \mathsf{swk})$. Thus, $b_{\mathsf{rl}} = \lfloor B'/P \rceil$ and $a_{\mathsf{rl}} = \lfloor A'/P \rceil$ are uniform over $\mathcal{R}_{qQ}$ and independent of $(b, a, b', a', \mathsf{swk})$. Now Lemma 1 implies that $d_0 + b_{\mathsf{rl}}$ and $d_1 + a_{\mathsf{rl}}$ are each uniform over $\mathcal{R}_{qQ}$.

Corollary 1 applies to the rescaling of $(d_0 + b_{\mathsf{rl}}, d_1 + a_{\mathsf{rl}})$: no randomization key is consumed, b'' and a'' over $\mathcal{R}_Q$, and

$$\Pr\left[|T_{\mathsf{rs},i}| > \frac{\lambda_{\mathsf{rs}}(\epsilon/2)}{\Delta}(|\widehat{s}_i| + 1)\right] < \epsilon/2,$$

i.e. the announced bound with $\alpha = 0$.

Rescaling, case 2 If $\mathsf{rdk}_{\mathsf{rl}}$ is only assumed independent of (a, a', swk), then b_{rl} may depend on (b, b'), and Corollary 1 is not available. In that case, the rescaling is performed by Algorithm 1 with a fresh $\mathsf{rdk}_{\mathsf{rs}}$ independent of $(b, a, b', a', \mathsf{swk}, \mathsf{rdk}_{\mathsf{rl}})$.

Here, $\mathsf{rdk}_{\mathsf{rs}}$ is independent of $(d_0 + b_{\mathsf{rl}}, d_1 + a_{\mathsf{rl}})$, which is a deterministic function of $(b, a, b', a', \mathsf{swk}, \mathsf{rdk}_{\mathsf{rl}})$, and $\mathsf{rdk}_{\mathsf{rs}}$ is admissible for the rescaling. Theorem 2, applied with target probability $\epsilon/2$ gives that b'' and a'' are uniform over $\mathcal{R}_Q$ and

$$\Pr\left[|T_{\mathsf{rs},i}| > \frac{\lambda_{\mathsf{rs}}(\epsilon/2)}{\Delta}(|\widehat{s}_i| + 1 + \alpha_{\mathsf{rs}}(q))\right] < \epsilon/2,$$

Combining the two tail bounds via the union bound yields the expected result, in both cases.

Lemma 6. *Let s be a uniform random variable over the set of ternary polynomials of degree $\leq N-1$ with a Hamming weight of h. Then, for all $0 \leq i < N/2$,*

$$\Pr(|\widehat{s}_i| \geq t) \leq 4\exp(-t^2/(2h)).$$

Proof. For fixed i, we write $\widehat{s}_i = \sum_{j=0}^{N-1} c_j s_j$, where c_j is a complex value of modulus 1.

We use Hoeffding inequality for the real and imaginary part. The real part is $\sum_{j=0}^{N-1} \mathrm{Re}(c_j)s_j$, a sum of independent random variables. Let $S = \{j/s_j \neq 0\}$. For $j \in S$, there exists a real θ_j such that the random variable takes values in $[-\cos\theta_j, \cos\theta_j]$; a similar setting holds for the imaginary part. We introduce the quantities $\tau_1 = \left(\sum_{j\in S}\cos(\theta_j)^2\right)^{1/2}$ and $\tau_2 = \left(\sum_{j\in S}\sin(\theta_j)^2\right)^{1/2}$.

Conditionally on S, the s_j are i.i.d. uniform on $-1, 1$ and Hoeffding's inequality then shows that

$$\Pr(|\mathrm{Re}(\widehat{s}_i)| \geq u|S) \leq 2\exp\left(-\tfrac{u^2}{2\tau_1^2}\right),$$
$$\Pr(|\mathrm{Im}(\widehat{s}_i)| \geq u|S) \leq 2\exp\left(-\tfrac{u^2}{2\tau_2^2}\right).$$

In view of $\tau_1^2 + \tau_2^2 = h$, for $\mathsf{DFT}_N(s)_i$ to have modulus greater than t, either $|\mathrm{Re}(\widehat{s}_i)| \geq h^{-1/2}\tau_1 \cdot t$ or $|\mathrm{Im}(\widehat{s}_i)| \geq h^{-1/2}\tau_2 \cdot t$. Using the union bound, we obtain

$$\Pr(|\widehat{s}_i| \geq t|S) \leq 4\exp(-t^2/(2h)).$$

Taking conditional expectation with respect to S yields the result.

Theorem 6 (DFT tail of an h-sparse secret key). *Let $s \in \mathcal{R}$ be chosen uniformly among ternary polynomials with Hamming weight h. Then,*

$$\Pr\left[\|\widehat{\mathbf{s}}\|_\infty > \sqrt{2h\ln\frac{2N}{\epsilon}}\right] \leq \epsilon \ .$$

Proof. The Theorem follows by using the union bound and Lemma 6. Explicitly, with $t = \sqrt{2h\ln(2N/\epsilon)}$,

$$\Pr\left[\|\widehat{\mathbf{s}}\|_\infty \geq t\right] \leq \sum_{i=0}^{N/2-1} \Pr\left[|\widehat{s}_i| \geq t\right] \leq \frac{N}{2} \cdot 4\exp\left(-\frac{t^2}{2h}\right) = \epsilon.$$

Theorem 7 (RDK reuse). *Let C be the fixed circuit in which every multiplication is performed as in Theorem 5.*[8] *Number the* ModDown *steps of C as $1, \ldots, k_{\mathsf{md}}$ in the evaluation order, and let $K = \{\mathsf{rdk}_1, \ldots, \mathsf{rdk}_{k_{\mathsf{md}}}\}$ be randomization keys. For $n \geq 2$, evaluation n times sequentially the circuit C, where in every iteration* ModDown *step t consumes rdk_t, every* Rescale *consumes a fresh randomization key, and a fresh randomization key ρ_i is added to the input of the i-th evaluation for $i \geq 2$. Then every rounding step of every evaluation consumes an admissible key.*

Proof. We first record the following fact about the elementary CKKS operations. The a-part of the output is a function of the a-parts of the inputs alone and does not involve their b-parts for the basic operations like Add, Sub, PtMult, and ModRaise. For key switching, it is a function of the a-part of the input, of swk and of the rdk consumed. The only exception is the tensor product, whose output a-part $d_1 = ab' + a'b$ (the component multiplied by s) does depend on the b-parts of the inputs. Note however that the input of the relinearization that follows a tensor product is $(0, d_2) = (0, aa')$, only depends on the a-part.

First, every rescale consumes an admissible key since it uses a fresh randomization key. The main thing we have to show is that every ModDown consumes an admissible key.

Let (b_i, a_i) be the input ciphertext of the i-th iteration, to which the ρ_i is added for $i \geq 2$. Then a_i is uniform and independent of every previous information, including $\mathcal{K}$ and swk: for $i = 1$, since the encryption is fresh, and for $i \geq 2$ by Lemma 2 applied to ρ_i. Note that b_i does depend on the $\mathcal{K}$, through the errors of the previous iterations.

Fix i. For $1 \leq j \leq k_{\mathsf{md}}$, let $a_i^{(j)}$ be the a-part of the input ciphertext of j-th key switching, and let $\mathcal{A}_i^{(j)}$ be the tuple of a-parts of all intermediate ciphertexts computed in iteration i, before the j-th keyswitching, starting with a_i. We show by induction that:

$$(\mathcal{A}_i^{(j)}, \mathsf{swk}, \mathsf{rdk}_1, \ldots, \mathsf{rdk}_{j-1}) \text{ is independent of } (\mathsf{rdk}_j, \ldots, \mathsf{rdk}_{k_{\mathsf{md}}}). \qquad (P_j)$$

Since $a_i^{(j)} \in \mathcal{A}_i^{(j)}$, (P_j) implies that rdk_j is independent of $(a_i^{(j)}, \mathsf{swk})$, that is, that rdk_j is admissible for the j-th key switching, which proves the theorem.

Induction

1. Base case $j = 1$. The tuple $\mathcal{A}_i^{(1)}$ starts with a_i, which is independent of $(\mathcal{K}, \mathsf{swk})$ as shown above. Before the first key switching, no element of $\mathcal{K}$ is consumed, and by the fact recorded above the a-part of the output of every operation is a function of the a-parts of its inputs and of swk. Independence of $\mathcal{K}$ is therefore preserved and (P_1) holds.
2. Inductive step. Assume (P_j) and consider the j-th key switching. Write $\mathsf{rdk}_{<j} = (\mathsf{rdk}_1, \ldots, \mathsf{rdk}_{j-1})$ and similarly for $\mathsf{rdk}_{\leq j}$ and $\mathsf{rdk}_{>j}$. Since $(\mathcal{A}_i^{(j)}, \mathsf{swk}, \mathsf{rdk}_{<j})$

[8] The rescaling may be deferred past additions and plaintext multiplications, as in the giant step of Section 5.2; the proof is unchanged.

is independent of $(\mathsf{rdk}_{\geq j})$ by (P_j), and rdk_j is independent of $(\mathsf{rdk}_{>j})$, the tuple $(\mathcal{A}_i^{(j)}, \mathsf{swk}, (\mathsf{rdk}_{\leq j}))$ is independent of $(\mathsf{rdk}_{>j})$. The output of j-th $\mathsf{ModDown}$ is a function of $(a_i^{(j)}, \mathsf{swk}, (\mathsf{rdk}_{\leq j}))$, hence is independent of $(\mathsf{rdk}_{>j})$ as well. We now distinguish two cases.

(a) If the $\mathsf{ModDown}$ belongs to a key switching that is not part of a ciphertext-ciphertext multiplication, the result is added to $(b, 0)$ and the a-part of the new ciphertext is the $\mathsf{ModDown}$ output itself, which is covered by the above.

(b) If the $\mathsf{ModDown}$ belongs to a ciphertext-ciphertext multiplication, the a-part of the relinearized ciphertext is obtained by adding $d_1 = ab' + a'b$ to the $\mathsf{ModDown}$ output. As b, b' depend on the errors of the previous iterations, hence on $(\mathsf{rdk}_{>j})$, independence is lost at this point. However, the rescaling that immediately follows adds a fresh randomization key, so the a-part of the output of multiplication is uniform and independent of everything sampled before, by Lemma 2.

In both cases, the a-part of the ciphertext resulting after the j-th key switching are independent of $\mathsf{rdk}_{>j}$. Between this operation and $(j+1)$-th key switching, no key of $\mathcal{K}$ is consumed and, exactly as in the base case, every a-parts of the intermediate ciphertexts are independent of $(\mathsf{rdk}_{>j})$, which is (P_{j+1}).

Remark 4. The theorem above makes the bounds of Section 4 available at every rounding step of every iteration, so that the error produced within one iteration is analyzed exactly as for a single evaluation of C; the errors of *distinct* iterations, however, are not independent, since they are produced by the same set $\mathcal{K}$ of shared randomization keys. Thus, the errors from each iteration should be combined by a union bound.

B.4 Proofs of Section 5

Theorem 8 (Error of the BSGS algorithm for PCMv). *Let* ct *be a ciphertext with the underlying message* $\mathbf{v}$ *and scaling factor* $\Delta_{\mathbf{v}}$*. For each* $\ell \in L$*, let* diag_ℓ *be the plaintext encoding* $M^{(\ell)}$ *with scale* Δ_M*, and write:*

$$\widetilde{M}^{(\ell)} = \frac{1}{\Delta_M}\mathsf{DFT}(\mathsf{diag}_\ell).$$

Then the homomorphic error introduced by the BSGS algorithm can be represented as:

$$\widehat{\mathbf{e}}_{\mathsf{mv}}^{(\Delta_M \Delta_v)} = \sum_{j \in \mathsf{gs}} \widehat{\mathbf{e}}_{\mathsf{gs},j}^{(\Delta)} + \sum_{i \in \mathsf{bs}, j \in \mathsf{gs}} \mathsf{rot}_j(\widehat{\mathbf{e}}_{\mathsf{bs},i}^{(\Delta)}) \odot \widetilde{M}^{(i+j)},$$

where $\widehat{\mathbf{e}}_{\mathsf{bs},i}^{(\Delta)}$ *and* $\widehat{\mathbf{e}}_{\mathsf{gs},j}^{(\Delta)}$ *are the key switching errors with scale* Δ_v *and* $\Delta_M \cdot \Delta_v$ *respectively. The tail satisfies:*

$$\Pr\left[|\widehat{\mathbf{e}}_{\mathsf{mv},k}^{(\Delta_M \Delta_v)}| > \frac{\lambda_{\mathsf{ks}}(\epsilon/2)}{\Delta_{\mathbf{v}}} \left(B_{\mathsf{bs},k} + B_{\mathsf{gs},k}\right)\right] < \epsilon,$$

where

$$B_{\mathsf{bs},k} = \sqrt{\sum_{i\in\mathsf{bs},j\in\mathsf{gs}} \left|\widetilde{M}_k^{(i+j)} \cdot \widehat{s}_{j+k}\right|^2} + \sqrt{\sum_{i\in\mathsf{bs},j\in\mathsf{gs}} \left|\widetilde{M}_k^{(i+j)}\right|_2^2} \cdot (1+\alpha_{\mathsf{ks}}(P)),$$

$$B_{\mathsf{gs},k} = \frac{\sqrt{|\mathsf{gs}|}}{\Delta_M}\left(|\widehat{s}_k| + 1 + \alpha_{\mathsf{ks}}(P)\right).$$

Proof. Baby-step rotations. For each $i \in \mathsf{bs}$, applying Rot_i to ct yields a ciphertext at scale $\Delta_{\mathbf{v}}$ whose slot vector is

$$\mathsf{rot}_i(v) + \widehat{\mathbf{e}}_{\mathsf{bs},i}^{(\Delta)},$$

where $\widehat{\mathbf{e}}_{\mathsf{bs},i}^{(\Delta)}$ is the key-switching error at scale $\Delta_{\mathbf{v}}$ given by Theorem 3.

Inner accumulation. For each $j \in \mathsf{gs}$,

$$\mathsf{ct}_j \;:=\; \sum_{i\in\mathsf{bs}} \mathsf{Rot}_i(\mathsf{ct}) \odot \mathsf{diag}_{i+j}$$

is at scale $\Delta_M\Delta_{\mathbf{v}}$ and, since $\widetilde{M}^{(\ell)}$ denotes the slot vector of the encoded diagonal (so that the encoding error is already accounted for in $\widetilde{M}$) and plaintext–ciphertext products are exact on the slots, has slot vector

$$\sum_{i\in\mathsf{bs}} \big(\mathsf{rot}_i(v) + \widehat{\mathbf{e}}_{\mathsf{bs},i}^{(\Delta)}\big) \odot \mathsf{rot}_{-j}\big(\widetilde{M}^{(i+j)}\big).$$

No additional error is introduced here.

Giant-step rotations. Applying Rot_j to ct_j and summing over $j \in \mathsf{gs}$ yields a ciphertext at scale $\Delta_M\Delta_{\mathbf{v}}$ which encrypts:

$$\sum_{j\in\mathsf{gs}}\Big[\sum_{i\in\mathsf{bs}} \big(\mathsf{rot}_{i+j}(v) + \mathsf{rot}_j(\widehat{\mathbf{e}}_{\mathsf{bs},i}^{(\Delta)})\big) \odot \widetilde{M}^{(i+j)} \;+\; \widehat{\mathbf{e}}_{\mathsf{gs},j}^{(\Delta)}\Big],$$

where $\widehat{\mathbf{e}}_{\mathsf{gs},j}^{(\Delta)}$ is the KS error of the j-th giant-step rotation at scale $\Delta_M\Delta_{\mathbf{v}}$.

The first term reconstructs $\widetilde{M}v$. The remaining computation error at slot k is therefore

$$\widehat{\mathbf{e}}_{\mathsf{mv},k}^{(\Delta)} \;=\; \underbrace{\sum_{i\in\mathsf{bs},j\in\mathsf{gs}} \widetilde{M}_k^{(i+j)} \cdot (\widehat{\mathbf{e}}_{\mathsf{bs},i}^{(\Delta)})_{k+j}}_{=:\,T_{\mathsf{bs}}} \;+\; \underbrace{\sum_{j\in\mathsf{gs}} (\widehat{\mathbf{e}}_{\mathsf{gs},j}^{(\Delta)})_k}_{=:\,T_{\mathsf{gs}}},$$

using $(\mathsf{rot}_j(\widehat{\mathbf{e}}_{\mathsf{bs},i}^{(\Delta)}))_k = (\widehat{\mathbf{e}}_{\mathsf{bs},i}^{(\Delta)})_{k+j}$.

Bounding T_{gs}. The $|\mathsf{gs}|$ giant-step rotations use independent RDKs, so the $\widehat{\mathbf{e}}_{\mathsf{gs},j}^{(\Delta)}$ are independent KS errors at scale $\Delta_M\Delta_{\mathbf{v}}$. Applying Theorem 4 with weights

$c^{(j)} = e_k$ (the k-th standard basis vector of $\mathbb{C}^{N/2}$), we have $\sum_j \|c^{(j)}\|_2^2 = |\mathsf{gs}|$ and $\sum_j \|c^{(j)} \odot \widehat{\mathbf{s}}\|_2^2 = |\mathsf{gs}| \cdot |\widehat{s}_k|^2$, giving

$$\Pr\Big[\, |T_{\mathsf{gs}}| > \frac{\lambda_{\mathsf{ks}}(\epsilon/2)}{\Delta_M \Delta_{\mathbf{v}}} \sqrt{|\mathsf{gs}|}\, \big(|\widehat{s}_k| + 1 + \alpha_{\mathsf{ks}}(P)\big)\Big] < \epsilon/2.$$

Define

$$B_{\mathsf{gs},k} := \sqrt{|\mathsf{gs}|}/\Delta_M \cdot (|\widehat{s}_k| + 1 + \alpha_{\mathsf{ks}}(P)),$$

the bound can be represented by

$$\lambda_{\mathsf{ks}}(\epsilon/2)/\Delta_{\mathbf{v}} \cdot B_{\mathsf{gs},k}$$

Bounding T_{bs}. The $|\mathsf{bs}|$ baby-step rotations use independent RDKs, so $\{\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{bs},i}\}_{i\in\mathsf{bs}}$ are independent KS errors at scale $\Delta_{\mathbf{v}}$. For each $i \in \mathsf{bs}$, define $c^{(i)} \in \mathbb{C}^{N/2}$ by

$$c^{(i)}_\ell \;=\; \begin{cases} \overline{\widetilde{M}^{(i+j)}_k} & \text{if } \ell \equiv k + j \pmod{N/2} \text{ for some } j \in \mathsf{gs}, \\ 0 & \text{otherwise.} \end{cases}$$

Since $j \mapsto k + j \pmod{N/2}$ is injective on $\mathsf{gs} \subseteq \mathbb{Z}_{N/2}$, $c^{(i)}$ is well-defined, and $\langle c^{(i)}, \widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{bs},i}\rangle = \sum_{j\in\mathsf{gs}} \widetilde{M}^{(i+j)}_k (\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{bs},i})_{k+j}$. We compute

$$\sum_{i\in\mathsf{bs}} \|c^{(i)}\|_2^2 = \sum_{\substack{i\in\mathsf{bs}\\ j\in\mathsf{gs}}} \big|\widetilde{M}^{(i+j)}_k\big|^2,$$
$$\sum_{i\in\mathsf{bs}} \|c^{(i)} \odot \widehat{\mathbf{s}}\|_2^2 = \sum_{\substack{i\in\mathsf{bs}\\ j\in\mathsf{gs}}} \big|\widetilde{M}^{(i+j)}_k \cdot \widehat{s}_{k+j}\big|^2.$$

Theorem 4 then yields

$$\Pr\Big[\, |T_{\mathsf{bs}}| > \frac{\lambda_{\mathsf{ks}}(\epsilon/2)}{\Delta_{\mathbf{v}}}\, B_{\mathsf{bs},k}\Big] < \epsilon/2.$$

Union bound. Combining the two events with failure probability $\epsilon/2$ each,

$$\Pr\Big[\, |\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{mv},k}| > \frac{\lambda_{\mathsf{ks}}(\epsilon/2)}{\Delta_{\mathbf{v}}}\big(B_{\mathsf{bs},k} + B_{\mathsf{gs},k}\big)\Big] < \epsilon.$$

Corollary 2 (Uniform PCMv bound). *Under the assumptions of Theorem 8, for every $0 < \epsilon < 1$,*

$$\Pr\left[\|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{mv}}\|_\infty > \frac{\lambda^{(N/2)}_{\mathsf{ks}}(\epsilon)}{\Delta_{\mathbf{v}}} \cdot (\beta_h + 1 + \alpha_{\mathsf{ks}}(P)) \cdot \Big(\frac{\sqrt{|\mathsf{gs}|}}{\Delta_M} + B_M\Big)\right] < \epsilon,$$

where

$$B_M := \max_{0\le k<N/2} \sqrt{\sum_{i\in\mathsf{bs}, j\in\mathsf{gs}} |\widetilde{M}^{(i+j)}_k|^2}.$$

Giant-step hoisting. With hoisting [6], the $|\mathsf{gs}|$ giant-step rotations of Eq. (3) share a single $\mathsf{ModDown}$: for each $j \in \mathsf{gs}$, the automorphism $X \mapsto X^{5^j}$ is applied to $\mathsf{ct}_j = (b_j, a_j)$, giving (b_j^*, a_j^*) under the rotated key s_j^*; the products $\sum_i \overline{a_{j,i}^*} \cdot \mathsf{swk}_j^{(i)}$ with the switching key swk_j from s_j^* to s are accumulated in $\mathcal{R}_{PQ}^2$ over $j \in \mathsf{gs}$, a single randomization key $\mathsf{rdk}_{\mathsf{gs}}$ is added, and a single $\mathsf{ModDown}$ by P is performed. That is, the giant step outputs

$$(b'', a'') = \Big(\sum_{j\in\mathsf{gs}} b_j^* + \Big\lfloor \frac{B'}{P} \Big\rceil,\ \Big\lfloor \frac{A'}{P} \Big\rceil\Big), \quad (B', A') = \mathsf{rdk}_{\mathsf{gs}} + \sum_{j\in\mathsf{gs}} \sum_{i=0}^{\mathsf{dnum}-1} \overline{a_{j,i}^*} \cdot \mathsf{swk}_j^{(i)} \pmod{PQ}.$$

For $n \geq 1$, we define the hoisted key error term

$$\alpha_{\mathsf{ks}}^{(n)}(P) := \frac{2\sqrt{3}\sigma}{P} \left(1 + \sqrt{n} \cdot \max_{0\leq i<\mathsf{dnum}} D_i \cdot \frac{N \cdot \mathsf{dnum}}{2}\right),$$

so that $\alpha_{\mathsf{ks}}^{(1)}(P) = \alpha_{\mathsf{ks}}(P)$ and $\alpha_{\mathsf{ks}}^{(n)}(P) \leq \sqrt{n}\, \alpha_{\mathsf{ks}}(P)$.

Corollary 3 (PCMv with giant-step hoisting). *Under the assumptions of Theorem 8, assume that the giant step is hoisted as above, that the switching keys $(\mathsf{swk}_j)_{j\in\mathsf{gs}}$ are distinct and independently generated, and that $\mathsf{rdk}_{\mathsf{gs}}$ is independent of $\big((a_j^*)_{j\in\mathsf{gs}}, (\mathsf{swk}_j)_{j\in\mathsf{gs}}\big)$; in particular a fresh $\mathsf{rdk}_{\mathsf{gs}}$, independent of the input ciphertext and of all baby-step keys, is admissible. Then a'' is uniform over $\mathcal{R}_Q$ and, if moreover $\mathsf{rdk}_{\mathsf{gs}}$ is independent of $(b_j^*)_{j\in\mathsf{gs}}$ as well, so is b''. The error decomposes as*

$$\widehat{\mathbf{e}}_{\mathsf{mv}}^{(\Delta_M \Delta_v)} = \widehat{\mathbf{e}}_{\mathsf{gs}}^{(\Delta)} + \sum_{i\in\mathsf{bs}, j\in\mathsf{gs}} \mathsf{rot}_j(\widehat{\mathbf{e}}_{\mathsf{bs},i}^{(\Delta)}) \odot \widetilde{M}^{(i+j)},$$

where $\widehat{\mathbf{e}}_{\mathsf{gs}}^{(\Delta)}$ is the error of the single hoisted key switching at scale $\Delta_M \Delta_{\mathbf{v}}$, and for every $0 < \epsilon < 1$ and $0 \leq k < N/2$,

$$\Pr\left[|\widehat{\mathbf{e}}_{\mathsf{mv},k}^{(\Delta_M\Delta_v)}| > \frac{\lambda_{\mathsf{ks}}(\epsilon/2)}{\Delta_{\mathbf{v}}} \left(B_{\mathsf{bs},k} + B_{\mathsf{gs},k}^{\mathsf{hoist}}\right)\right] < \epsilon, \qquad B_{\mathsf{gs},k}^{\mathsf{hoist}} := \frac{1}{\Delta_M}\left(|\widehat{s}_k| + 1 + \alpha_{\mathsf{ks}}^{(|\mathsf{gs}|)}(P)\right),$$

with $B_{\mathsf{bs},k}$ as in Theorem 8. The uniform bound of Corollary 2 holds with $\sqrt{|\mathsf{gs}|}/\Delta_M$ replaced by $1/\Delta_M$ and $\alpha_{\mathsf{ks}}(P)$ replaced by $\alpha_{\mathsf{ks}}^{(|\mathsf{gs}|)}(P)$.

Compared with Theorem 8, the rounding contribution $(|\widehat{s}_k| + 1)$ of the giant step is incurred once instead of $|\mathsf{gs}|$ times, which removes the factor $\sqrt{|\mathsf{gs}|}$ from it; only the switching-key contribution, which is already negligible for a suitable choice of P, keeps a $\sqrt{|\mathsf{gs}|}$ factor. Hoisting thus reduces both the number of randomization keys of the giant step from $|\mathsf{gs}|$ to 1 and its error, while the dominant baby-step term $B_{\mathsf{bs},k}$ is unchanged.

Proof. The baby steps and the inner accumulation are unchanged, so the slot-k error is $T_{\mathsf{bs}} + (\widehat{\mathbf{e}}_{\mathsf{gs}}^{(\Delta)})_k$ with T_{bs} as in the proof of Theorem 8, and the bound on

T_{bs} with failure probability $\epsilon/2$ applies verbatim. It remains to bound $(\widehat{\mathbf{e}}_{\mathsf{gs}}^{(\Delta)})_k$, the message-level error of the hoisted key switching, with failure probability $\epsilon/2$.

Uniformity. Write $\mathsf{rdk}_{\mathsf{gs}} = (B_{\mathsf{rdk}}, A_{\mathsf{rdk}}) = (-A_{\mathsf{rdk}}s + E_{\mathsf{rdk}}, A_{\mathsf{rdk}})$. Since $\mathsf{rdk}_{\mathsf{gs}}$ is independent of $((a_j^*)_j, (\mathsf{swk}_j)_j)$ and s is invertible in $\mathcal{R}_{PQ}$, both A_{rdk} and B_{rdk} are uniform over $\mathcal{R}_{PQ}$ and independent of $((a_j^*)_j, (\mathsf{swk}_j)_j)$; the sums $\sum_{j,i} \overline{a_{j,i}^*} A_{\mathsf{swk}_j}^{(i)}$ and $\sum_{j,i} \overline{a_{j,i}^*} B_{\mathsf{swk}_j}^{(i)}$ are functions of these, so Lemma 1 gives that A' and B' are uniform over $\mathcal{R}_{PQ}$. Hence $a'' = \lfloor A'/P \rceil$ is uniform over $\mathcal{R}_Q$; if $\mathsf{rdk}_{\mathsf{gs}}$ is also independent of $(b_j^*)_j$, then $\lfloor B'/P \rceil$ is uniform and independent of $\sum_j b_j^*$, and b'' is uniform by Lemma 1. This is Step 1 of the proof of Theorem 3 with (a, swk) replaced by $((a_j^*)_j, (\mathsf{swk}_j)_j)$.

Error identity. Using $B_{\mathsf{swk}_j}^{(i)} + A_{\mathsf{swk}_j}^{(i)} s = P g_i s_j^* + E_{\mathsf{swk}_j}^{(i)}$, $\sum_i \overline{a_{j,i}^*} g_i = a_j^*$ and $B_{\mathsf{rdk}} + A_{\mathsf{rdk}} s = E_{\mathsf{rdk}}$, exactly as in Step 2 of that proof,

$$b'' + a''s - \sum_{j \in \mathsf{gs}} \big(b_j^* + a_j^* s_j^*\big) = \frac{1}{P}\Big(E_{\mathsf{rdk}} + \sum_{j \in \mathsf{gs}} \sum_{i=0}^{\mathsf{dnum}-1} \overline{a_{j,i}^*}\, E_{\mathsf{swk}_j}^{(i)}\Big) - \Big[\frac{B'}{P}\Big]_1 - \Big[\frac{A'}{P}\Big]_1 s.$$

Since $b_j^* + a_j^* s_j^*$ decrypts to the rotated plaintext of ct_j, applying DFT and dividing by $\Delta_M \Delta_{\mathbf{v}}$ yields the claimed decomposition, with

$$\widehat{\mathbf{e}}_{\mathsf{gs}}^{(\Delta)} = \frac{1}{\Delta_M \Delta_{\mathbf{v}}}\Big(\frac{1}{P}\Big(\mathsf{DFT}(E_{\mathsf{rdk}}) + \sum_{j,i} \mathsf{DFT}(\overline{a_{j,i}^*}) \odot \mathsf{DFT}(E_{\mathsf{swk}_j}^{(i)})\Big) - \mathsf{DFT}\Big(\Big[\frac{B'}{P}\Big]_1\Big) - \mathsf{DFT}\Big(\Big[\frac{A'}{P}\Big]_1\Big) \odot \widehat{\mathbf{s}}\Big).$$

Tail bound. Set $\epsilon' := (\epsilon/2)/(\mathsf{dnum}+3)$. The terms $\mathsf{DFT}(E_{\mathsf{rdk}})_k$, $\mathsf{DFT}([B'/P]_1)_k$ and $\mathsf{DFT}([A'/P]_1)_k \cdot \widehat{s}_k$ are bounded exactly as in Step 3 of the proof of Theorem 3: since B' and A' are uniform over $\mathcal{R}_{PQ}$, the coefficients of $[B'/P]_1$ and $[A'/P]_1$ are i.i.d. $\mathcal{U}_P$, and Theorem 1 with parameter ϵ' bounds the three terms by $\sigma\sqrt{2N\ln(4/\epsilon')}$, $\sqrt{\frac{N}{6}\ln(4/\epsilon')}$ and $\sqrt{\frac{N}{6}\ln(4/\epsilon')}\,|\widehat{s}_k|$ respectively, each with failure probability ϵ'. These three rounding and RDK terms appear *once*, whatever the size of gs: this is where hoisting gains over the $|\mathsf{gs}|$ independent $\mathsf{ModDown}$s of Theorem 8.

For the switching-key term, fix $i \in [\mathsf{dnum}]$ and consider $T^{(i)} := \sum_{j \in \mathsf{gs}} \mathsf{DFT}(\overline{a_{j,i}^*})_k \cdot \mathsf{DFT}(E_{\mathsf{swk}_j}^{(i)})_k$. As the swk_j are independently generated, the $E_{\mathsf{swk}_j}^{(i)}$, $j \in \mathsf{gs}$, are i.i.d. with $\mathcal{DG}_\sigma$ coefficients. As in the proof of Theorem 4, we use the worst-case bound $|\mathsf{DFT}(\overline{a_{j,i}^*})_k| \le ND_i/2$ to treat the weights $\mathbf{w}^{(j)} := \mathsf{DFT}(\overline{a_{j,i}^*})_k\, e_k$ as deterministic, so that $\sum_{j \in \mathsf{gs}} \|\mathbf{w}^{(j)}\|_2^2 \le |\mathsf{gs}|(ND_i/2)^2$, and Theorem 1 gives

$$\Pr\Big[\,|T^{(i)}| > \sigma\sqrt{2N\ln(4/\epsilon')} \cdot \sqrt{|\mathsf{gs}|} \cdot \frac{ND_i}{2}\Big] < \epsilon'.$$

A union bound over $i \in [\mathsf{dnum}]$ bounds $|\sum_i T^{(i)}|$ by $\sigma\sqrt{2N\ln(4/\epsilon')} \cdot \sqrt{|\mathsf{gs}|} \cdot \frac{N \cdot \mathsf{dnum} \cdot \max_i D_i}{2}$ with failure probability $\mathsf{dnum}\,\epsilon'$. Combining the $\mathsf{dnum}+3$ events by a union bound (total failure $\le (\mathsf{dnum}+3)\epsilon' = \epsilon/2$), and using $\sigma\sqrt{2N\ln(4/\epsilon')} = 2\sqrt{3}\sigma \cdot \lambda_{\mathsf{ks}}(\epsilon/2)$ and $\sqrt{\frac{N}{6}\ln(4/\epsilon')} = \lambda_{\mathsf{ks}}(\epsilon/2)$, we obtain

$$\Pr\Big[\,|(\widehat{\mathbf{e}}_{\mathsf{gs}}^{(\Delta)})_k| > \frac{\lambda_{\mathsf{ks}}(\epsilon/2)}{\Delta_M \Delta_{\mathbf{v}}}\big(|\widehat{s}_k| + 1 + \alpha_{\mathsf{ks}}^{(|\mathsf{gs}|)}(P)\big)\Big] < \epsilon/2,$$

i.e. $|(\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{gs}})_k| \le \lambda_{\mathsf{ks}}(\epsilon/2)/\Delta_{\mathbf{v}} \cdot B^{\mathsf{hoist}}_{\mathsf{gs},k}$ with probability at least $1-\epsilon/2$. A final union bound with the event on T_{bs} concludes. The uniform version follows by replacing $|\widehat{s}_k|$ with β_h and taking a union bound over the $N/2$ slots, as in Corollary 2.

Theorem 9 (Polynomial evaluation with RDK). *Let* ct *be a ciphertext with scaling factor* $\Delta_{\ell'}$ *decrypting to a plaintext* pt*, which encrypts real-valued message such that* $\|\mathsf{DFT}_N(\mathsf{pt})\|_\infty \le 1$.

Let S *be a degree* 2^k-1 *polynomial as in (4),* $R = 2^t + 2^{k-t}$*,* $\epsilon \in (0,1)$*. We assume that* $\Theta_R(\epsilon) < 4^{-k}$*. Then, using the homomorphic Paterson-Stockmeyer's algorithm with baby-step parameter* t *and RDK, one obtains an output ciphertext* ct' *decrypting to* pt' *such that the message-level error*

$$\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{poleval}} = \mathsf{DFT}_N(\mathsf{pt}') - (S(\mathsf{DFT}_N(\mathsf{pt})_i))_{0\le i<N/2}$$

verifies

$$\Pr\left[\|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{poleval}}(S)\|_\infty > \Theta_R(\epsilon)\ \cdot A_S + \Theta_R^2(\epsilon)\cdot B_S\right] < \epsilon,$$

where

$$A_S := 1 + \sum_{j=0}^{2^{k-t}-1}\Big(\sum_{i=0}^{2^t-1}|p_{i+j2^t}|\frac{i^2}{2} + |BS_j|_\infty (j2^t)^2\Big), \quad B_S := \sum_{j,i}|p_{i+j2^t}|\frac{(ij2^t)^2}{2}.$$

Proof. We estimate the message-level error on individual Chebyshev monomials. If η_i is the error on T_i, we have the induction

$$\eta_i = 2T_{\lceil i/2\rceil}\eta_{\lfloor i/2\rfloor} + 2T_{\lfloor i/2\rfloor}\eta_{\lceil i/2\rceil} + 2\eta_{\lfloor i/2\rfloor}\eta_{\lceil i/2\rceil} + \widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{mult},i} - \eta_{i \bmod 2}, \tag{6}$$

with $\eta_0 = 0$ and $\eta_1 = 0$: as described in Section 5.2, $T_1(x) = x$ is never rescaled but only brought to lower levels by exact modulus dropping, so that it carries no computation error. Rounding the constants that absorb the scaling-factor mismatch perturbs the computed monomials deterministically by at most $1/(2\Delta_{\ell+2})$ each; as for the rounding of the coefficients p_{i+j2^t}, we regard this as a perturbation of S (by at most $A_S/(2\Delta_{\ell+2}) + B_S/(4\Delta^2_{\ell+2})$, from the analysis below applied with $\Theta = 1/(2\Delta_{\ell+2})$), which is smaller than the bound of the theorem by a factor $\approx \lambda_{\mathsf{rs}}(\epsilon)(\beta_s+1)$ and is not tracked further. (Should the implementation instead rescale x to level $\ell+2$, one would have $|\eta_1| \le \Theta$; the lemma below is stated for a general $\overline{\eta}_1 \ge 0$ so as to cover this case as well, at the cost of larger constants.) We analyze this induction in the following lemma:

Lemma 7. *Let* $(\bar\eta_i)$ *be inductively defined by* $\bar\eta_0 = 0$*,* $\bar\eta_1 \ge 0$ *and, for* $i > 1$*,* $\bar\eta_i = 2\bar\eta_{\lceil i/2\rceil} + 2\bar\eta_{\lfloor i/2\rfloor} + 2\bar\eta_{\lceil i/2\rceil}\bar\eta_{\lfloor i/2\rfloor} + \Theta + \mathbf{1}_{i \bmod 2=1}\overline{\eta_1}$*, with* Θ *a positive constant. We assume that* $\Theta + \frac{10}{3}\bar\eta_1 < 2\cdot 4^{-s}$*. Then, for* $0 \le i \le 2^s$*, one has*

$$\overline{\eta}_i \le \frac{i^2}{2}\left(\Theta + \frac{10}{3}\overline{\eta_1}\right).$$

Proof (of Lemma 7). We let $a = \overline{\eta}_1$ and $B = \Theta + \frac{10}{3}a$, and first analyse the linearised recurrence $f_0 = 0$, $f_1 = a$,

$$f_i = 2f_{\lceil i/2 \rceil} + 2f_{\lfloor i/2 \rfloor} + \Theta + \mathbf{1}_{i \bmod 2=1} a.$$

For all positive i, we can write f_i as a linear combination of Θ and a, as $f_i = u_i\Theta + v_i a$. If $\mu = 2^{\lfloor \log_2 i \rfloor}$, we prove by induction that $u_i = \mu i - \frac{2\mu^2+1}{3}, v_i = -i^2 + 6\mu i - 4\mu^2$.

For $i = 1$, we observe $u_i = 0$ and $v_i = 1$. For $i \geq 2$, we have, using the induction hypothesis,

$$\begin{aligned} u_i &= \underbrace{2\,(\mu/2)\lfloor i/2 \rfloor - 2\frac{2(\mu/2)^2+1}{3}}_{2u_{\lfloor i/2 \rfloor}} + \underbrace{2(\mu/2)\lceil i/2 \rceil - 2\frac{2(\mu/2)^2+1}{3}}_{2u_{\lceil i/2 \rceil}} + 1, \\ &= \mu i - \frac{2\mu^2+1}{3}, \end{aligned}$$

as claimed.

Regarding v_i, we first note that $2\lfloor i/2 \rfloor^2 + 2\lceil i/2 \rceil^2 = i^2 + \mathbf{1}_{i \bmod 2=1}$. It follows:

$$\begin{aligned} v_i &= 2\left(\underbrace{-\lfloor i/2 \rfloor^2 + 6\mu/2\lfloor i/2 \rfloor - 4(\mu/2)^2}_{2v_{\lfloor i/2 \rfloor}}\right) + 2\left(\underbrace{-\lceil i/2 \rceil^2 + 6\mu/2\lceil i/2 \rceil - 4(\mu/2)^2}_{2v_{\lceil i/2 \rceil}}\right) + \mathbf{1}_{i \bmod 2=1}, \\ &= -i^2 + 6\mu i - 4\mu^2. \end{aligned}$$

We deduce that

$$u_i \leq i^2\left(\frac{\mu}{i} - \frac{2}{3}\left(\frac{\mu}{i}\right)^2\right) \leq \frac{3}{8}i^2, \quad v_i \leq i^2\left(-1 + 6\frac{\mu}{i} - 4\left(\frac{\mu}{i}\right)^2\right) \leq \frac{5}{4}i^2,$$

where we used $\mu/i \in [1/2, 1]$. Overall, we obtain $f_i \leq \frac{3}{8}i^2\Theta + \frac{5}{4}i^2\overline{\eta_1} \leq \frac{3}{8}i^2\left(\Theta + \frac{10}{3}\overline{\eta_1}\right) = \frac{3}{8}i^2 B$.

We now prove by induction that, for $0 \leq i \leq 2^s$,

$$\overline{\eta_i} \leq f_i + \frac{1}{16}B^2 i^4, \quad \overline{\eta_i} \leq \frac{1}{2}Bi^2.$$

This follows from the definitions for $i = 0, 1$, and assume both inequalities to be valid for $j < i$. Let $i \geq 2$; from the definition

$$\begin{aligned}
\overline{\eta}_i &\leq 2f_{\lfloor i/2 \rfloor} + 2f_{\lceil i/2 \rceil} + \Theta + \mathbf{1}_{i \bmod 2 = 1}\overline{\eta}_1 + \frac{B^2}{8}\left(\left\lfloor \frac{i}{2} \right\rfloor^4 + \left\lceil \frac{i}{2} \right\rceil^4\right) + 2\overline{\eta}_{\lfloor i/2 \rfloor}\overline{\eta}_{\lceil i/2 \rceil}, \\
&\leq f_i + \frac{B^2}{8}\left(\left\lfloor \frac{i}{2} \right\rfloor^4 + \left\lceil \frac{i}{2} \right\rceil^4\right) + 2\overline{\eta}_{\lfloor i/2 \rfloor}\overline{\eta}_{\lceil i/2 \rceil}, \\
&\leq f_i + \frac{B^2}{8}\left(\left\lfloor \frac{i}{2} \right\rfloor^4 + \left\lceil \frac{i}{2} \right\rceil^4\right) + \frac{B^2}{2}\left(\left\lfloor \frac{i}{2} \right\rfloor^2 + \left\lceil \frac{i}{2} \right\rceil^2\right), \\
&\leq f_i + \frac{i^4 B^2}{16}, \\
&\leq i^2 B\left(\frac{3}{8} + \frac{i^2 B}{16}\right),
\end{aligned}$$

which concludes the induction, under the assumption $2^{2s}B \leq 2$.

We estimate the error on individual Chebyshev monomials. Let $\Theta = \Theta_R(\epsilon)$, and we place ourselves in the event that the computation error at every stage of polynomial evaluation is at most Θ. The stages are the $2^t - 1$ multiplications producing $T_2, \ldots, T_{2^t}$, the $2^{k-t} - 2$ multiplications producing $T_{2\cdot 2^t}, \ldots, T_{(2^{k-t}-1)2^t}$, and the final double rescale, i.e. fewer than $R = 2^t + 2^{k-t}$ stages, each involving at most one rescaling and one key switching on $N/2$ slots; by Theorems 2 and 3 and a union bound, this event has probability at least $1 - \epsilon$. Note that bringing $T_1(x) = x$ to a lower level is not a stage: it is an exact modulus dropping and introduces no error.

For an individual slot $x \in [-1, 1]$, if η_i is the error on $T_i(x)$, Equation (6) holds with $\eta_1 = 0$ and $|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{mult},i}| \leq \Theta$. Bounding $|T_i(x)|$ by 1, we deduce that the sequence $\overline{\eta}_i$ defined in Lemma 7 with $\overline{\eta}_1 = 0$ upper bounds the sequence $|\eta_i|$, $0 \leq i \leq 2^t$. Since $\overline{\eta}_1 = 0$, the hypothesis of the lemma reads $\Theta < 2 \cdot 4^{-t}$, which follows from $\Theta < 4^{-k}$, and its conclusion is

$$|\eta_i| \leq \frac{i^2}{2}\Theta, \qquad 0 \leq i \leq 2^t.$$

The error on baby-step j is equal to $e_j = \sum_{i=0}^{2^t-1} p_{i+j2^t}\eta_i$, so that

$$|e_j| \leq \sum_{i=0}^{2^t-1} |p_{i+j2^t}|\frac{i^2}{2}\Theta_R(\epsilon).$$

We again make use of Lemma 7 for the giant step error, with $\overline{\eta}_1 = |\eta_{2^t}| \leq 2^{2t-1}\Theta$, so that

$$|\eta_{j2^t}| \leq j^2 2^{2t}\Theta$$

for $2 \leq j < 2^{k-t}$. This application is legitimate as $(\Theta + \frac{10}{3}\overline{\eta}_1)4^{k-t} \leq \Theta(1 + \frac{5}{3}4^t)4^{k-t} < 2 \cdot 4^k\Theta < 2$, under the assumption $\Theta < 4^{-k}$.

Hence, the error on $\mathrm{BS}_j T_{j2^t}$ (we recall that this term is not rescaled) is

$$\leq |e_j| + \|\mathrm{BS}_j\|_\infty |\eta_{j2^t}| + |e_j||\eta_{j2^t}|,$$

and, adding the error of the final double rescale of the sum $\sum_j \mathrm{BS}_j T_{j2^t}$, which is at most $\Theta_R(\epsilon)$ and accounts for the leading term 1 in A_S, the total error is upper bounded by

$$\|\hat{e}^{(\Delta)}_{\mathsf{poleval}}(S)\|_\infty \leq \Theta_R(\epsilon) + \Theta_R(\epsilon) \sum_{j=0}^{2^{k-t}-1} \left(\sum_{i=0}^{2^t-1} |p_{i+j2^t}| \frac{i^2}{2} + \|\mathrm{BS}_j\|_\infty (j2^t)^2 \right) + \Theta_R(\epsilon)^2 \left(\sum_{j=0}^{2^{k-t}-1} \sum_{i=0}^{2^t-1} |p_{i+j2^t}| \frac{(ij2^t)^2}{2} \right),$$

as claimed.

Theorem 10 (Double angle formula with RDK). *Under the same notations and assumptions as in Theorem 9 with $R = 2^t + 2^{k-t} + r$, and $\Theta_R(\epsilon) + 3\|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{poleval}}(S)\|_\infty < 4^{4-r}$, when the polynomial evaluation is followed by r iterations of the double angle formula the total error $\mathbf{e}_{\mathsf{EM}}$ can be estimated as*

$$\Pr\left[\|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{EM}}\|_\infty > 3 \cdot 2^{2r-1} \cdot (\Theta_R(\epsilon) \cdot A_S + \Theta_R(\epsilon)^2 \cdot B_S) + \Theta_R(\epsilon) \cdot (2^{2r-1}+1)\right] < \epsilon.$$

Proof. This induction obtaining S_1 has the same shape as the one used for the Chebyshev monomial computation, and Lemma 7 can still be utilized, with the total error divided by 2 to account for the fact that the last step is S_1^2 rather than $2S_1^2 - 1$, under the modified conditions $R \geq (2^t + 2^{k-t} + r)$, $\Theta_R(\epsilon) + 3\|\widehat{\mathbf{e}}^{(\Delta)}_{\mathsf{poleval}}(S)\|_\infty < 4^{4-r}$.

Theorem 11 (Error of binary bootstrapping). *Considering the above construction, for $0 \leq i < d_{\mathsf{StC}}$ and $0 \leq j < d_{\mathsf{CtS}}$, let $\mathbf{e}_{\mathsf{StC},i}$ and $\mathbf{e}_{\mathsf{CtS},j}$ be the message-side PCMv errors of the corresponding steps, bounded by Theorem 8, and let $\mathbf{e}'_{\mathsf{DtS}}$ and $\mathbf{e}'_{\mathsf{StD}}$ be the message side key switching error from the* DtS *step and* StD *step, bounded by Theorem 3. We let $\mathbf{e}_{\mathsf{EM}}$ denote the homomorphic error introduced by the* EvalMod *step (independent of the input message). Defining the accumulated* StC *and* CtS *errors by*

$$\mathbf{e}_{\mathsf{StC}} := \sum_{i=0}^{d_{\mathsf{StC}}-1} \left(\widetilde{M}_i \widetilde{M}_{i-1} \cdots \widetilde{M}_0\right)^{-1} \mathbf{e}_{\mathsf{StC},i}, \quad \mathbf{e}_{\mathsf{DtS}} := 2U_{N/2}^{-1} \mathbf{e}'_{\mathsf{DtS}},$$

$$\mathbf{e}_{\mathsf{CtS}} := \frac{2q_0(K+1)}{\Delta_{\mathsf{EM}}} \sum_{j=0}^{d_{\mathsf{CtS}}-1} \widetilde{N}_{d_{\mathsf{CtS}}-1} \cdots \widetilde{N}_{j+1}\, \mathbf{e}_{\mathsf{CtS},j}, \quad \mathbf{e}_{\mathsf{StD}} = 2U_{N/2}^{-1} \mathbf{e}'_{\mathsf{StD}}$$

and set $\mathbf{e} := \mathbf{e}_{\mathsf{In}} + \mathbf{e}_{\mathsf{StC}} + \mathbf{e}_{\mathsf{DtS}} + \mathbf{e}_{\mathsf{StD}} + \mathbf{e}_{\mathsf{CtS}}$. Then, the element-wise bootstrapping error $\mathbf{e}_{\mathsf{BTS}} := \mathbf{z}_{\mathsf{EM}} - \mathbf{z}$ satisfies, for every $0 \leq k < N/2$,

$$|\mathbf{e}_{\mathsf{BTS},k}| \;\leq\; |\mathbf{e}_{\mathsf{EM},k}| + \|f - p\|_\infty + \frac{\pi^2}{4} |\mathbf{e}_k|^2$$

Proof. We trace the message-level error along the four stages of the bootstrapping circuit. Throughout the proof, all scaling factors are taken in the idealized form described in Section 5.4; we denote $\mathbf{m}_{\mathsf{StC}}, \mathbf{m}_{\mathsf{MR}}, \mathbf{m}_{\mathsf{CtS}}, \mathbf{m}_{\mathsf{EM}}$ for the decrypted slot vectors at the end of each stage.

Step 1 (Error after StC). By Theorem 8, the i-th PCMv step ($0 \le i < d_{\mathsf{StC}}$) adds a message-level error $\mathbf{e}_{\mathsf{StC},i}$ to the output of $\widetilde{M}_i$. Composing the d_{StC} steps yields

$$\begin{aligned}&\widetilde{M}_{d_{\mathsf{StC}}-1} \cdots \widetilde{M}_0(\mathbf{z} + \mathbf{e}_{\mathsf{In}}) \; + \; \sum_{i=0}^{d_{\mathsf{StC}}-1} \widetilde{M}_{d_{\mathsf{StC}}-1} \cdots \widetilde{M}_{i+1}\, \mathbf{e}_{\mathsf{StC},i} \\ =&\widetilde{\mathsf{StC}}\left(\mathbf{z} + \mathbf{e}_{\mathsf{In}} + \sum_{i=0}^{d_{\mathsf{StC}}-1} (\widetilde{M}_i \cdots \widetilde{M}_0)^{-1} \mathbf{e}_{\mathsf{StC},i}\right),\end{aligned}$$

where in the second line we used the identity $\widetilde{\mathsf{StC}}^{-1} \widetilde{M}_{d_{\mathsf{StC}}-1} \cdots \widetilde{M}_{i+1} = (\widetilde{M}_i \cdots \widetilde{M}_0)^{-1}$. Recalling the definition of $\mathbf{e}_{\mathsf{StC}}$ in the theorem statement and that $\widetilde{\mathsf{StC}} = \frac{q_0}{2\Delta_{\mathsf{In}}} U_{N/2}$, this can be simplified to:

$$\frac{q_0}{2\Delta_{\mathsf{In}}} U_{N/2}\left(\mathbf{z} + \mathbf{e}_{\mathsf{In}} + \mathbf{e}_{\mathsf{StC}}\right), \text{with scale } \Delta_{\mathsf{In}}.$$

This is equivalent with the ciphertext encrypting:

$$\mathbf{m}_{\mathsf{StC}} := U_{N/2}\left(\frac{\mathbf{z} + \mathbf{e}_{\mathsf{In}} + \mathbf{e}_{\mathsf{StC}}}{2}\right), \text{with scale } q_0.$$

Step 2 (Error after ModRaise). The modulus lift adds the integer-valued vector $\mathbf{I}$ (with $\|\mathbf{I}\|_\infty \le K$ by the choice of K in line 5 of Algorithm 3) to the coefficients, while the two key switching operations DtS and StD contribute, by Theorem 3, message-level errors $\mathbf{e}'_{\mathsf{DtS}}$ and $\mathbf{e}'_{\mathsf{StD}}$ respectively. The modulus lift itself is exact in the message domain. Therefore

$$\begin{aligned}\mathbf{m}_{\mathsf{MR}} &= \mathbf{m}_{\mathsf{StC}} + U_{N/2}(\mathbf{I}) + \mathbf{e}'_{\mathsf{DtS}} + \mathbf{e}'_{\mathsf{StD}} \\ &= U_{N/2}\left(\mathbf{I} + \frac{1}{2}(\mathbf{z} + \mathbf{e}_{\mathsf{In}} + \mathbf{e}_{\mathsf{StC}} + \mathbf{e}_{\mathsf{DtS}} + \mathbf{e}_{\mathsf{StD}})\right), \text{with scale } q_0.\end{aligned}$$

Step 3 (Error after CtS). The same composition argument as in Step 1 applied to Theorem 8 for $\widetilde{N}_0, \ldots, \widetilde{N}_{d_{\mathsf{CtS}}-1}$ gives

$$\begin{aligned}&\widetilde{N}_{d_{\mathsf{CtS}}-1} \cdots \widetilde{N}_0 \mathbf{m}_{\mathsf{MR}} + \sum_{j=0}^{d_{\mathsf{CtS}}-1} \widetilde{N}_{d_{\mathsf{CtS}}-1} \cdots \widetilde{N}_{j+1}\, \mathbf{e}_{\mathsf{CtS},j} \\ =&\widetilde{\mathsf{CtS}}\, \mathbf{m}_{\mathsf{MR}} + \frac{\Delta_{\mathsf{EM}}}{2q_0(K+1)} \mathbf{e}_{\mathsf{CtS}},\end{aligned}$$

using the definition of $\mathbf{e}_{\mathsf{CtS}}$ in the theorem statement. Substituting Steps 1–2 and recalling $\widetilde{\mathsf{CtS}} = \frac{\Delta_{\mathsf{EM}}}{q_0(K+1)} U_{N/2}^{-1}$, we obtain the ciphertext encrypting

$$\mathbf{m}_{\mathsf{CtS}} = \frac{1}{K+1}\left(\mathbf{I} + \frac{\mathbf{z}+\mathbf{e}}{2}\right), \text{with scale } \Delta_{\mathsf{EM}}.$$

where the accumulated input-equivalent error

$$\mathbf{e} \;:=\; \mathbf{e}_{\mathsf{In}} + \mathbf{e}_{\mathsf{StC}} + \mathbf{e}_{\mathsf{DtS}} + \mathbf{e}_{\mathsf{StD}} + \mathbf{e}_{\mathsf{CtS}}$$

is exactly the quantity defined in the theorem statement.

Step 4 (Error after EvalMod and final scale adjustment). Set

$$\mathbf{w} = \frac{1}{K+1}\left(\mathbf{I} + \frac{\mathbf{z}+\mathbf{e}}{2}\right)$$

EvalMod step first homomorphically computes p, which results:

$$\begin{aligned}\mathbf{e}_{\mathsf{BTS}} :=& \mathbf{m}_{\mathsf{EM}} - \mathbf{z}\\ =& p(\mathbf{w}) + \mathbf{e}_{\mathsf{EM}} - \mathbf{z}\\ =& \big(p(\mathbf{w}) - f(\mathbf{w})\big) + \big(f(\mathbf{w}) - \mathbf{z}\big) + \mathbf{e}_{\mathsf{EM}}\end{aligned}$$

Therefore,

$$\begin{aligned}|\mathbf{e}_{\mathsf{BTS}}, k| &\le |\mathbf{e}_{\mathsf{EM},k}| + |f(\mathbf{w}_k) - \mathbf{z}_k| + |p(\mathbf{w}_k) - f(\mathbf{w}_k)|\\ &\le |\mathbf{e}_{\mathsf{EM},k}| + \frac{\pi^2}{4}|\mathbf{e}_k|^2 + \|p - f\|_\infty,\end{aligned}$$

where the final inequality is from Lemma 1 of [1].

Theorem 11 has the following uniform version. For $M \in \mathbb{R}^{d\times d}$, let the operator norm be defined as $|||M||| = \max_{x\in\mathbb{R}^d-\{0\}} \|M\cdot x\|_\infty / \|x\|_\infty$. Define the norms

$$\kappa_{\mathsf{StC},i} := \left|\left|\left|(\widetilde{M}_i \cdots \widetilde{M}_0)^{-1}\right|\right|\right|, \quad \kappa_{\mathsf{CtS},j} := \left|\left|\left|\widetilde{N}_{d_{\mathsf{CtS}}-1} \cdots \widetilde{N}_{j+1}\right|\right|\right|, \quad \kappa_{U^{-1}} := \left|\left|\left|U_{N/2}^{-1}\right|\right|\right|.$$

Theorem 11 then has the uniform version:

Corollary 4 (Uniform BinBoot bound). *For $0 < \epsilon < 1$, let $\epsilon' = \frac{1}{d_{\mathsf{StC}}+d_{\mathsf{CtS}}+3}\,\epsilon$. Let $B_{\mathsf{StC},i}(\epsilon')$ and $B_{\mathsf{CtS},j}(\epsilon')$ denote the uniform PCMv bounds of Corollary 2 for the i-th layer of* StC *and the j-th layer of* CtS*, let $B_{\mathsf{DtS}}(\epsilon')$ and $B_{\mathsf{StD}}(\epsilon')$ denote the uniform message-side key-switching bounds of Theorem 3 for the* DtS *and* StD *steps, and let $B_{\mathsf{EM}}(\epsilon')$ be the public EvalMod bound. Then*

$$\Pr\left[\|\mathbf{e}_{\mathsf{BTS}}\|_\infty \ge B_{\mathsf{EM}}(\epsilon') + \|p - f\|_\infty + \frac{\pi^2}{4}E^2\right] < \epsilon,$$

where

$$E := \|\mathbf{e}_{\mathsf{In}}\|_\infty + \sum_{i=0}^{d_{\mathsf{StC}}-1} \kappa_{\mathsf{StC},i} \cdot B_{\mathsf{StC},i}(\epsilon') + 2\,\kappa_{U^{-1}}\big(B_{\mathsf{DtS}}(\epsilon') + B_{\mathsf{StD}}(\epsilon')\big) + \frac{2q_0(K+1)}{\Delta_{\mathsf{EM}}} \sum_{j=0}^{d_{\mathsf{CtS}}-1} \kappa_{\mathsf{CtS},j} \cdot B_{\mathsf{CtS},j}(\epsilon').$$

Theorem 12 (Cleaning error). *Let* ct *encrypt* $\mathbf{z} = \mathbf{b} + \mathbf{e}$ *with scaling factor* Δ *with* $\mathbf{b} \in \{0,1\}^{N/2}$. *Define* $\widetilde{\Delta} = \lfloor \Delta \rceil$ *and*

$$\widetilde{\mathbf{e}} = \frac{\Delta}{\widetilde{\Delta}}\mathbf{z} - \mathbf{b} = \frac{\Delta}{\widetilde{\Delta}}\mathbf{e} + \frac{\Delta - \widetilde{\Delta}}{\widetilde{\Delta}}\mathbf{b}.$$

so that

$$|\widetilde{\mathbf{e}}_k| \le \frac{\Delta}{\widetilde{\Delta}}|\mathbf{e}_k| + \frac{1}{2\widetilde{\Delta}}.$$

Let $\mathbf{e}_{\mathsf{ks},1}, \mathbf{e}_{\mathsf{ks},2}$ *denote the key switching errors of steps 1–2 and* $\mathbf{e}_{\mathsf{rs},3}$ *the rescale error of step 4. Then the output of the cleaning algorithm encrypts* $\mathbf{b} + \mathbf{e}_{\mathsf{cln}}$ *at scale* $\widetilde{\Delta}^3/q$ *where*

$$|(\mathbf{e}_{\mathsf{cln}})_k| \le 3|(\widetilde{\mathbf{e}})_k|^2 + 2|(\widetilde{\mathbf{e}})_k|^3 + (3 + 2|(\widetilde{\mathbf{e}})_k|) \cdot |(\mathbf{e}_{\mathsf{ks},1})_k| + 2|(\mathbf{e}_{\mathsf{ks},2})_k| + |(\mathbf{e}_{\mathsf{rs},3})_k|$$

Proof. Set $\widetilde{\mathbf{z}} := (\Delta/\widetilde{\Delta})\,\mathbf{z} = \mathbf{b} + \widetilde{\mathbf{e}}$ with

$$\widetilde{\mathbf{e}} \;=\; \tfrac{\Delta}{\widetilde{\Delta}}\,\mathbf{z} - \mathbf{b} \;=\; \tfrac{\Delta}{\widetilde{\Delta}}\,\mathbf{e} \;+\; \tfrac{\Delta - \widetilde{\Delta}}{\widetilde{\Delta}}\,\mathbf{b},$$

so that, slot-wise, $|\widetilde{\mathbf{e}}_k| \le \frac{\Delta}{\widetilde{\Delta}}|\mathbf{e}_k| + \frac{1}{2\widetilde{\Delta}}$, using $|b_k| \le 1$ and $|\Delta - \widetilde{\Delta}| \le 1/2$.

We trace the message-level slot vectors through the algorithm.

Step 1 (Tensor + Relin). Encrypts $\widetilde{\mathbf{z}}^{\odot 2} + \mathbf{e}_{\mathsf{ks},1}$ at scale $\widetilde{\Delta}^2$, where $\mathbf{e}_{\mathsf{ks},1}$ is the key-switching error of the key switching (Theorem 3).

Step 2 (Tensor + Relin against $\widetilde{\mathbf{z}}$*).* Encrypts

$$(\widetilde{\mathbf{z}}^{\odot 2} + \mathbf{e}_{\mathsf{ks},1}) \odot \widetilde{\mathbf{z}} + \mathbf{e}_{\mathsf{ks},2} \;=\; \widetilde{\mathbf{z}}^{\odot 3} + \widetilde{\mathbf{z}} \odot \mathbf{e}_{\mathsf{ks},1} + \mathbf{e}_{\mathsf{ks},2}$$

at scale $\widetilde{\Delta}^3$. The error introduced is the key-switching error $\mathbf{e}_{\mathsf{ks},2}1$.

Step 3 (Affine combination). The combination $3\widetilde{\Delta} \cdot$ (step 1 ct) $- 2 \cdot$ (step 2 ct) at scale $\widetilde{\Delta}^3$ encrypts

$$\begin{aligned} &3\widetilde{\Delta}\,(\widetilde{\mathbf{z}}^{\odot 2} + \mathbf{e}_{\mathsf{ks},1}) - 2(\widetilde{\mathbf{z}}^{\odot 3} + \widetilde{\mathbf{z}} \odot \mathbf{e}_{\mathsf{ks},1} + \mathbf{e}_{\mathsf{ks},2}) \\ &= f_{\mathsf{cln}}(\widetilde{\mathbf{z}}) + (3 - 2\widetilde{\mathbf{z}}) \odot \mathbf{e}_{\mathsf{ks},1} - 2\,\mathbf{e}_{\mathsf{ks},2}, \end{aligned}$$

where $f_{\mathsf{cln}}(x) = 3x^2 - 2x^3$. This step is exact since $3\widetilde{\Delta}$ and 2 are both an integer.

Step 4 Recaling by q*.* Rescaling by q adds the rescale error $\mathbf{e}_{\mathsf{rs},3}$ (Theorem 2), producing a ciphertext at scale $\widetilde{\Delta}^3/q$ encrypting

$$f_{\mathsf{cln}}(\widetilde{\mathbf{z}}) + (3 - 2\widetilde{\mathbf{z}}) \odot \mathbf{e}_{\mathsf{ks},1} - 2\,\mathbf{e}_{\mathsf{ks},2} + \mathbf{e}_{\mathsf{rs},3}.$$

The cleaning error is the deviation from $\mathbf{b}$:

$$\mathbf{e}_{\mathsf{cln}} \;=\; \big(f_{\mathsf{cln}}(\widetilde{\mathbf{z}}) - \mathbf{b}\big) + (3 - 2\widetilde{\mathbf{z}}) \odot \mathbf{e}_{\mathsf{ks},1} - 2\,\mathbf{e}_{\mathsf{ks},2} + \mathbf{e}_{\mathsf{rs},3}. \tag{7}$$

Bounding $f_{\mathsf{cln}}(\widetilde{\mathbf{z}})_k - \mathbf{b}_k$. Fix k and write $\widetilde{\mathbf{z}}_k = b_k + \widetilde{e}_k$ with $b_k \in \{0,1\}$. Expanding,

$$f_{\mathsf{cln}}(b_k + \widetilde{e}_k) - b_k \;=\; 3(b_k + \widetilde{e}_k)^2 - 2(b_k + \widetilde{e}_k)^3 - b_k.$$

Since $b_k^2 = b_k$ and $b_k^3 = b_k$ for $b_k \in \{0,1\}$, the remaining terms are $(3 - 6b_k)\widetilde{e}_k^2 - 2\widetilde{e}_k^3$, so, using $|3 - 6b_k| = 3$,

$$|f_{\mathsf{cln}}(\widetilde{\mathbf{z}})_k - b_k| \;\le\; 3|\widetilde{e}_k|^2 + 2|\widetilde{e}_k|^3.$$

Bounding the coefficient of $e_{\mathsf{ks},1}$. For $b_k \in \{0,1\}$, $|3 - 2b_k| \le 3$, so

$$|3 - 2\widetilde{z}_k| \;=\; |3 - 2b_k - 2\widetilde{e}_k| \;\le\; |3 - 2b_k| + 2|\widetilde{e}_k| \;\le\; 3 + 2|\widetilde{e}_k|.$$

Combining. Taking absolute values slot-wise in (7),

$$|(\mathbf{e}_{\mathsf{cln}})_k| \;\le\; 3|\widetilde{e}_k|^2 + 2|\widetilde{e}_k|^3 + (3 + 2|\widetilde{e}_k|)\,|(\mathbf{e}_{\mathsf{ks},1})_k| + 2\,|(\mathbf{e}_{\mathsf{ks},2})_k| + |(\mathbf{e}_{\mathsf{rs},3})_k|.$$

This bound can be made uniform; we obtain the following corollary.

Corollary 5 (Uniform Cleaning bound). *Under the assumptions of Theorem 12, fix* $0 < \epsilon < 1$ *and let* $\epsilon' := \epsilon/3$. *Define*

$$B_1(\epsilon') := \frac{\lambda_{\mathsf{ks}}^{(N/2)}(\epsilon')}{\widetilde{\Delta}^2}\big(\beta_h + 1 + \alpha_{\mathsf{ks}}(P)\big), \qquad B_2(\epsilon') := \frac{\lambda_{\mathsf{ks}}^{(N/2)}(\epsilon')}{\widetilde{\Delta}^3}\big(\beta_h + 1 + \alpha_{\mathsf{ks}}(P)\big),$$

$$B_3(\epsilon') := \frac{\lambda_{\mathsf{rs}}^{(N/2)}(\epsilon')}{\widetilde{\Delta}^3/q}\big(\beta_h + 1 + \alpha_{\mathsf{rs}}(q)\big),$$

and let $\widetilde{B} := \frac{\Delta}{\widetilde{\Delta}}\,\|\mathbf{e}\|_\infty + \frac{1}{2\widetilde{\Delta}}$ *be the deterministic upper bound on* $\|\widetilde{\mathbf{e}}\|_\infty$ *given by Theorem 12. Then the following holds:*

$$\Pr\Big[\|\mathbf{e}_{\mathsf{cln}}\|_\infty > 3\,\widetilde{B}^{\,2} + 2\,\widetilde{B}^{\,3} \;+\; \big(3 + 2\widetilde{B}\big)\,B_1(\epsilon') \;+\; 2\,B_2(\epsilon') \;+\; B_3(\epsilon')\Big] < \epsilon$$

C RDK footprint for bootstrapping

Estimates of RDK footprint for bootstrapping can be found in Table 3.

D Experiments

D.1 Circuit bounds for PCMV in Section 5

We validate the PCMv bound of Theorem 8 varying diagonal scaling factors of $\Delta_M \in \{2^{20}, 2^{25}, 2^{30}\}$ and $\Delta_{\mathbf{v}} \approx 2^{40}$, separating the two error contributions: the encoding error of the diagonals (ecd) and the key-switching error (ks). Instead of bounding the encoding errors, we measured it exactly after the encoding, and used the exact value. Table 8 shows that for small and moderate Δ_M, ecd dominates and the overall bound is essentially flat in the tail probability, matching the empirical error to within 0.05 bits. At $\Delta_M = 2^{30}$, ks starts to take over and the dependence on the tail probability becomes visible, but our bound remains within roughly one bit of the Gaussian extrapolation.

Table 3. Key Size Comparison: SWK, shared and per-invocation RDKs (MB). $\mathsf{rdk}_{\mathsf{md}}$: randomization keys consumed at ModDown steps; by Theorem 7 they are shared by all invocations of the circuit. $\mathsf{rdk}_{\mathsf{rs}}$: randomization keys consumed at Rescale steps, which must be fresh at every invocation; one per level consumed by the step, and living at the ciphertext modulus of that level (no auxiliary modulus P). ρ is the fresh key added to the input of every iteration after the first.

	SWK			$\mathsf{rdk}_{\mathsf{md}}$ (shared)		$\mathsf{rdk}_{\mathsf{rs}}$ (per invocation)	
	# keys	dnum	size	# keys	size	# keys	size
CtS	10	4	2120	8	424	1	47
	10	4	2080	8	416	1	46
	10	4	2040	8	408	1	45
EM	1	3	150	9	420	9	366
StC	10	1	60	8	48	1	4
	10	1	50	8	40	1	3
	10	1	40	8	32	1	2
ρ	–	–	–	–	–	1	5
Total			6540		1788 (27.3%)		518 (7.9%)

Table 4. Output precision of binary bootstrapping with parameter given in Table 1 as a function of the input noise $\|\mathbf{e}_{\mathsf{In}}\|_\infty$. All entries are in $\log_2$.

$\|\mathbf{e}_{\mathsf{In}}\|_\infty$	-15.0	-12.0	-9.0	-7.5	-6.0
$\|\mathbf{e}_{\mathsf{BTS}}\|_\infty$	-9.05	-9.04	-9.13	-8.81	-8.27

E Encoding error of the diagonal plaintexts

For each nonzero diagonal $\ell \in \mathcal{L}$, let diag_ℓ be the plaintext encoding of $M^{(\ell)}$ with scale Δ_M. We let ρ_ℓ be the encoding rounding error, namely $\rho_\ell := \mathsf{diag}_\ell - \Delta_M \mathsf{iDFT}(M^{(\ell)})$, a polynomial with coefficients in $(-1/2, 1/2]^N$, and

$$\eta_\ell := \frac{1}{\Delta_M}\mathsf{DFT}(\rho_\ell) = \frac{1}{\Delta_M}\mathsf{DFT}(\mathsf{diag}_\ell) - M^{(\ell)}.$$

Thus, replacing the exact diagonal $M^{(\ell)}$ by its encoded plaintext amounts to replacing M by a deterministic perturbed matrix $\widetilde{M} = M + E_M$, where the diagonal representation of E_M is given by $(E^{(\ell)})_{\ell\in\mathcal{L}} = (\eta_\ell)_{\ell\in\mathcal{L}}$.

Therefore, even before accounting for computation errors, the evaluated linear transform is

$$\widetilde{M}\mathbf{v} = M\mathbf{v} + E_M\mathbf{v} = M\mathbf{v} + \sum_{\ell\in\mathcal{L}} \eta_\ell \odot \mathsf{rot}_\ell(\mathbf{v}).$$

For an arbitrary input vector $\mathbf{v}$, this gives the slot-wise deterministic bound

$$|(E_M\mathbf{v})_k| \le \sum_{\ell\in\mathcal{L}} |(\eta_\ell)_k|\, \|\mathsf{rot}_\ell(\mathbf{v})\|_\infty \le \|\mathbf{v}\|_\infty \sum_{\ell\in\mathcal{L}} |(\eta_\ell)_k|.$$

Table 5. Cleaning chain configuration. Each row i shows the input scale Δ_i of cleaning i, the rescaling prime q_i, the output scale $\widetilde{\Delta}_i^3/q_i$ (or $\widetilde{\Delta}_i^3$ when no rescale is performed), and the certified output precision $-\log_2\|\mathbf{e}_{\mathsf{cln},i}\|_\infty$ given by Corollary 5. Cleaning 4 is run without step 4.

i	$\log_2 \Delta_i$	$\log_2 q_i$	$\log_2(\widetilde{\Delta}_i^3/q_i)$	precision (bits)
1	30	30	60	16.41
2	60	120	60	31.24
3	60	90	90	60.90
4	90	–	270	120.21

Table 6. Per-step error decomposition of Corollary 5 (all entries in $\log_2$, computed at $\epsilon = \epsilon_{\mathsf{BTS}}/4$ and $\epsilon' = \epsilon/3$). $\widetilde{B}$ is the deterministic bound on the input $\|\widetilde{\mathbf{e}}\|_\infty$ of cleaning i; $3\widetilde{B}^2 + 2\widetilde{B}^3$ is the quadratic floor; B_1, B_2, B_3 are the key-switching and rescaling tails. The last column is the total Corollary bound on $\|\mathbf{e}_{\mathsf{cln},i}\|_\infty$ (step 3 only for cleaning 4).

i	$\widetilde{B}$	$3\widetilde{B}^2+2\widetilde{B}^3$	$(3+2\widetilde{B})B_1$	$2B_2$	B_3	$\|\mathbf{e}_{\mathsf{cln},i}\|_\infty$
1	−9.00	−16.41	−41.33	−71.92	−42.92	−16.41
2	−16.41	−31.24	−101.33	−161.91	−42.92	−31.24
3	−31.24	−60.90	−101.33	−161.91	−72.92	−60.90
4	−60.90	−120.21	−161.33	−251.91	–	−120.21

This bound is deterministic once the matrix M, the diagonal decomposition, and the plaintext scale Δ_M are fixed. In Section 7, we discuss how this deterministic encoding budget is instantiated in parameter selection. In particular, one may always use the universal worst-case bound $\|\eta_\ell\|_\infty \leq \frac{N}{2\Delta_M}$, whereas for concrete bootstrapping matrices we can compute the exact values of η_ℓ.

Table 7. Comparison of the empirical tail, Gaussian fit, and our bounds for the rescaling (Theorem 2) and key-switching (Theorem 3) errors. Entries are $\log_2$ of the error magnitude (bits).

	Rescaling			Key switching		
Tail-prob	Empirical	Gaussian	Thm 2	Empirical	Gaussian	Thm 3
2^{-5}	−17.88	−17.88	−16.93	−17.89	−17.90	−16.96
2^{-10}	−17.39	−17.38	−16.61	−17.44	−17.40	−16.63
2^{-64}	—	−16.05	−15.47	—	−16.06	−15.47
2^{-128}	—	−15.55	−14.99	—	−15.56	−14.99

Table 8. Validating PCMv bound from Theorem 8, per scaling factor Δ_M. Entries are $\log_2$ of the error magnitude (bits).

$\log_2 \Delta_M$	Tail-prob	Empirical	Gaussian	ecd	ks	total
20	2^{-5}	−17.49	−17.49	−17.49	−24.30	−17.48
20	2^{-10}	−17.48	−17.48	−17.49	−24.02	−17.47
20	2^{-64}	—	−17.47	−17.49	−22.95	−17.46
20	2^{-128}	—	−17.46	−17.49	−22.48	−17.45
25	2^{-5}	−20.56	−20.55	−20.59	−24.30	−20.48
25	2^{-10}	−20.54	−20.53	−20.59	−24.02	−20.46
25	2^{-64}	—	−20.44	−20.59	−22.95	−20.33
25	2^{-128}	—	−20.38	−20.59	−22.48	−20.25
30	2^{-5}	−24.73	−24.55	−25.50	−24.30	−23.78
30	2^{-10}	−24.38	−24.29	−25.50	−24.02	−23.58
30	2^{-64}	—	−23.39	−25.50	−22.95	−22.72
30	2^{-128}	—	−22.99	−25.50	−22.48	−22.31

Table 9. Validating Polynomial evaluation error including polynomial evaluation algorithm and double-angle formula introduced in Theorem 9 and 10. Entries are $\log_2$ of the error magnitude (bits).

Tail-prob	Empirical	Gaussian	Thm 9+ 10
2^{-5}	−16.51	−16.51	−13.55
2^{-10}	−16.05	−16.01	−13.22
2^{-64}	—	−14.67	−12.06
2^{-128}	—	−14.17	−11.58